People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research

University of Abdelhamid Mehri – Constantine 2

Faculty of New Technologies of Information and Communication (NTIC)

Department of Software Technologies and Information Systems (TLSI)
Department of Fundamental Computer Science and its Applications (IFA)

MASTER'S THESIS

to obtain the diploma of Master's degree in Computer Science

Option: Software Engineering and Intelligent Systems (ILSI)
Networks and Distributed Systems(RSD)

RequAI: Smart Recruitment Assistant – Chatbot Extension for Wassit Online

Realized by:

Benameur Khaoula

Nedjem Mohamed Anis

Under supervision of:

Telghamti Samira

June 2026

بسم الله الرحمن الرحيم

Acknowledgments

Before all and after all, and because everything seeks your supreme face, thank God, the Almighty, who gave us this dream and made it possible for us to realize it, gave us the effort, patience, and intelligence, and who subjugated it to us, and without him we would not have been victorious.

To Mrs. TELGHAMTI SAMIRA, our supervisor, who guided and supported us throughout the realization of this project. We sincerely thank her for her trust, her encouragement, and the autonomy she allowed us in developing our ideas. Her presence and advice were valuable throughout the different stages of this work.

We would also like to express our sincere thanks to Dr. ZAKARIA LAKHDARA for his valuable help and guidance, particularly in structuring this thesis and supporting us in the artificial intelligence aspects of the project. His advice and remarks were of great value in improving the quality and direction of our work.

We would like to express our sincere thanks to the members of the jury for your participation and valuable contributions to our study. Find, here, the expression of our sincere thanks for the honor you have given us by accepting and taking the time to read and evaluate this work.

Our gratitude also goes to all the teachers and professors who contributed to our academic training throughout our university journey. Their knowledge, dedication, and support have played an important role in shaping our skills and preparing us for this achievement.

Finally, we would like to thank everyone who supported us, directly or indirectly, during the completion of this thesis.

Dedication

To my beloved parents, Youcef and Siham, for their unconditional love, faith, encouragement, and endless support.

To my brothers, Oussama and Borhan, and my sisters, Oumaima and Manel, my best companions and forever friends.

To my dearest cousin and sister, Roudaina, and to my little cousin and brother, Skandar.

To the people who went out of their way to help me and be there for me, even from hundreds of kilometers away.

To my dear friends, my second family, each one of you by name.

To all of you, with a heart full of love and warmth.

To everyone who helped me stand up after each fall, and to everyone who has been part of this journey: thank you.

This accomplishment is as much yours as it is mine.

BENAMEUR KHAOULA

Dedication

In the name of Allah, the Most Gracious, the Most Merciful

All praise belongs to Allah, who granted me the ability, the blessing, and the strength to reach this moment who shaped me through every trial, tested me through every hardship, and taught me through every struggle. Today I do not merely close an academic chapter; I open an entirely new one.

I carry with me every challenge that shaped my character and strengthened my determination to continue forward. Every challenge along this journey became a lesson that shaped who I am today. The path was not always easy, but it taught me resilience, patience, and the value of hard work. I chose to challenge myself, grow beyond limitations, and pursue a path shaped by dedication and ambition. Like many meaningful journeys, this one came with moments of doubt, sacrifice, and perseverance that strengthened my resolve. My motivation has always extended beyond personal achievement, rooted in gratitude, growth, and the desire to contribute meaningfully. I hope this achievement serves as a reminder that perseverance, patience, and belief in oneself can open new paths, regardless of where one begins. This work is dedicated to everyone striving to build a better future through persistence and hope.

To My Father — Nedjem Belkacem Your name is my name. Your spirit walks with me in every step I take — a fighter, a poet. You taught me that words matter, that vision matters, even in absence. Your silence gave me strength; your struggle gave me purpose. You showed me what it means to carry fire and still speak with grace. Your blood runs in my veins; your dreams live on through me.

To My Sisters — Yassmin Sara My protectors, my voice of reason, my anchor in every storm. You saw me before anyone else did. You believed in me when I couldn't believe in myself, loved me without conditions, and held this family together with strength I cannot measure.

To My Brothers — Yasser, Mouiz Arkan The real ones. My equals. You held me down when the world turned its back. When critics grew loud, your loyalty grew louder. Every text every call, every laugh — you reminded me of who I was.

We made it To My Mother — Halima Ben Khdim Allah A name that carries its own prayer. You taught me that love is action, sacrifice is quiet, and faith is louder than fear. You don't just shape homes — you shape legacies. No spotlight, no applause; just love in its purest, most selfless form. The sacrifices you made, the tears you held back, the mornings you rose before everyone else — I feel the weight of all of it, and I honour every bit. You are my hero, and now, as a man, I finally understand the full measure of your strength. Everything I am is rooted in everything you gave.

To My Friends — Yasser Mohamed Not just part of this journey — you are its foundation. When the weight was heaviest, you were there. You gave me laughter in the darkest seasons, silence when I needed peace, and honesty, love and loyalty when the world gave me only noise.

To Haithem — A Special Friend Companion in Work We met during my internship — and what began as a professional encounter grew into something I didn't expect: a work partnership built on trust, and a friendship that feels like brotherhood. You understood the pressure without needing it explained, shared the load without being asked, and stood by me the way only a brother would. I am grateful for the work we did together, and even more grateful for the person you turned out to be.

The real honour: I never walked alone. You walked with me. Thank you for believing in me. My story is yours. We are here — and we are only getting started. — Peace —

NEDJEM MOHAMED ANIS

ملخص

تهدف هذه المذكرة إلى تصميم وتطوير مساعد ذكي للتوظيف تحت اسم RequAI ، موجه لدعم الباحثين عن عمل في سياق منصة Wassit Online وخدمات الوكالة الوطنية للتشغيل ANEM . يعتمد النظام على روبوت محادثة يسمى Requi ، يسمح للمستخدمين بطرح أسئلتهم بلغة طبيعية، والحصول على توجيه حول عروض العمل، إجراءات التسجيل، الوثائق المطلوبة، وخطوات الترشح. يعالج المشروع بعض حدود أنظمة البحث التقليدية، مثل الاعتماد على الكلمات المفتاحية والمرشحات الثابتة، من خلال إدماج تقنيات معالجة اللغة الطبيعية، تحليل السيرة الذاتية، استخراج المهارات، والمطابقة بين ملفات المترشحين وعروض العمل. كما يعتمد النظام على قاعدة معرفة خاصة بمجال التوظيف من أجل تقديم إجابات أكثر وضوحا وموثوقية. يهدف RequAI إلى تحسين تجربة الباحث عن عمل، تسهيل الوصول إلى المعلومات، ودعم رقمنة خدمات التوظيف في الجزائر.

الكلمات المفتاحية:

التوظيف الذكي، روبوت محادثة، معالجة اللغة الطبيعية، Wassit Online ، مطابقة الوظائف، RequAI

Abstract

This thesis aims to design and develop an intelligent recruitment assistant called RequAI, intended to support job seekers in the context of Wassit Online and ANEM-related services. The system includes a chatbot named Requi, which allows users to ask questions in natural language and receive guidance about job offers, registration procedures, required documents, and application steps. The project addresses some limitations of traditional job search systems, such as dependence on keywords and fixed filters, by integrating natural language processing, CV analysis, skill extraction, and candidate-job matching. The system also relies on a recruitment knowledge base in order to provide clearer and more reliable answers. RequAI aims to improve the job seeker experience, facilitate access to employment information, and support the digital transformation of recruitment services in Algeria.

Keywords: Intelligent recruitment, Chatbot, Natural Language Processing, Wassit Online, Job matching, RequAI

Résumé

Ce mémoire vise à concevoir et développer un assistant intelligent de recrutement appelé RequAI, destiné à accompagner les chercheurs d'emploi dans le contexte de la plateforme Wassit Online et des services liés à l'ANEM. Le système intègre un chatbot nommé Requi, qui permet aux utilisateurs de poser des questions en langage naturel et de recevoir des orientations concernant les offres d'emploi, les procédures d'inscription, les documents requis et les étapes de candidature. Le projet traite certaines limites des systèmes classiques de recherche d'emploi, notamment la dépendance aux mots-clés et aux filtres fixes, en intégrant le traitement automatique du langage naturel, l'analyse du CV, l'extraction des compétences et la mise en correspondance entre les profils des candidats et les offres d'emploi. Le système s'appuie également sur une base de connaissances dédiée au recrutement afin de fournir des réponses plus claires et plus fiables. RequAI vise à améliorer l'expérience du chercheur d'emploi, faciliter l'accès à l'information et soutenir la transformation numérique des services de recrutement en Algérie.

Mots clés : Recrutement intelligent, Chatbot, Traitement automatique du langage naturel, Wassit Online, Correspondance emploi-profil, RequAI

Table of Contents

List of Figures

List of Tables

General Introduction

Context and Motivation

In recent years, digital transformation has become an important direction for governments around the world. Public administrations are increasingly using digital technologies to improve the accessibility, efficiency, transparency, and quality of public services. This transformation aims to simplify administrative procedures, reduce delays, and offer citizens easier access to information and services.

Algeria is also involved in this global movement through several initiatives related to the modernization and digitalization of public services. In this context, digital transformation is considered an important step toward improving communication between citizens and public institutions, as well as enhancing the quality of administrative services [1]. However, this transformation still faces several challenges, such as unequal access to digital services, infrastructure limitations, and the digital divide [2].

The employment sector is also affected by this digital transformation. Recruitment and job search services are no longer limited to traditional procedures, since online employment platforms have become important tools for connecting job seekers and employers. These platforms can facilitate access to job opportunities, improve communication between candidates and recruiters, and reduce the time and cost of recruitment processes [3]. However, the effectiveness of such platforms depends not only on the availability of job offers, but also on the quality of the user experience, the ease of use, and the relevance of the results provided to users [4].

In Algeria, Wassit Online represents one of the main digital platforms related to public employment services. It is associated with the Agence Nationale de l'Emploi (ANEM) and aims to support job seekers and employers by facilitating access to employment offers and recruitment services. Through this platform, job seekers can search for job opportunities and access employment-related services online. However, like many classical employment platforms, the search process can still depend on manual input, keywords, and predefined filters.

These classical search and filtering methods may remain limited when users do not know the exact keywords to use, when job descriptions and user profiles use different terms, or when the system cannot understand the real intention behind the user's request. Traditional keyword-based search can therefore lead to irrelevant results or missed opportunities because it does not always capture semantic relationships between candidates' needs and job offers [5]. In addition, rigid filtering criteria can reduce the flexibility of job matching, especially in a dynamic labor market [6].

In addition to these general limitations, field information collected through an interview with an ANEM worker shows that some parts of the recruitment process still require manual intervention. Candidate lists generated by the system may sometimes include profiles that do not fully match the employer's requirements, which obliges ANEM agents to review them manually. The interview also highlighted the importance of considering not only technical qualifications, but also soft skills and candidate suitability for a given position. Moreover, although the platform acts as an intermediary between job seekers and employers, several steps of the process may still depend on agency-level procedures and presencial interactions.

For these reasons, there is a need for a more interactive and intelligent solution that can improve the way users access job information. An intelligent chatbot can provide a conversational interface that allows users to express their needs in natural language instead of relying only on manual search and fixed filters. Such a system can guide users during the job search process, help them clarify their preferences, and provide more personalized assistance. In the recruitment context, conversational AI can also help automate repetitive interactions and support users throughout the recruitment process [7]. Therefore, this project is motivated by the need to improve the job search and recruitment experience through the development of an intelligent chatbot inspired by the Wassit Online platform.

Problem Statement

Although digital employment platforms such as Wassit Online facilitate access to job offers and support the connection between job seekers and employers, the recruitment process can still face several limitations. In many classical platforms, job search and candidate matching depend mainly on manual input, predefined filters, and keyword-based search. These methods can be useful for basic search tasks, but they may not always provide accurate or personalized results when users express their needs in different ways or when job offers require more specific matching criteria.

This situation can make the job search process less efficient for job seekers. A candidate may not always know the exact keywords to use, or may describe a job need using general expressions, synonyms, or incomplete information. As a result, relevant job offers may be missed, while other results may not correspond well to the candidate's profile, skills, location, experience, or expectations. Previous studies also show that traditional keyword-based systems can fail to capture semantic relationships between job descriptions and candidate profiles, which may lead to inaccurate filtering and missed opportunities [5]. In addition, rigid filtering criteria can reduce the flexibility of job matching in a dynamic labour market [6].

These limitations can also affect employers and ANEM agents. From the employer's perspective, receiving candidate profiles that do not fully match the required criteria can increase the effort needed to identify suitable applicants. From the ANEM side, field information collected through an interview with an ANEM worker indicates that candidate lists generated by the system may still require manual review when the proposed profiles do not fully correspond to employer needs. This shows that the recruitment process is not only a matter of displaying job offers, but also of improving the relevance of matching and reducing unnecessary manual intervention.

Another important gap is that classical platforms may not fully understand the user's real intention, context, preferences, soft skills, professional attitudes, or suitability indicators for a given position. In practice, recruitment decisions may depend on more than technical qualifications alone. However, classical search and filtering systems generally remain limited to explicit criteria and do not provide a fully conversational or intelligent interaction that helps users clarify their needs step by step.

Therefore, the main problem addressed in this project is the lack of an intelligent and conversational support mechanism that can help users express their job search needs naturally, improve access to relevant employment information, and support better matching between job seekers and job offers in a recruitment context inspired by the Wassit Online platform.

The main research question of this project can be formulated as follows:

> How can an intelligent chatbot improve job search assistance, user guidance, and the relevance of recruitment matching in a context inspired by the Wassit Online platform?

Objectives of the Project

The main objective of this project is to design and develop an intelligent chatbot that supports job search and recruitment management in a context inspired by the Wassit Online platform. The proposed system aims to improve user assistance by offering a more interactive way to access employment information, express job search needs, and support better communication between job seekers, employers, and ANEM-related services.

More specifically, this project aims to:

- Assist job seekers through a conversational interface that makes the job search process simpler and more interactive;
- Allow users to express their needs using natural language instead of depending only on manual search, fixed keywords, and predefined filters;
- Identify important job search criteria such as professional domain, location, experience, salary expectations, skills, and user preferences;
- Support better matching between candidate profiles and job offers by considering both explicit criteria and user context;
- Provide users with reliable guidance about recruitment procedures, job search steps, and platform-related information;
- Help structure candidate information in a clearer way in order to support employers and reduce unnecessary manual effort for ANEM agents;
- Take into account, when relevant, additional suitability indicators such as soft skills, professional attitudes, and general compatibility with the requirements of a position;
- Improve the overall user experience of digital employment services by making access to job information more personalized, accessible, and efficient.

Project Challenges

Although the proposed project aims to improve job search assistance and recruitment management through an intelligent chatbot, its implementation in the Algerian employment context presents several challenges. These challenges are related not only to the technical development of the system, but also to the nature of recruitment data, the specificities of the local context, and the practical constraints of deploying an intelligent solution connected to public employment services.

One of the main challenges is the limited availability of structured and accessible recruitment data. An intelligent chatbot requires reliable information about job offers, candidate profiles, required skills, locations, procedures, and recruitment rules. However, such data may not always be available in a clean, complete, or directly usable format. In addition, access to real data from public employment platforms may be restricted for administrative, privacy, or security reasons. This makes the development and evaluation of the system more difficult, especially when the goal is to provide relevant and personalized responses.

Another important challenge is the complexity of natural language understanding in the Algerian context. Users may interact with the chatbot in different languages, including Arabic, French, English, Algerian dialect, or even mixed language forms. This multilingual and informal communication can make intent recognition, entity extraction, and response generation more difficult. The system must therefore be able to understand different ways of expressing the same need, such as searching for a job, asking about required documents, or requesting information about recruitment procedures.

The project also faces challenges related to job matching. Recruitment is not limited to matching keywords between a candidate profile and a job offer. A good matching process should consider several factors, such as skills, experience, location, education level, salary expectations, and professional preferences. In addition, field information collected from ANEM shows that candidate lists may still require manual review when the proposed profiles do not fully match employer needs. This means that the system must go beyond simple filtering and try to improve the relevance of recommendations while remaining realistic and explainable.

Another difficulty concerns the integration of soft skills and candidate suitability. In practice, employers may look not only for technical qualifications, but also for communication skills, motivation, professional behavior, and compatibility with the position. However, these elements are difficult to measure automatically and may require careful design to avoid inaccurate or unfair conclusions. Therefore, the chatbot can support the collection and organization of such information, but the final evaluation should remain controlled and interpreted carefully.

Privacy and data protection represent another major challenge. Recruitment systems deal with sensitive personal information, such as identity details, education, work experience, skills, and career preferences. For this reason, the chatbot must be designed with attention to data security, confidentiality, and responsible use of candidate information. This is especially important if the system is intended to be connected to real employment services or public platforms.

Finally, the practical deployment of such a project in Algeria may face organizational and infrastructural constraints. Some recruitment procedures still depend on agency-level validation and presencial interactions. Therefore, the chatbot cannot completely replace existing administrative processes. Instead, it should be seen as a support tool that improves access to information, guides users, reduces repetitive questions, and helps structure recruitment-related data. These challenges show that the implementation of the proposed system is not a simple task, but they also justify the importance of developing an intelligent assistant adapted to the needs and realities of the Algerian employment context.

Structure of the Thesis

This thesis is organized into two main chapters, followed by a general conclusion and additional annexes.

The first chapter presents the state of the art related to the main concepts of the project. It introduces recruitment and e-recruitment, online employment platforms, and the Wassit Online platform. It also discusses chatbots, conversational artificial intelligence, natural language processing, transformer-based models, large language models, and retrieval-augmented generation. This chapter provides the theoretical background needed to understand the technologies and concepts used in the proposed system.

The second chapter presents the proposed contribution. It describes the general idea of the RequAI system, its architecture, main functional modules, methodology, data requirements, and implementation choices. It also presents the development environment, the two chatbot approaches explored during experimentation, the system implementation, and the evaluation results obtained from both approaches. The first approach is presented as a structured classical NLP-based pipeline, while the second approach extends the system with multilingual language detection, intent classification, and transformer-based entity extraction.

Finally, the thesis ends with a general conclusion that summarizes the work carried out, highlights the main contributions of the project, discusses the limitations of the current implementation, and presents possible future perspectives for improving and extending the RequAI system.

Chapter 1

State of the Art

Introduction

Recruitment management has evolved with the development of digital platforms and intelligent information systems. Online employment platforms centralize job offers, candidate profiles, and recruitment procedures in a single environment. However, many systems still depend on structured forms, keyword-based search, fixed filters, and manual processing. These limitations can reduce search flexibility, affect candidate-job matching, and limit user interaction.

In this context, conversational agents offer a relevant solution for improving access to recruitment services. A chatbot can act as an intelligent interface between the user and the system by interpreting natural language queries, identifying user intentions, extracting useful information, and generating appropriate responses. In a recruitment platform such as Wassit Online, this type of system can assist job seekers in searching for opportunities, understanding procedures, and receiving guidance.

This chapter presents the main concepts, technologies, and existing works related to the development of an intelligent chatbot for recruitment management. It first introduces recruitment, e-recruitment, online employment platforms, and the Wassit Online platform. It then discusses chatbots and conversational AI, starting with traditional rule-based systems and their limitations.

The chapter also presents Natural Language Processing (NLP), including text representation, intent recognition, entity extraction, and multilingual processing. Transformer-based models such as BERT and GPT are introduced because of their role in contextual understanding and text generation. Large Language Models (LLMs) are then discussed, with attention to their advantages and limitations, including hallucinations, lack of grounding, and privacy concerns.

Finally, Retrieval-Augmented Generation (RAG) is presented as a technique that combines information retrieval with text generation to produce more reliable and knowledge-grounded answers. The chapter concludes with relevant work on recruitment chatbots, AI-based matching, skill extraction, resume-job matching, multilingual support, and RAG-based systems.

The objective of this chapter is to understand the technologies used in intelligent recruitment systems, analyze their advantages and limitations, and justify the use of an intelligent chatbot to improve job search and recruitment processes.

1.1 Recruitment and E-Recruitment

This section introduces the concept of recruitment and explains its evolution from traditional methods to digital recruitment practices. It also discusses the role of online employment platforms and the main limitations of classical job search and matching systems.

1.1.1 Definition of Recruitment

Recruitment is a fundamental human resource management process through which an organization attracts, identifies, and selects suitable candidates for available job positions. It connects organizational needs with the skills, qualifications, experience, and profiles of potential candidates. Recruitment is therefore not limited to announcing vacancies or collecting applications; it also involves analyzing job requirements, reaching appropriate candidates, and supporting hiring decisions.

In modern organizations, recruitment plays an important role in matching the right candidates with the right positions. The quality of recruitment affects human resource management, since unsuitable selection may lead to poor performance, high turnover, and additional administrative costs. Therefore, recruitment can be considered both an administrative and strategic process.

With digital technologies, recruitment practices have progressively evolved from traditional manual procedures toward electronic recruitment methods. Online employment platforms now connect job seekers and employers, facilitate access to opportunities, and reduce time and cost in the recruitment process [3]. However, the effectiveness of recruitment systems also depends on usability, trust, and the relevance of matching results [4]. This evolution prepares the transition toward e-recruitment, online employment platforms, and intelligent recruitment systems.

1.1.2 From Traditional Recruitment to E-Recruitment

Traditional recruitment was generally carried out through physical agencies, newspaper advertisements, paper-based CVs, direct contact, and manual screening. Job seekers often had to visit employment offices or companies in person, while employers relied on administrative staff to collect applications, review documents, and communicate with candidates. Although this approach allowed direct interaction, it was time-consuming and limited by geographical distance, slow communication, and administrative workload.

These limitations became more visible as labor markets expanded and the number of applications increased. Manual screening makes it difficult to process large volumes of applications efficiently, while physical procedures can reduce access for candidates living far from agencies or employers. Traditional methods may also delay the matching process when communication and document processing depend mainly on face-to-face interactions or paper-based records.

E-recruitment emerged as a response to these limitations by using digital tools and online platforms to support recruitment activities. It allows employers to publish job offers online, receive digital applications, store CVs, communicate electronically with candidates, and filter profiles more rapidly. Online recruitment platforms now play an important role in connecting employers and job seekers by expanding access to opportunities and reducing time and cost [3].

However, e-recruitment does not remove all recruitment challenges. Large numbers of online applications may create information overload for recruiters. In addition, many digital recruitment systems still depend on keyword-based search and rigid filters, which may fail to capture the real relationship between a candidate's profile and the requirements of a job offer [5]. The effectiveness of e-recruitment also depends on user experience, trust, and the relevance of results [4]. In some contexts, unequal access to digital services and internet infrastructure may limit inclusiveness [2].

Therefore, the evolution from traditional recruitment to e-recruitment represents an important step toward more accessible and efficient recruitment services. Nevertheless, remaining limitations show the need for more intelligent approaches capable of improving interaction, filtering, and matching quality.

1.1.3 Online Employment Platforms

Online employment platforms are digital systems designed to connect job seekers, employers, and, in some cases, public employment-service institutions. They represent a main form of e-recruitment by providing a centralized environment where job offers, candidate profiles, applications, and recruitment-related information can be managed. Through these platforms, employers publish vacancies, while job seekers search for opportunities, submit applications, and update their profiles.

Their role goes beyond publishing job advertisements. Online platforms can store CVs, facilitate communication between candidates and employers, and help recruiters manage large numbers of applications. By digitizing these tasks, they improve vacancy visibility, reduce administrative workload, and make recruitment faster and more accessible. Shrivastav et al. [3] highlight the role of online platforms in modern recruitment by facilitating communication, expanding access, and reducing time and cost.

Online employment platforms may be operated by private companies, recruitment agencies, public employment services, or government institutions. In the public sector, they support employment policies and improve access to labor-market services by helping job seekers access official offers and administrative information, while allowing institutions to organize recruitment activities more efficiently.

Despite their advantages, these platforms still present limitations. Large numbers of online applications can create information overload, and many platforms rely on keyword-based search and rigid filtering mechanisms, which may fail to capture the semantic relationship between candidate skills and job requirements [5]. As a result, relevant candidates may be ignored, while unsuitable profiles may appear in results. Their effectiveness also depends on usability, trust, and matching relevance [4]. Unequal access to digital services may further limit inclusion in contexts where the digital divide remains a challenge [2].

These limitations show that online employment platforms improve recruitment accessibility and efficiency, but they may still require manual verification, better user assistance, and more intelligent matching mechanisms. In Algeria, Wassit Online represents an important public online employment platform related to the National Employment Agency (ANEM). Therefore, the next section focuses on Wassit Online, its role, services, and challenges.

1.1.4 Limitations of Traditional Job Search and Matching Systems

Traditional job search methods and basic online matching systems have improved access to employment information, but they still present several limitations. In traditional recruitment, job seekers often depend on physical agencies, printed advertisements, paper documents, and direct contact with employers or administrative services. This process can be slow, geographically limited, and difficult for users who need continuous guidance during application procedures.

Although online employment platforms have reduced some constraints, many basic digital recruitment systems remain limited by their search and filtering mechanisms. Job search often depends on keywords, fixed criteria, and predefined filters. These mechanisms can be useful for simple queries, but they may fail to understand the semantic relationship between candidate skills, experience, profile, and job requirements [5]. Consequently, relevant candidates may be excluded because they do not use the exact expected terms, while unsuitable candidates may still appear.

Another limitation is the mismatch between candidate profiles and employer needs. Traditional matching systems often focus on explicit information such as diploma, location, job title, or years of experience, while ignoring implicit skills, transferable competencies, professional context, or personal suitability. The OECD highlights that traditional job-matching systems can be limited when they rely on rigid criteria in dynamic labor markets [6].

Online recruitment can also generate information overload. Employers may receive many applications, while job seekers may face many offers without enough personalized guidance. This increases the need for manual verification. Studies on online job search engines show that usability and user experience affect platform effectiveness [8], while e-recruitment adoption depends on trust, perceived value, and matching relevance [4].

In public employment platforms, these limitations can be more visible because they serve a large and diverse population. They improve access to job offers and administrative services, but may still require human intervention, manual checking, and in-person procedures. In Algeria, unequal access to digital services and infrastructure can also reduce the inclusiveness of online public services [2].

Field observations related to the ANEM/Wassit recruitment process also show that filters may sometimes be too strict or not fully aligned with employer needs. Candidate lists may include unsuitable profiles, requiring manual review by agency staff. Moreover, soft skills, psychological suitability, and personality-job fit remain difficult to evaluate automatically using basic filtering systems. Some procedures may also remain manual or in-person, which limits full automation.

These limitations can reduce matching quality, slow recruitment procedures, and affect user satisfaction. Therefore, more intelligent recruitment support tools are needed to understand user queries, extract relevant information, improve matching, and provide immediate guidance. This motivates the study of conversational AI, NLP-based systems, LLMs, and RAG for recruitment support.

1.2 Wassit Online Platform

This section focuses on Wassit Online as the main Algerian public employment platform related to ANEM. It presents the platform, its main services, its role in the Algerian employment context, and the limitations that motivate the need for intelligent assistance tools.

1.2.1 Overview of Wassit Online

Wassit Online is an official Algerian online employment platform related to the National Employment Agency, known as ANEM. It represents a public digital service dedicated to employment intermediation, unlike private job portals operated mainly by commercial recruitment companies. The platform is part of the broader digitalization of employment services in Algeria, where public institutions increasingly use digital tools to improve access to administrative and labor-market services [1, 2].

As a public employment platform, Wassit Online facilitates access to employment services for both job seekers and employers. It supports the publication and consultation of job offers, job seeker registration, employer registration, and interaction with employment services. The official Wassit Online website provides spaces for job seekers and employers, as well as public access to available job offers [9]. This shows that the platform acts as an intermediary between labor supply and demand.

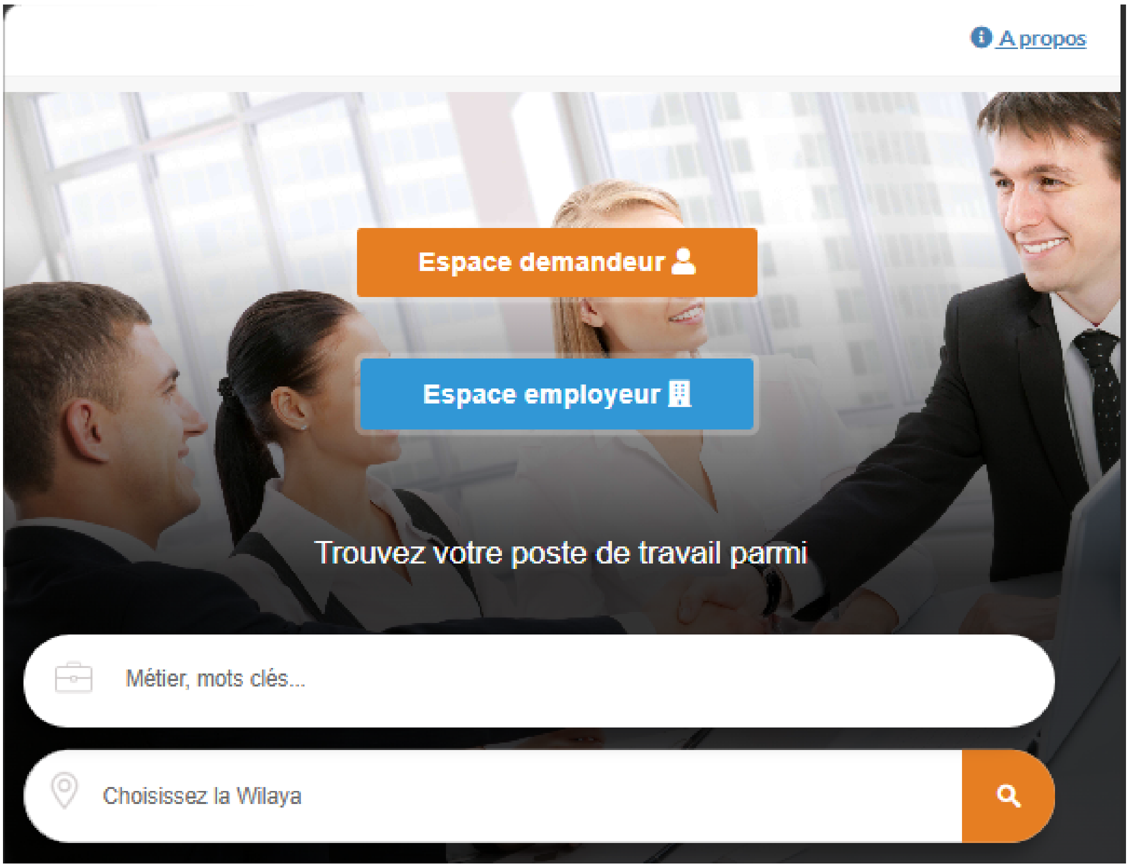

Figure 1.1: Official homepage of the Wassit Online platform. [9].

Wassit Online is also connected to the institutional role of ANEM in the Algerian labor market. According to official information from the Ministry of Labour, ANEM monitors the national employment market and provides recruitment services for job seekers and employers [10]. In this context, Wassit Online can be understood as a digital extension of public employment services.

For this thesis, Wassit Online represents the target context in which an intelligent recruitment chatbot could be integrated or used as a complementary support tool. Such a chatbot could help users navigate the platform, understand procedures, access information more easily, and interact with employment services in a more guided way.

1.2.2 Main Services and Functionalities

Wassit Online provides digital services that support the relationship between job seekers, employers, and ANEM employment services. The platform gives public access to available job offers and allows users to search for vacancies by occupation, keyword, domain, and wilaya [11]. It also presents recent offers and offers available for online application, showing its role as a digital access point for employment opportunities [9].

For job seekers, Wassit Online provides account creation, authentication, and access to a personal space. It includes a registration interface and a login page that allow users to connect to their accounts [12, 13]. Through these functions, job seekers can access employment services online, consult offers, and interact with the platform without depending only on physical agency visits. The platform also supports administrative follow-up services, such as renewing a job request and updating a phone number [9].

For employers, Wassit Online provides registration and authentication services. The employer registration interface is organized in several steps, including general information, employer information, and CNAS-related information [14]. This shows that the platform supports employers in their interaction with ANEM employment services and helps organize recruitment-related exchanges between employers, candidates, and public institutions.

The platform also supports online postulation through the "FORSATI" service, which allows autonomous job seekers to apply for available job offers on Wassit Online [15]. This functionality moves part of the recruitment interaction from a fully physical process toward an online process. However, digital services do not completely remove the role of ANEM agencies or human verification, since public employment platforms may still require institutional supervision and administrative validation.

Overall, Wassit Online acts as a public digital employment service. It facilitates access to job offers, supports job seeker and employer accounts, enables online consultation and postulation, and structures the interaction between users and ANEM services.

1.2.3 Role of Wassit Online in the Algerian Employment Context

Wassit Online plays an important role within the Algerian public employment system because it represents a digital extension of ANEM services. As a public employment platform, it is not only a job portal for publishing offers, but also a tool that supports employment intermediation between job seekers, employers, and public employment agencies. Through this platform, part of the relationship between labor supply and demand is transferred to a digital environment while remaining connected to ANEM's institutional framework.

This role can be understood within the broader digital transformation of public services in Algeria. Public institutions increasingly rely on digital tools to modernize administrative services, improve accessibility, and reduce dependence on fully physical procedures [1, 2]. In this context, Wassit Online contributes to the digitalization of employment services by allowing users to access employment-related information and services online.

Wassit Online supports the connection between job seekers and employers. For job seekers, it facilitates access to job offers and employment information. For employers, it provides a digital channel to interact with ANEM services and publish or manage recruitment needs. This intermediary role is important because employment services in Algeria are linked to public institutions and national employment policies. According to official information, ANEM is responsible for organizing knowledge about the national employment market and supporting the connection between job seekers and employers [10].

By providing online access to employment services, Wassit Online can help reduce some geographical and administrative barriers. Job seekers may consult offers and access certain services without being fully dependent on physical movement to agencies, while employers can use the platform as a digital access point for recruitment procedures.

For this thesis, Wassit Online is important because it represents the target environment in which an intelligent recruitment chatbot could provide added value. Since the platform already connects job seekers, employers, and ANEM services, a chatbot could complement its role by improving guidance, simplifying access to information, and making user interaction more accessible.

1.2.4 Limitations and Challenges of the Current Platform

Wassit Online represents an important step toward the digitalization of employment services in Algeria. However, like many public online employment platforms, it can still face functional, organizational, and user-experience challenges.

One possible limitation is the dependence on structured forms and predefined search criteria. For example, the public job search interface allows users to search using occupation, keyword, domain, and wilaya [11]. These fields are useful for organizing job offers, but they may not capture the full context of a candidate's profile or the detailed expectations of an employer. Recruitment matching often depends not only on explicit criteria such as diploma, job title, or location, but also on skills, experience, motivation, and professional context.

Basic digital matching systems may also suffer from keyword-based search and rigid filtering. Such systems can fail to identify semantic relationships between candidate profiles and job requirements, which may lead to inaccurate filtering or missed opportunities [5]. The OECD also notes that traditional matching approaches based on rigid criteria can be limited in dynamic labor markets [6]. As a result, a platform may return profiles that do not fully correspond to employer needs or exclude candidates who do not use the exact expected terms.

Another challenge concerns user guidance and interaction. Online platforms provide access to services, but users may still need help understanding procedures, choosing the correct service, following application steps, or obtaining answers to specific questions. Without immediate conversational assistance, job seekers may still depend on agency staff or external explanations. Usability and user experience are important factors in the effectiveness of employment platforms [8], while e-recruitment adoption also depends on trust, perceived usefulness, and result relevance [4].

Public employment platforms may also require human verification and institutional intervention. Some administrative procedures in the Algerian employment system remain linked to employment agencies and official validation processes [10]. Therefore, the digital platform does not replace ANEM agents, but supports and extends public employment services while maintaining human supervision.

Field observations related to the ANEM/Wassit recruitment process suggest that filters may sometimes be too strict or not fully aligned with employer needs. Candidate lists may include unsuitable profiles, requiring manual review. Soft skills, motivation, psychological suitability, and personality-job fit also remain difficult to evaluate automatically using basic forms and filters.

Another broader challenge is the digital divide. Although online services can improve access to employment information, unequal access to internet infrastructure and digital tools may limit inclusion. In Algeria, digitalization supports public service modernization, but access inequalities and infrastructure limitations can still affect inclusiveness [2].

Overall, these limitations can affect matching quality, slow parts of the recruitment process, and reduce user satisfaction. They do not reduce the importance of Wassit Online, but they show that additional intelligent support tools could improve its effectiveness.

Conversational AI, NLP, LLMs, and RAG can therefore be studied as possible approaches to improve guidance, understand user requests, support relevant matching, and provide reliable answers.

1.3 Chatbots and Conversational AI

This section presents chatbots and conversational AI as technologies that can improve user interaction with digital services. It first introduces their general role, then discusses different chatbot approaches and the NLP techniques used to support more flexible conversations.

1.3.1 Definition and Role of Chatbots

A chatbot is a conversational software system designed to interact with users through natural language. It allows users to ask questions, request information, or perform simple tasks using text or voice-based communication. Unlike traditional graphical interfaces, where users navigate menus and forms, chatbots provide a more interactive mode of communication by simulating dialogue between the user and the system.

Chatbots play an important role in human-computer interaction because they make digital services easier to access. They can answer frequently asked questions, guide users through procedures, provide assistance, and automate repetitive tasks. They also improve service availability because they can support users at any time and respond to several users simultaneously.

The use of chatbots has expanded across many domains, including customer service, education, healthcare, public services, and recruitment [16]. In public services, chatbots can help citizens access administrative information and understand procedures more easily. This diversity of applications shows that chatbots can be adapted to contexts where users need fast and guided access to information.

In recruitment, chatbots can support both job seekers and employers. For job seekers, they can provide information about job offers, application procedures, required documents, and digital employment platforms. For employers, they can assist with frequent questions and support the management of interactions with candidates. Conversational AI in recruitment can also reduce repetitive communication tasks and improve recruitment service efficiency [7].

However, chatbot effectiveness depends on its design and underlying technology. Early chatbots were often based on predefined rules and scripted responses, which made them useful for repetitive interactions but limited when users asked unexpected or complex questions [16]. More advanced chatbots integrate NLP techniques to better interpret user input and provide relevant responses. This evolution is important for recruitment platforms, where users may express their needs in different ways.

In this thesis, chatbots are relevant because they can act as an intelligent interface between users and Wassit Online. By helping users search for information, understand procedures, and interact more naturally with digital employment services, a chatbot can improve accessibility and user experience.

1.3.2 Traditional Rule-Based Chatbots

This subsection presents traditional chatbot systems that rely on predefined rules and fixed dialogue paths. It explains how these systems work, their advantages in controlled contexts, and their limitations when user requests become more complex.

1.3.2.1 Rule-Based Dialogue Flow and Decision Trees

Traditional rule-based chatbots rely on predefined rules and scripted dialogue paths. The conversation is designed in advance, and the chatbot follows a fixed structure to decide which response should be given. The interaction is usually organized through a dialogue flow, where each step guides the user toward a specific answer, service, or action.

A common way to represent this interaction is the decision tree. In a decision tree, the conversation is structured as branches where each user input, selected option, keyword, or answer leads to a specific next step. For example, if the user selects registration, documents, or job search, the chatbot follows the corresponding branch and provides the predefined response.

Rule-based dialogue flows often depend on menus, buttons, keywords, pattern matching, or if–then rules. When the input matches a predefined condition, the chatbot returns the associated response. This makes chatbot behavior predictable and easy to control when expected requests are limited. Caldarini et al. [16] explain that early chatbot systems were mainly based on predefined rules and suited to simple interactions. Similarly, Dogan and Gurcan [17] show that rule-based chatbots remain useful in restricted domains where requests are routine.

This structure is commonly used in FAQ systems, basic customer support, service navigation, appointment booking, and simple information retrieval. In recruitment, a rule-based chatbot could guide users through fixed questions related to registration steps, required documents, application procedures, or agency working hours.

Therefore, rule-based dialogue flows and decision trees provide a simple and controlled way to design chatbot interactions in predictable service environments.

1.3.2.2 Advantages of Rule-Based Chatbots

Despite their simple design, rule-based chatbots can be useful in well-defined contexts. Since responses are based on predefined rules or scripts, their behavior is predictable, making them suitable when user requests are known in advance and answers must remain consistent.

One main advantage is control. Each response can be written, verified, and aligned with the intended service procedure before deployment. This is useful in administrative and public services, where information must remain accurate and consistent. Caldarini et al. [16] explain that rule-based chatbots are effective for simple and repetitive tasks, especially when requests are predictable.

Rule-based chatbots are also easier to design and implement than advanced AI-based systems. They do not require large datasets, complex model training, or high computational resources. Their dialogue logic is transparent because each rule can be inspected and modified manually, which makes testing and maintenance easier when the domain is limited.

Another advantage is consistency. Since the chatbot follows predefined rules, it can provide the same answer to the same type of question. Dogan and Gurcan [17] show that rule-based chatbots can be effective in restricted service contexts with routine questions. This makes them useful for FAQs, form guidance, service navigation, appointment scheduling, and simple procedural assistance.

In recruitment, a rule-based chatbot can answer fixed questions such as how to register, which documents are required, how to consult job offers, what steps are needed to apply, or when an employment agency is open. In public employment services, this controlled behavior is useful because answers must remain aligned with official procedures.

1.3.2.3 Limitations of Rule-Based Chatbots

Although rule-based chatbots are useful in controlled contexts, their performance depends strongly on predefined rules, keywords, and dialogue paths. They can respond correctly only when user input matches scenarios anticipated by the designer. When users ask questions unexpectedly, use incomplete expressions, or leave the predefined flow, the system may fail.

A main limitation is the weak ability to understand natural language variations. Users may express the same intention using different words, spelling mistakes, informal expressions, mixed languages, or dialectal forms. A system based on exact keywords or patterns may not recognize the real intention. Caldarini et al. [16] explain that rule-based chatbots are limited when handling complex or unexpected queries.

Another limitation is the lack of adaptability. Rule-based chatbots do not learn automatically from new interactions and cannot easily adapt to new questions without manual updates. As the domain grows, the number of rules and branches increases, making maintenance more difficult. Dogan and Gurcan [17] also highlight that rule-based systems are mainly suitable for routine interactions.

Rule-based chatbots also have difficulty managing complex conversations. They usually treat each input according to predefined conditions and may not preserve enough context from previous exchanges. This limits personalized or context-aware guidance.

In recruitment, these limitations are important because job seekers may ask diverse questions about job offers, registration, documents, application status, eligibility, and agency services. They may also use different languages, informal wording, or incomplete questions. In Wassit Online, a purely rule-based chatbot may help with fixed questions, but it may be insufficient when users need flexible answers, semantic understanding, or guidance based on their situation.

Therefore, while rule-based chatbots remain useful for simple tasks, their limitations motivate more advanced conversational approaches based on NLP and artificial intelligence.

1.3.3 Natural Language Processing for Chatbots

This subsection explains how NLP allows chatbots to process user language more intelligently than rule-based systems. It presents the main NLP tasks needed in a recruitment chatbot, including text preprocessing, intent recognition, entity extraction, and multilingual processing.

1.3.3.1 Text Preprocessing and Representation

Natural Language Processing (NLP) allows chatbots to process and interpret user language more flexibly than rule-based systems. Instead of depending only on fixed keywords or dialogue paths, NLP-based chatbots can analyze user input and prepare it for automatic understanding. Before identifying intent or extracting entities, the input text is usually cleaned, normalized, and transformed into a computational form.

Text preprocessing prepares raw user input by reducing noise and standardizing text. Common operations include lowercasing, tokenization, punctuation removal, stop-word handling, stemming or lemmatization, and normalization of spelling variations. Depending on the application, preprocessing may also handle numbers, abbreviations, special characters, and informal expressions.

The choice of preprocessing techniques depends on the language, task, and chatbot objective. For example, a chatbot for administrative procedures may need to preserve numbers, dates, or document names. In a recruitment chatbot, preprocessing can help analyze questions related to job offers, registration, documents, skills, locations, experience, and application procedures.

After preprocessing, text must be represented in a machine-readable form. Traditional methods include bag-of-words and TF-IDF, where text is represented according to word frequency or term importance. Word embeddings represent words as dense vectors that capture semantic relationships. More recent NLP systems use contextual representations, where the meaning of a word depends on its surrounding context [18].

Good text representation directly affects downstream tasks. If user input is poorly represented, the chatbot may fail to understand the request. This is important in recruitment, where users may express similar needs in different ways, such as asking for a job in a specific city, requesting documents, or asking about application conditions.

In a multilingual context such as Algeria, preprocessing and representation are more challenging. Users may write in Arabic, French, English, Algerian Darja, or mixed forms. They may also use informal spelling, transliteration, or code-switching. Therefore, NLP-based recruitment chatbots need methods that support linguistic variation and prepare input for later analysis.

Once the text is cleaned and represented, the chatbot can perform tasks such as intent recognition and entity extraction.

1.3.3.2 Intent Recognition

Intent recognition is the NLP task of identifying the user's goal from an input message. In task-oriented chatbots, it helps the system understand what the user wants and decide which response, service, or action should follow. It is therefore central to chatbots designed for structured assistance.

After preprocessing and representation, the user message can be classified into predefined intent categories. These categories depend on the application domain and supported services. In a recruitment chatbot, possible intents may include searching for job offers, asking about registration, required documents, application procedures, agency location or opening hours, chatbot capabilities, greeting, goodbye, and fallback cases.

Several approaches can be used for intent recognition. Simple systems rely on rule-based matching, while more advanced systems use traditional machine learning classifiers, deep learning models, or language-model-based methods. Modern NLP models improve the ability to capture contextual meaning, which can support more accurate classification [18, 19].

Correct intent recognition improves the chatbot's ability to provide relevant answers and guide users efficiently. For example, "I want to find a job in Constantine" should be recognized as a job search request, while "What documents do I need to register?" should be identified as a request about required documents. In both cases, the correct intent allows the chatbot to select an appropriate response or ask relevant follow-up questions.

Errors in intent recognition can lead to irrelevant answers, misunderstanding of user needs, or incorrect guidance. This can reduce user satisfaction, especially in public employment services where users need clear and reliable assistance. Therefore, intent recognition must be designed according to the target domain, expected questions, and available services.

In addition to recognizing the general purpose of a message, the chatbot may also need to identify specific information such as location, job title, skill, experience, or document type. This is related to entity extraction.

1.3.3.3 Entity Extraction

Entity extraction is an NLP task that identifies important information units from user input. While intent recognition determines the general purpose of the message, entity extraction identifies details needed to answer the user or execute an action. For example, if the user asks for job offers in a specific city, the intent may be job search, while the city name is extracted as an entity.

In task-oriented chatbots, entity extraction is related to named entity recognition and slot filling. Named entity recognition detects elements such as names, dates, locations, numbers, organizations, and document names. Slot filling uses these elements to complete the information required by the system. Thus, entity extraction allows the chatbot to move from general understanding to a more precise interpretation.

In recruitment, useful entities may include job title, location, skills, years of experience, education level, contract type, salary expectations, required documents, agency name, agency location, and application status. For example, in "I am looking for a developer job in Algiers with two years of experience", the chatbot may extract *developer* as job title, *Algiers* as location, and *two years* as experience.

Extracting entities can improve response relevance. It helps the system filter job offers, provide precise information, ask only for missing details, guide users through procedures, and personalize recommendations. In recruitment systems, information extraction is important because matching quality depends on identifying and structuring relevant information from job offers and candidate profiles [20]. Intelligent job-matching approaches also require skills and profile characteristics to compare candidates with job requirements [21].

Entity extraction can use rule-based patterns, dictionaries, or gazetteers for terms such as city names, document types, or job categories. More advanced systems use machine learning or deep learning models trained on labeled examples. Language-model-based approaches can improve extraction by using context, especially when the same word has different meanings depending on the sentence [18].

However, entity extraction is difficult in real chatbot interactions. User messages are often short, incomplete, informal, or misspelled, and may contain abbreviations, mixed languages, or dialectal expressions. In a recruitment chatbot for Wassit Online, users may write in Arabic, French, English, Algerian Darja, or mixed forms, making extraction more complex.

1.3.3.4 Multilingual NLP

Multilingual NLP refers to the ability of NLP systems to process and respond to input in different languages. In chatbots, this is important because users do not always communicate in one language. They may use formal language, informal expressions, dialects, or mixed-language forms.

This is particularly relevant in Algeria, where users may communicate in Arabic, French, English, Algerian Darja, Arabizi, or mixed expressions. For a recruitment chatbot, job seekers may ask about job offers, registration, documents, skills, locations, agency services, or application steps using different linguistic forms. Supporting multilingual interaction can improve accessibility and user experience.

Multilingual NLP in chatbots can involve language detection, multilingual preprocessing, intent recognition, and entity extraction across languages. In some cases, translation or cross-lingual understanding can allow the chatbot to process input in one language and retrieve or generate an answer in another. Multilingual models such as multilingual BERT have shown the ability to transfer knowledge across languages [22].

However, multilingual NLP introduces challenges. Some languages and dialects have fewer datasets and NLP tools. Dialectal Arabic and Algerian Darja are often written informally and without standardized spelling. Arabizi adds another difficulty because Arabic or Darja expressions may be written using Latin characters and numbers. Code-switching between Arabic, French, English, and Darja can also affect language detection, intent recognition, and entity extraction.

For a recruitment chatbot linked to Wassit Online, multilingual NLP is important because the system should remain accessible to diverse users. If the chatbot cannot handle linguistic variation, it may misunderstand requests or fail to provide useful guidance. Therefore, multilingual NLP contributes to more inclusive and realistic recruitment chatbots.

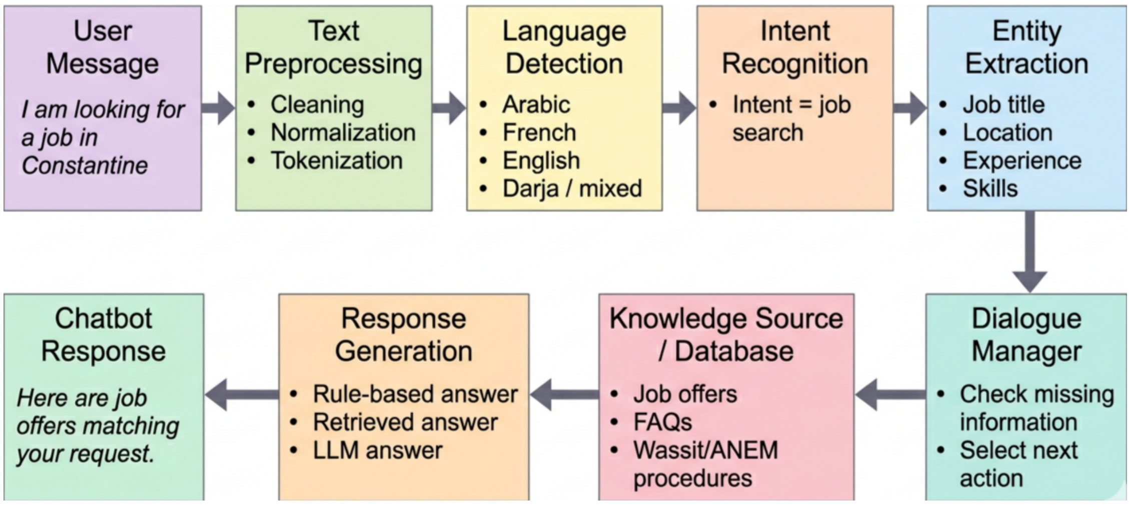

Figure 1.2: NLP pipeline for a recruitment chatbot, showing the main steps from user message processing to response generation.

1.3.4 Transformer-Based Language Models

This subsection presents Transformer-based models as an important evolution in modern NLP. It explains the Transformer architecture and introduces models such as BERT and GPT, which are widely used for contextual understanding and text generation.

1.3.4.1 Transformer Architecture

The Transformer architecture is one of the main foundations of modern NLP and recent language models. It was introduced by Vaswani et al. [23] as an alternative to earlier sequence-processing models based mainly on recurrent structures. Instead of processing text word by word in strict sequence, the Transformer uses attention mechanisms to model relationships between elements of an input sequence.

The central idea of the Transformer is self-attention. It allows the model to assign different levels of importance to different words in the same sequence. This helps capture long-distance dependencies and contextual relationships, which are important for natural language understanding.

The original Transformer follows an encoder-decoder structure. The encoder receives the input sequence and transforms it into contextual representations, while the decoder uses these representations to generate an output sequence. This structure was originally designed for sequence-to-sequence tasks such as machine translation, but its principles are now widely used in NLP.

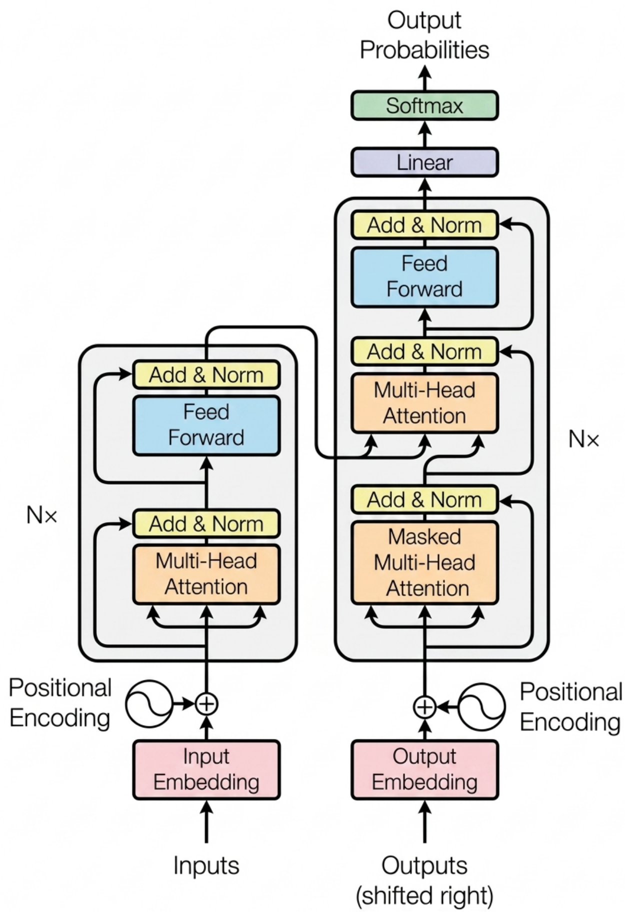

Figure 1.3: Transformer model architecture. Adapted from Vaswani et al. [23].

Several components make the Transformer effective. Multi-head attention allows the model to analyze input from different perspectives. Positional encoding provides word-order information, since the Transformer does not process tokens sequentially. Feed-forward layers refine the internal representations produced by attention. Together, these components allow parallel and context-aware language processing.

Transformers are useful for language understanding and generation tasks, including classification, question answering, information extraction, and text generation. In chatbot systems, Transformer-based models improve intent understanding, interpretation of language variation, and response generation. In recruitment chatbots, they can help understand questions about job offers, skills, locations, documents, and procedures.

1.3.4.2 BERT and Contextual Understanding

BERT, or Bidirectional Encoder Representations from Transformers, is a Transformer-based language model designed mainly for language understanding tasks. It was introduced by Devlin et al. [18] and represents an important advancement in NLP because it interprets words according to their surrounding context.

The main idea behind BERT is bidirectional contextual understanding. Unlike earlier word representation methods, where a word often had the same representation regardless of use, BERT considers both left and right context. For example, the word "bank" has different meanings in "the bank approved the loan" and "the river bank was covered with trees". By analyzing the full sentence, BERT produces more accurate word representations.

BERT is based on the encoder part of the Transformer. During pre-training, it uses tasks such as masked language modeling and next sentence prediction [18]. In masked language modeling, some words are hidden and the model learns to predict them from surrounding context. This helps BERT learn relationships between words and sentences.

After pre-training, BERT can be fine-tuned for tasks such as text classification, question answering, named entity recognition, and intent recognition [18]. This makes it useful for chatbots that need to understand user messages, classify intents, and extract information.

In a recruitment chatbot, contextual understanding can help interpret questions about job offers, skills, job titles, locations, experience, required documents, and application procedures. By considering context, BERT-like models can support more accurate language understanding in recruitment interactions.

1.3.4.3 GPT and Generative Models

GPT, or Generative Pre-trained Transformer, is a Transformer-based language model designed mainly for text generation. Unlike models focused mainly on understanding or classification, GPT models generate natural language by predicting the next word or token based on previous context. This makes them important in conversational systems, where the system must produce coherent and relevant responses.

GPT models are mainly based on the decoder part of the Transformer. Their generation process is autoregressive: the model produces text step by step, using previous tokens to predict the next one. This mechanism allows GPT-style models to generate paragraphs, answer questions, complete sentences, and participate in dialogue-like interactions.

These models are pre-trained on large text corpora to learn language patterns, grammar, factual associations, and forms of expression. After pre-training, they can be adapted or prompted for different NLP tasks. Brown et al. [19] show that large GPT-style models can perform tasks with few examples or instructions, making them flexible for language-based applications.

A simple distinction can be made between BERT and GPT. BERT is mainly oriented toward language understanding, while GPT is mainly oriented toward text generation. Therefore, GPT-style models are especially useful when a system needs to generate complete natural language answers instead of only classifying or extracting information.

In chatbots, generative models can produce flexible responses instead of selecting only predefined answers. In recruitment, GPT-like models can help answer questions about job offers, procedures, required documents, skills, platform guidance, and general recruitment information.

However, in sensitive domains such as public employment services, generative flexibility must be controlled. Answers should remain accurate, clear, and aligned with official information. For this reason, GPT-style models are important for conversational systems, but their use in recruitment requires appropriate design and supervision.

1.3.5 Large Language Models

This subsection focuses on Large Language Models and their role in conversational systems. It presents their definition, their advantages for flexible dialogue, and the main limitations that must be considered in sensitive domains such as recruitment.

1.3.5.1 Definition of Large Language Models

Large Language Models (LLMs) are advanced language models trained on very large text corpora to process, understand, and generate natural language. They learn statistical and contextual patterns from large-scale data, allowing them to predict, complete, classify, summarize, and generate text. Many recent LLMs are based on Transformer architectures [23].

The main characteristic of LLMs is their scale. Compared with earlier NLP models, they are trained on larger datasets and contain many parameters, which enables broader language capabilities. As a result, LLMs can support tasks such as text generation, question answering, summarization, translation, classification, and dialogue. According to the OECD, recent AI language models are powerful generative systems that support interaction, personalization, and content creation across domains [24].

LLMs can also adapt to tasks through prompting or fine-tuning. Instead of building a separate model for each task, an LLM can often be guided by instructions, examples, or task-specific data. GPT-style models demonstrated that large generative language models can perform several tasks with limited examples through prompting [19].

In recruitment chatbots, LLMs can help understand and answer questions about job offers, registration, documents, skills, locations, experience, and platform guidance. Their natural language generation makes interaction more flexible than predefined answers. However, LLMs should be considered support tools that require control, especially in public employment-service contexts where answers must be accurate and aligned with official information.

1.3.5.2 Advantages of LLMs in Conversational Systems

LLMs provide several advantages for conversational systems compared with rule-based chatbots and earlier NLP approaches. While rule-based chatbots require predefined paths and scripted answers, LLMs can process a wider variety of natural language inputs. This allows users to express the same request in different ways.

One important advantage is the ability to generate fluent and coherent responses. Instead of selecting only from predefined answers, an LLM can produce responses adapted to the user's question and context. This makes interactions more natural, especially when users ask open-ended questions or need explanations. The OECD notes that AI language models can support interaction, personalization, and content generation across domains [24].

LLMs can also support short conversational context by considering previous user messages. This can help maintain continuity and reduce the need for users to repeat information. For example, if a user first asks about job offers in a city and then asks about required documents, the system may use the previous context to answer more relevantly.

Another advantage is task flexibility. LLMs can answer questions, summarize information, explain procedures, reformulate requests, and assist with decision support. This reduces the need to manually script every possible dialogue path. GPT-style models have shown strong flexibility through instructions or limited examples [19].

LLMs may also support personalization when reliable user information and data are available. They can adapt responses according to the user's question, profile, preferences, or context. They can also support multilingual or mixed-language interaction more naturally than many earlier systems.

In recruitment chatbots, these advantages are relevant because job seekers may ask natural questions about offers, registration, documents, skills, locations, procedures, or platform guidance. In a public employment-service context such as Wassit Online, LLMs can improve accessibility and interaction when used with appropriate controls and reliable sources.

1.3.5.3 Limitations of LLMs

Although LLMs offer important advantages, they also present limitations, especially in sensitive domains such as recruitment and public employment services. In this context, information should be accurate, fair, updated, and aligned with official procedures. The main limitations include hallucinations, lack of grounding, and privacy concerns.

Hallucinations Hallucinations are outputs that appear fluent and plausible but are factually incorrect, unsupported, or fabricated. This occurs because LLMs generate text based on learned statistical patterns rather than guaranteed factual correctness. Zhang et al. [25] explain that hallucination is a major issue in LLMs. In a recruitment chatbot, hallucinations can be problematic when answering questions about job offers, documents, eligibility, deadlines, or administrative procedures, because incorrect information may mislead users.

Lack of Grounding Another limitation is lack of grounding. The model may generate answers without reliably connecting them to verified sources, official documents, or updated knowledge bases. In Wassit Online and ANEM services, a recruitment assistant should provide information grounded in official job offers, administrative rules, registration procedures, and reliable platform data. Without grounding, the chatbot may generate outdated or non-verifiable answers. This shows the importance of connecting conversational systems to reliable information sources [26].

Privacy Concerns LLM-based systems may raise privacy concerns because they can process sensitive user information. In recruitment, users may share names, contact details, CV content, education, experience, skills, location, and employment status. A recruitment chatbot must therefore limit unnecessary data collection, secure stored data, control access, and anonymize data when possible. Miranda et al. [27] highlight privacy risks related to memorization, leakage, and exposure of sensitive information.

These limitations do not remove the value of LLMs, but they show that recruitment chatbots require careful design, reliable information sources, verification mechanisms, and appropriate safeguards.

1.3.6 Retrieval-Augmented Generation

This subsection introduces Retrieval-Augmented Generation as an approach that connects language generation with external knowledge sources. It explains how RAG can help recruitment chatbots provide more reliable, updated, and domain-grounded answers.

1.3.6.1 Definition and Principle of RAG

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. Instead of relying only on the internal knowledge learned by a language model, RAG retrieves relevant information from external sources and uses it as additional context when generating an answer. This makes the response more connected to available documents, databases, or knowledge bases.

The principle of RAG is simple. When a user asks a question, the system searches a knowledge base or document collection, retrieves relevant passages, and provides them to the language model to generate a grounded answer. Lewis et al. [26] introduced RAG as a method that combines a neural retriever with a generation model for knowledge-intensive NLP tasks.

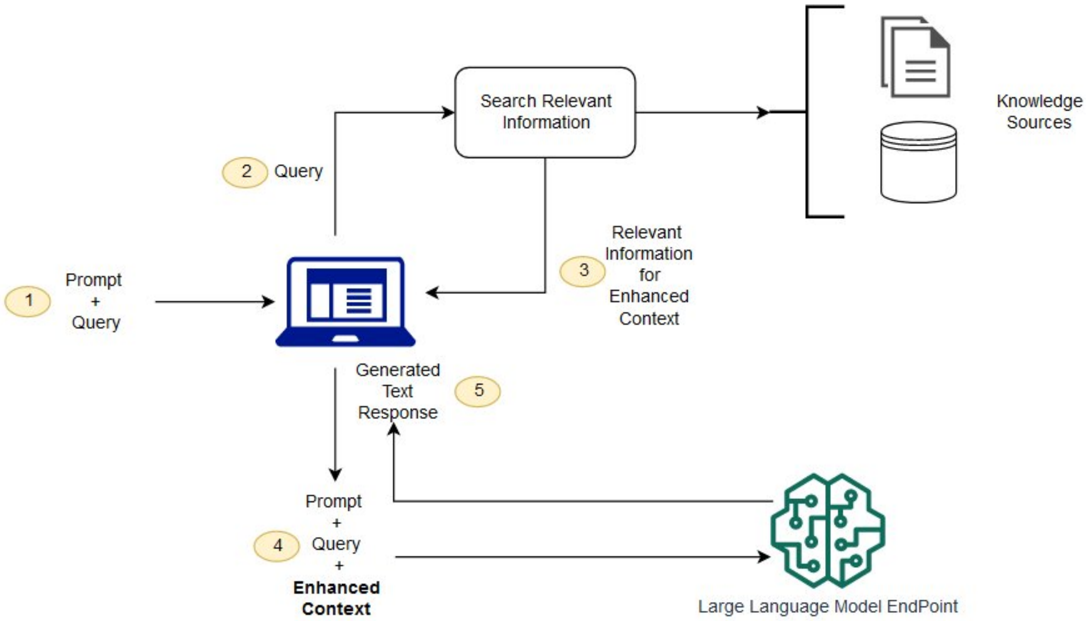

Figure 1.4: Conceptual flow of using Retrieval-Augmented Generation with Large Language Models. [28].

RAG reduces dependence on the model's internal training knowledge. This is important because language models may not contain the most recent or specific information needed to answer a user question. By using external sources, RAG can improve factual accuracy and reduce unsupported answers when retrieved documents are reliable and up to date.

This approach is useful in domains where information changes over time. In recruitment and public employment services, users may ask about job offers, registration procedures, documents, eligibility, application steps, or platform instructions. A RAG-based chatbot could retrieve information from official Wassit Online or ANEM sources, FAQs, job-offer databases, procedures, or platform documentation before generating an answer.

However, RAG is not a complete solution by itself. The final answer depends on the quality, relevance, and freshness of retrieved information. If retrieval returns incomplete or outdated documents, the response may still be inaccurate. Therefore, RAG requires careful source selection and retrieval design.

1.3.6.2 RAG Architecture

A RAG architecture is usually organized as a pipeline connecting a user query, a retrieval component, an external knowledge source, and a generative language model. Its objective is to provide relevant information before generating the final answer, so the response is supported by retrieved documents rather than only by the model's internal knowledge [26].

A typical RAG system begins with a document collection or knowledge base. This collection may contain structured or unstructured information, such as documents, FAQs, procedures, or database records. Before retrieval, documents are cleaned, divided into chunks, and converted into vector representations using an embedding model. These vectors are stored in a vector index or search system.

When the user submits a question, the query is transformed into a representation that can be compared with indexed documents. The retriever selects the most relevant chunks or passages, which serve as external evidence for the answer. The quality of this step is important because the generator depends on retrieved information.

After retrieval, the selected passages are combined with the user question to build a prompt or context for the language model. The generator then uses both the query and retrieved information to produce the final answer. Some systems may also include ranking, filtering, citation generation, or answer verification.

In a recruitment chatbot, the knowledge base could include official Wassit Online or ANEM procedures, FAQs, job offer descriptions, required documents, application instructions, agency information, and platform guidance. For example, if a job seeker asks about registration documents, the system could retrieve the official list before generating an answer.

Therefore, a RAG-based recruitment chatbot depends on the careful organization of retrieval and generation components. The knowledge base must be reliable and domain-specific, the retrieval process must select relevant information, and the generator must use the retrieved content appropriately.

1.3.6.3 Advantages of RAG for Recruitment Chatbots

RAG is particularly useful for recruitment chatbots because recruitment information is specific, procedural, and frequently updated. Users may ask about job offers, registration, documents, eligibility, application procedures, agency information, or platform guidance. A RAG-based chatbot can answer using retrieved information instead of relying only on internal model knowledge.

One main advantage is grounding responses in external and reliable sources. By retrieving relevant passages from official documents, FAQs, job descriptions, or platform data, the chatbot can produce answers connected to the target domain. This is important in public employment services, where responses should remain accurate, traceable, and aligned with official information. Lewis et al. [26] show that RAG improves knowledge-intensive NLP tasks by combining retrieval with generation.

RAG can also reduce hallucination risks compared with purely generative responses. Since the model receives retrieved information before producing the answer, it can rely more on available evidence. This is important in recruitment, where incorrect information about deadlines, documents, eligibility, or procedures may mislead users. However, answer quality still depends on the reliability of retrieved sources.

Another advantage is that RAG can support updated and domain-specific information without retraining the whole language model. Recruitment information may change over time, especially job offers, platform instructions, and administrative procedures. By updating the knowledge base, the chatbot can access newer information while keeping the same language model.

RAG can also improve user assistance by allowing job seekers and employers to ask natural-language questions instead of manually searching several pages or documents. For example, a job seeker could ask how to apply, what documents are required, or where the nearest agency is located. The chatbot can retrieve relevant information and generate a clearer answer.

In addition, RAG can improve transparency because answers may be linked to retrieved sources or supporting passages. Nevertheless, it must be designed carefully because its effectiveness depends on the quality, freshness, and organization of the knowledge base and retrieval strategy.

Overall, RAG provides a strong basis for building a more reliable and domain-grounded recruitment chatbot, especially when the system must provide guidance based on official and frequently updated information.

1.4 Relevant Work

This section reviews existing research and systems related to recruitment chatbots, AI-based job matching, skill extraction, multilingual interaction, and RAG-based assistance. Its purpose is to position the proposed approach in relation to previous work and identify the gap addressed by this thesis.

1.4.1 Chatbots and Conversational AI in Recruitment

Conversational AI has been increasingly explored in recruitment to improve communication between job seekers, employers, and recruitment services. Chatbots are used as interactive assistants that answer candidate questions, explain application procedures, collect basic candidate information, and support repetitive communication tasks. Instead of requiring users to manually search pages or contact staff for simple questions, recruitment chatbots can provide faster assistance.

Existing work shows that early chatbots were often based on predefined rules and scripted dialogue flows. These systems are useful when interaction is simple and predictable, such as answering FAQs or guiding users through fixed procedures [16]. Dogan and Gurcan [17] also show that rule-based chatbots can be effective in restricted service contexts. In recruitment, they can answer questions about registration steps, documents, agency hours, or basic application procedures.

More recent work focuses on conversational AI systems using NLP and generative models for flexible interaction. In recruitment, conversational AI can support candidate engagement by automating repetitive communication and allowing recruiters to focus on complex tasks [7]. Generative AI recruitment chatbots are also presented as tools for more personalized support during hiring [29]. These systems help job seekers ask questions in natural language and receive guidance adapted to their needs.

Recruitment chatbots offer availability, speed, and improved interaction. They can provide immediate responses, reduce repetitive questions, assist users outside working hours, and improve access to recruitment information. They can also support HR communication by collecting initial information, explaining procedures, and guiding users through application steps.

However, existing recruitment chatbot approaches also have limitations. Rule-based chatbots may fail with complex questions or unexpected wording, while generative systems may produce unsupported information if not connected to reliable sources. AI-based recruitment systems may also raise fairness, bias, and privacy concerns when processing candidate information or supporting decision-making. Research on AI fairness in hiring highlights the need to avoid unfair or discriminatory outcomes [30]. Privacy concerns must also be considered when systems process personal or professional data [27].

For a public employment platform such as Wassit Online, these works show the potential of conversational support, but also the need for reliability and control. A recruitment chatbot should provide information aligned with official procedures, support multilingual users, and help job seekers navigate employment services. Thus, existing work supports the proposed approach by showing that conversational assistance can improve accessibility and user interaction.

1.4.2 AI-Based Job Matching and Recommendation Systems

AI-based job matching and recommendation systems aim to improve the connection between job seekers and job offers by analyzing candidate profiles, job requirements, and recruitment preferences. Unlike basic search systems based on exact keywords or fixed filters, AI-based approaches attempt to identify more relevant relationships between candidates and vacancies. They can recommend jobs to candidates, suggest candidates to employers, rank applications, and reduce manual screening effort.

Several approaches have been explored in the literature. Content-based methods compare job seeker characteristics with job offer content, such as skills, education, experience, location, and category. Collaborative filtering uses previous interactions or preferences to recommend jobs or candidates based on similar users or patterns. Hybrid systems combine techniques to improve recommendation quality. Other works use machine learning classification, ranking models, semantic similarity, knowledge-based matching, or deep learning.

Recent studies show that AI can improve recruitment by moving beyond keyword matching. Deepthi et al. [5] propose an intelligent skill-based matching framework that highlights the limitations of keyword-based recruitment. Similarly, the OECD [31] explains that AI can support labor-market matching by identifying skills, improving personalization, and connecting job seekers with suitable opportunities.

AI-based recommendation systems can improve personalization, filtering, and decision support. They can help job seekers discover offers that match their profiles and help employers identify candidates corresponding to position requirements. Ranking models can also reduce screening time by prioritizing relevant applications.

However, these systems depend strongly on data quality and completeness. Incomplete CVs, unclear job descriptions, missing skills, or inconsistent profiles can reduce accuracy. Recommendation systems may also reproduce bias if trained on historical or unbalanced data. Raghavan et al. [30] emphasize fairness concerns in AI-based hiring systems. Lack of explainability can also make it difficult to understand why a candidate or offer was recommended.

Another limitation is the difficulty of capturing contextual and human factors. Recruitment decisions may depend on soft skills, motivation, work preferences, and employer expectations, which are difficult to represent using structured data or historical patterns. Therefore, AI-based matching systems should remain decision-support tools rather than autonomous decision-makers.

In Wassit Online, AI-based matching could improve the relevance of job recommendations and candidate lists. However, because Wassit is related to public employment services, such systems must remain transparent, fair, and aligned with institutional requirements. This group of works supports the proposed approach by showing that intelligent matching can improve recruitment support when combined with reliable data and human supervision.

1.4.3 Skill Extraction and Resume-Job Matching

Skill extraction and resume-job matching are important research directions in AI-based recruitment. They improve how candidate profiles and job offers are analyzed, structured, and compared. Skill extraction identifies information from resumes, CVs, job posts, or candidate profiles, such as technical skills, soft skills, qualifications, education, experience, and job-related attributes. This information can then support matching, recommendation, and ranking tasks.

Several studies use NLP and machine learning for extracting skills from job posts and resumes. Senger et al. [20] present skill extraction and classification as central tasks in NLP-based recruitment because matching quality depends on identifying candidate competencies and job requirements. These approaches transform unstructured text, such as CV descriptions or job advertisements, into structured information that can be compared automatically.

Resume-job matching compares candidate profiles with job requirements to estimate suitability. Traditional systems often rely on exact keyword overlap, which is limited because the same skill may be expressed using different words, abbreviations, or related terms. Modern approaches use semantic similarity, contextual representations, and learning-based methods. Yu et al. [21] show that stronger representations and semantic matching can capture deeper relationships between candidate profiles and job descriptions.

Other works emphasize multilingual skill identification. Kavas et al. [32] focus on multilingual skill extraction for vacancy–job seeker matching, showing that skill identification becomes more difficult when candidates and job offers use different languages or terminology. This is relevant for multilingual recruitment environments where the same competency may appear in Arabic, French, English, or mixed forms.

The main strength of skill extraction and resume-job matching is reducing manual screening and improving matching relevance. By identifying candidate competencies and comparing them with job requirements, these systems help recruiters prioritize relevant profiles and help job seekers find suitable offers. They also support structured analysis of resumes and job descriptions.

However, these approaches depend on input quality. Incomplete resumes, vague job offers, missing skills, and ambiguous skill names can reduce accuracy. Distinguishing hard skills from soft skills can also be difficult, especially when soft skills are expressed indirectly. Recruitment decisions may also depend on motivation, personality, communication ability, and psychological suitability, which are difficult to evaluate automatically from text alone.

Bias and explainability are also important challenges. If matching models are trained on biased or incomplete data, they may reproduce unfair patterns. If the system does not explain recommendations, users may have difficulty trusting results. These concerns are important in public employment services, where transparency and fairness are essential [30].

In Wassit Online, skill extraction and resume-job matching could improve candidate-job matching and reduce manual review. They could support an intelligent recruitment chatbot by helping it understand candidate skills, interpret job requirements, ask for missing information, and guide users more precisely. However, such systems should remain transparent, fair, and adapted to public employment-service requirements.

1.4.4 Multilingual and RAG-Based Recruitment Systems

Multilingual and RAG-based systems are increasingly relevant for recruitment chatbots because users may interact in different languages, while recruitment information must remain accurate, updated, and aligned with official sources. A recruitment chatbot may need to understand formal language, informal language, dialectal expressions, or mixed-language forms, and answer questions about procedures, documents, job offers, platform services, and agency information.

Research on multilingual NLP shows that language models can support cross-lingual understanding and transfer knowledge across languages. Pires et al. [22] show that multilingual BERT can perform cross-lingual transfer, which is useful when systems operate in more than one language. In recruitment, this matters because job seekers may use different languages to describe skills, locations, job titles, or administrative needs. In Algeria, users may communicate in Arabic, French, English, Algerian Darja, Arabizi, or mixed expressions.

Multilingual support can improve accessibility and inclusion. If users can interact using the language they are most comfortable with, the chatbot becomes easier to use and more adapted to real communication practices. For Wassit Online, multilingual support could help job seekers ask about offers, registration, documents, agency locations, skills, and procedures in a natural way.

RAG-based approaches address another important requirement: reliable and grounded answers. Lewis et al. [26] introduced RAG as an approach that combines retrieval with generation for knowledge-intensive NLP tasks. In recruitment, many answers should depend on external and updated information, such as official procedures, FAQs, job descriptions, required documents, and platform instructions. A RAG-based chatbot can retrieve information from trusted sources before generating a response.

Recent work also explores RAG in recruitment-related tasks. Afzal et al. [33] apply RAG to candidate profile summarization, showing that retrieval-based methods can ground recruitment outputs in relevant data. Ranaldi et al. [34] discuss multilingual retrieval-augmented language models, which is relevant for systems that manage information across languages. These works suggest that combining multilingual processing with retrieval-based grounding can support more reliable recruitment assistance.

For Wassit Online and ANEM services, a multilingual RAG-based chatbot could retrieve information from official platform guidance, administrative procedures, job offers, required documents, agency information, and FAQs. This would allow more factual, traceable, and official answers, while reducing the need for users to manually search multiple pages.

However, these approaches have limitations. Multilingual NLP remains difficult with dialectal Arabic, Arabizi, informal spelling, and code-switching. RAG also depends on the quality and freshness of the knowledge base; outdated or irrelevant retrieved documents can still lead to inaccurate answers. Privacy is also important because users may provide personal data, CV information, contact details, skills, education, or work experience. Therefore, multilingual and RAG-based recruitment systems require reliable sources, careful evaluation, privacy-aware design, and domain-specific adaptation.

Overall, existing works show that multilingual NLP and RAG can contribute to more accessible, reliable, and knowledge-grounded recruitment chatbots. For the proposed approach, they support the need for a chatbot that handles multilingual input while grounding answers in official Wassit Online and ANEM-related information.

1.4.5 Discussion and Positioning of the Proposed Approach

The reviewed works show that conversational AI and intelligent recruitment systems can improve different aspects of recruitment. Recruitment chatbots support communication, answer frequent questions, and guide candidates through procedures. AI-based matching systems improve the connection between job seekers and offers. Skill extraction and resume-job matching support structured profile analysis. Multilingual NLP and RAG improve accessibility and reliability by supporting different languages and grounding answers in external sources [5, 7, 20, 26].

Despite these advances, several gaps remain. Many approaches are developed for general recruitment or private hiring platforms, with limited attention to public employment-service platforms in Algeria. Existing systems do not always consider the procedures, services, and institutional role of Wassit Online and ANEM. Multilingual support also remains challenging, especially with Arabic, French, English, Algerian Darja, Arabizi, and mixed expressions. Previous work shows the potential of cross-lingual understanding, but adaptation to local language use requires additional design and evaluation [22, 32].

Another gap concerns answer reliability. Generative models improve flexibility, but responses may not always be grounded in official or updated information. This is problematic in public employment services, where users may ask about procedures, documents, eligibility, job offers, and application steps. RAG provides a useful direction by combining retrieval with generation, but its effectiveness depends on the quality, freshness, and organization of the knowledge base [26, 33].

The proposed approach is positioned as a chatbot-based recruitment assistant designed for Wassit Online. It does not aim to replace ANEM agents or the existing platform, but to provide an intelligent support layer that improves guidance, accessibility, and user interaction. The chatbot aims to help users ask questions naturally, understand recruitment procedures, access information about required documents, and navigate employment-related services.

Technically, the proposed approach combines conversational interaction with NLP and LLM-based capabilities. When possible, retrieval-based grounding can connect the chatbot to official or domain-specific information, such as Wassit Online guidance, ANEM procedures, job offers, FAQs, and administrative documents. This can reduce unsupported answers and make the system more suitable for public employment services.

The approach may also support future extensions, such as skill extraction, job recommendation, CV analysis, and soft-skill-related assistance. However, these features should be considered progressive improvements rather than replacements for human recruitment decisions. Motivation, personality, psychological suitability, and contextual fit remain difficult to evaluate automatically and may require human supervision.

Overall, the proposed work fits within the state of the art by combining ideas from recruitment chatbots, AI-based matching, multilingual NLP, and retrieval-based grounding. Its main contribution is to adapt these directions to the Wassit Online context through an intelligent recruitment assistant focused on user guidance, accessibility, and reliable employment-related information.

Conclusion

This chapter reviewed the main concepts, technologies, and related works needed to justify the development of an intelligent chatbot for recruitment management on Wassit Online. It first presented recruitment and e-recruitment to show the importance of digital platforms in employment services. It then introduced Wassit Online as the Algerian public employment context of this thesis.

The chapter also discussed chatbots and conversational AI, starting from rule-based chatbots and their limited flexibility. It then presented NLP as a key component for improving chatbot understanding through preprocessing, intent recognition, entity extraction, and multilingual processing. Transformer-based models, LLMs, and RAG were examined as advanced approaches for improving language understanding, response generation, and grounding in reliable information.

The relevant work showed that intelligent recruitment systems can support conversational assistance, job matching, skill extraction, resume-job matching, multilingual interaction, and knowledge-grounded responses. However, the reviewed works also revealed gaps related to public employment platforms, multilingual user needs, official procedure guidance, and reliable domain-specific information.

Based on this analysis, the proposed approach is justified as an intelligent recruitment chatbot that can support users of Wassit Online by improving guidance, accessibility, and interaction. Such a system should remain realistic, domain-aware, multilingual when possible, and grounded in reliable information without replacing ANEM agents or the existing platform.

Based on these findings, the next chapter presents the design of the proposed chatbot system.

Chapter

2

Contribution

Introduction

After presenting the general context of digital employment services and reviewing the main concepts related to recruitment, e-recruitment, conversational AI, natural language processing, large language models, and retrieval-augmented generation, this chapter presents the proposed contribution of this work. The previous chapter showed that online employment platforms such as Wassit Online play an important role in improving access to employment services, but may still face limitations related to keyword-based search, rigid filtering, manual verification, multilingual interaction, and the need for more personalized user guidance.

In this context, the proposed contribution consists of an intelligent recruitment chatbot designed to support job seekers in the context of Wassit Online and ANEM-related recruitment services. The system, named RequAI, includes a conversational assistant called Requi, whose role is to help users access employment-related information, understand recruitment procedures, express their job search needs in natural language, and receive more tailored guidance. The proposed solution also aims to support profile-based services such as CV analysis, skill extraction, and job matching in order to improve the relevance of recruitment assistance.

This chapter first provides an overview of the proposed solution and its main objectives. It then presents the conceptual architecture and methodology of the system. After that, the chapter describes the implementation and experimentation process. Finally, the evaluation methodology and results are presented and discussed in order to analyze the effectiveness, limits, and possible improvements of the proposed system.

2.1 Proposed Solution, Architecture, and Methodology

This section presents the proposed RequAI solution from a conceptual point of view. It describes the general idea of the system, its architecture, its main functional modules, the methodology followed, the required data, and the justification of the chosen approach.

2.1.1 Overview of the Proposed Solution

The proposed solution in this thesis is an intelligent recruitment assistant named RequAI, designed to support job seekers in the context of Wassit Online and ANEM-related services. The system includes a conversational chatbot called Requi, which acts as an interactive interface between the user and the recruitment support system. Its main purpose is to help users access employment-related information, understand recruitment procedures, express their job search needs naturally, and receive adapted guidance.

This solution is motivated by the limitations identified in the previous chapters. Although Wassit Online contributes to the digitalization of employment services in Algeria, the job search and recruitment process may still depend on structured forms, predefined filters, keyword-based search, and manual verification. These mechanisms are useful for basic search tasks, but they may not always capture the real intention of the user, the semantic relationship between candidate profiles and job offers, or additional suitability factors such as skills, preferences, experience, and professional context.

RequAI aims to address these limitations by providing a more interactive and intelligent support layer. Instead of forcing users to rely only on fixed fields or exact keywords, the chatbot allows job seekers to communicate using natural language. Through this interaction, the system can guide users step by step, answer common questions, ask for clarification when the request is incomplete, and support job search assistance. The chatbot is therefore not intended to replace Wassit Online or ANEM agents, but to complement existing services by improving accessibility, user guidance, and the organization of recruitment-related information.

The proposed system focuses mainly on the job seeker side. It helps users ask questions about job offers, required documents, registration steps, application procedures, and platform-related guidance. Since the Algerian context is multilingual, the system is also designed with attention to different forms of user expression, including Arabic, French, English, Algerian Darja, Arabizi, and mixed-language messages. This orientation aims to make the chatbot more accessible to users who may not formulate their requests using formal or technical language. Although the current scope is centered on job seekers, the architecture remains extensible, allowing recruiter-side functionalities to be considered in future versions of the system.

In addition to conversational assistance, RequAI is designed to support recruitment-oriented functionalities such as CV/profile analysis, skill extraction, and job matching. The general idea is to extract relevant information from the user profile, such as skills, education, experience, location, and preferences, then compare this information with job offer requirements. This allows the system to provide suggestions that are more relevant than simple keyword-based search results.

The system also relies on a structured knowledge base containing recruitment-related information, frequently asked questions, platform guidance, and ANEM-related procedures. This knowledge base supports more reliable answers by grounding chatbot responses in domain-specific information instead of depending only on ungrounded generated text. This is especially important in public employment services, where answers about procedures, required documents, and application steps must remain clear and trustworthy.

From a functional point of view, RequAI receives user input through a web-based interface, whether as a natural language message, profile information, or CV-related data. The system analyzes this input, identifies the user's intention, extracts useful information when needed, and activates the appropriate module to provide an answer, request clarification, extract skills, or suggest relevant job offers. Therefore, the main objective of RequAI is to provide a simple, accessible, and intelligent recruitment assistance experience that improves user guidance and supports more relevant job search in the Algerian employment context.

2.1.2 General System Architecture

The proposed architecture of RequAI is designed to support the main objectives of this work: improving job seeker guidance, enabling natural language interaction, supporting CV/profile analysis, extracting relevant skills, and providing more relevant job recommendations in the context of Wassit Online and ANEM-related services. Figure 2.1 provides a high-level overview of the proposed system architecture.

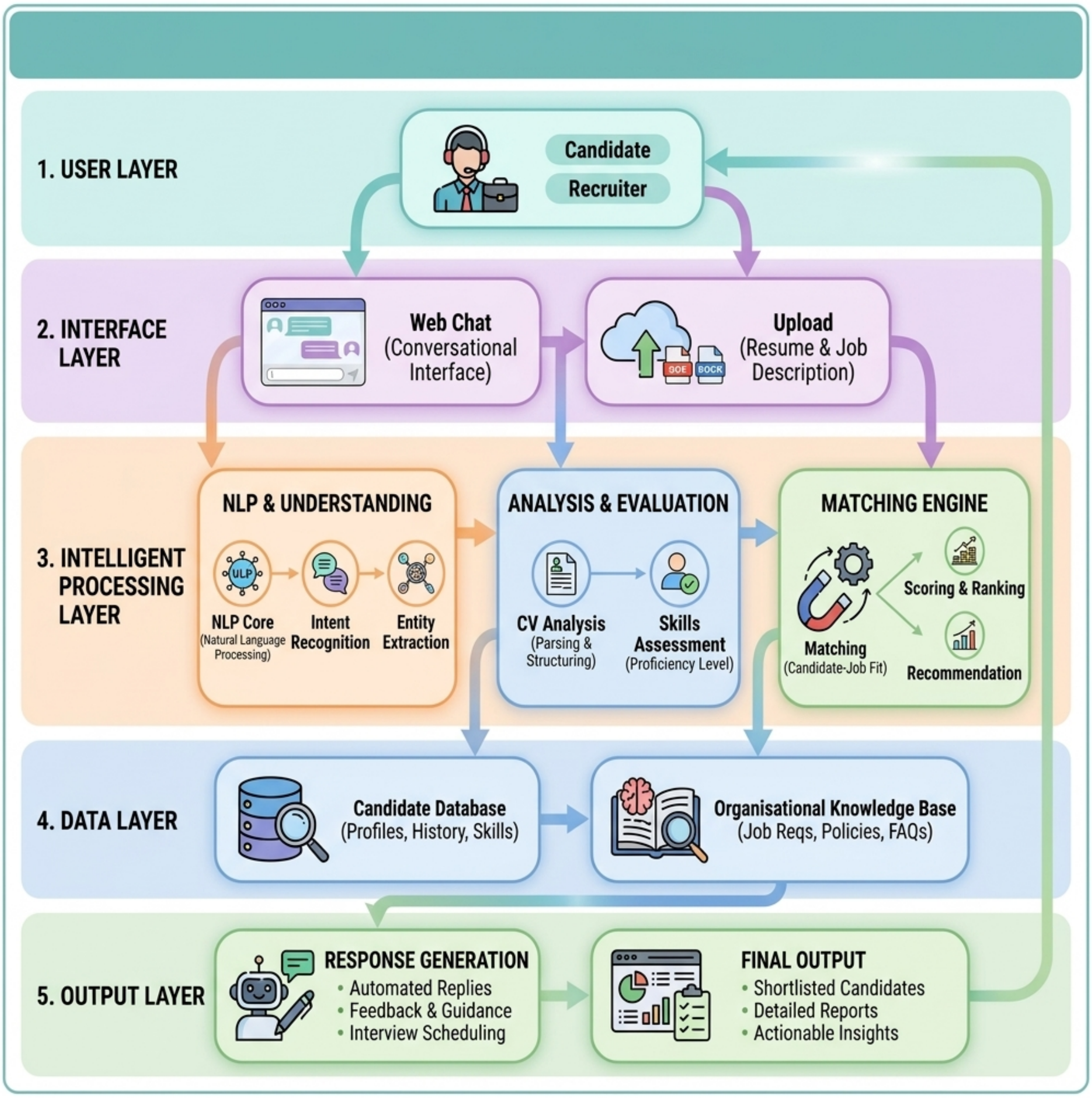

Figure 2.1: General architecture of the proposed intelligent recruitment chatbot.

2.1.2.1 General Description of the Architecture

The proposed RequAI architecture is organized into five main layers: the user layer, the interface layer, the intelligent processing layer, the data and knowledge layer, and the output layer. This layered organization helps separate the main responsibilities of the system, from user interaction to request processing, data management, and final response generation.

The user and interface layers represent the access point of the system, where job seekers interact with Requi through a web-based chatbot interface. The intelligent processing layer represents the core of the architecture, where user requests and profile information are interpreted and prepared for recruitment assistance. The data and knowledge layer provides the information required for profile analysis, job matching, and knowledge-based answers. Finally, the output layer presents the generated answers, guidance, or recommendations to the user.

This organization allows the system to process user input progressively and transform it into useful recruitment support. Each layer is described in the following subsections.

2.1.2.2 User and Interface Layer

The user and interface layer represents the entry point of the RequAI system. In the current scope of this work, the main user is the job seeker, who interacts with the system through a web-based interface. This interface provides access to the Requi chatbot, allowing users to ask employment-related questions, request guidance about Wassit Online and ANEM procedures, describe their job search needs, and receive personalized assistance. It also allows the user to provide profile or CV-related information when needed. The role of this layer is therefore to collect user input, transmit it to the intelligent processing layer, and display the generated answers, guidance, or recommendations returned by the system.

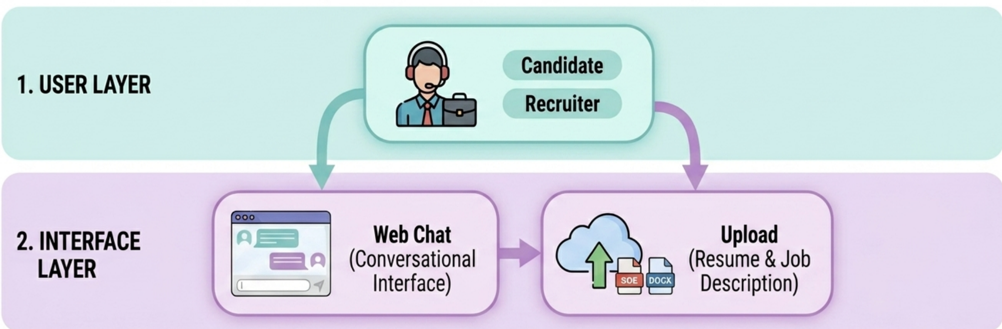

Figure 2.2: The user/interface layer

2.1.2.3 Intelligent Processing Layer

The intelligent processing layer represents the core of the RequAI architecture. Its role is to transform user input into meaningful information that can be used to generate guidance, analyze candidate profiles, and support job matching. This layer receives different types of input from the interface layer, such as natural language messages, CV-related information, and job search preferences.

The first function of this layer is to understand the user request. When the job seeker sends a message, the system analyzes its meaning in order to identify the general intention behind it. For example, the user may ask about available job offers, required documents, registration procedures, application steps, or platform-related guidance. The layer also identifies useful information contained in the message, such as job title, location, skills, education level, experience, or preferences.

In addition to understanding conversational messages, this layer supports the analysis of candidate information. When the user provides CV or profile data, the system uses this information to identify relevant elements such as education, professional experience, competencies, and career preferences. These elements help build a clearer representation of the job seeker profile and prepare it for the matching process.

This preparation is important because recruitment matching depends not only on job titles or keywords, but also on the candidate's actual skills, qualifications, and profile characteristics. By organizing this information, the system can better compare the candidate profile with the requirements of available job offers.

Finally, the intelligent processing layer prepares the information needed by the response generation and matching components. It connects the user's request, profile information, and extracted skills with the data and knowledge layer. This allows the system to provide answers, guidance, or recommendations that are more relevant to the user's needs.

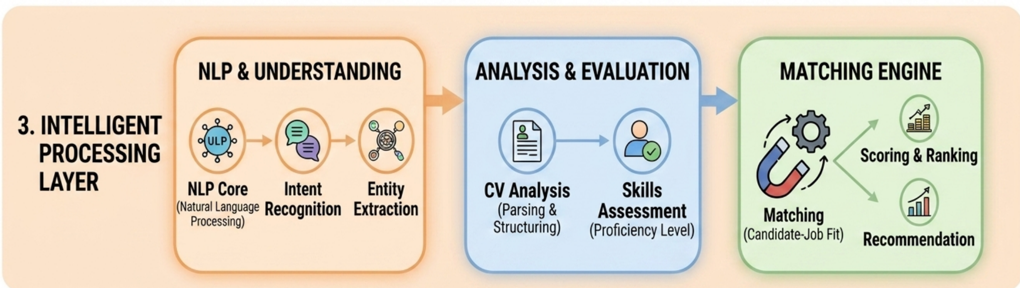

Figure 2.3: Intelligent Processing Layer of the RequAI system.

2.1.2.4 Data and Knowledge Layer

The data and knowledge layer represents the part of the architecture responsible for storing, organizing, and providing the information required by the RequAI system. This layer supports both the intelligent processing layer and the response generation process by making recruitment-related data available when needed.

In the proposed system, this layer includes candidate-related information such as profile data, CV content, extracted skills, education, experience, preferences, and interaction history. These elements help the system build a clearer representation of the job seeker and support more relevant assistance during the conversation.

This layer also includes job-related information, such as job offers, required skills, job titles, locations, experience requirements, and other criteria that may be useful for matching. By organizing this information, the system can compare candidate profiles with job offer requirements and provide more adapted recommendations.

In addition to candidate and job data, the data and knowledge layer contains recruitment-related knowledge. This may include frequently asked questions, Wassit Online guidance, ANEM-related procedures, required documents, application steps, and other employment-service information. This knowledge helps the chatbot provide clearer and more reliable answers to users.

Overall, the data and knowledge layer plays an important role in connecting user requests with useful recruitment information. It supports profile analysis, job matching, and knowledge-based responses, while helping the chatbot generate answers that are more relevant to the user's needs.

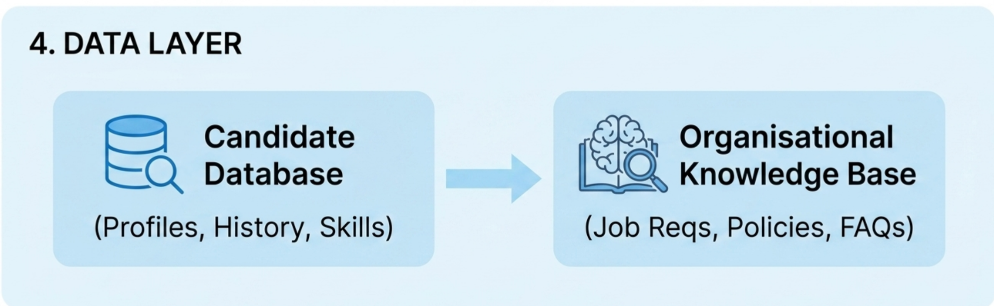

Figure 2.4: Data and knowledge layer of the RequAI system.

2.1.2.5 Output Layer

The output layer represents the final stage of the RequAI architecture. It is responsible for presenting the results produced by the system in a clear and useful form for the job seeker. After the user request has been processed and the necessary information has been retrieved or analyzed, this layer delivers the final response through the chatbot interface.

Depending on the user request, the output may take different forms. If the user asks an employment-related question, the system provides a direct answer or guidance based on the available recruitment knowledge. If the user provides profile or CV-related information, the output may include extracted skills, relevant profile elements, or suggestions to improve the clarity of the user profile. If the request concerns job search, the output may include job recommendations that correspond to the candidate's profile, preferences, and job offer requirements.

This layer also plays an important role in making the interaction understandable and user-friendly. The response should be simple, organized, and adapted to the user's needs. In this way, the output layer ensures that the results generated by the intelligent processing and data layers are transformed into practical assistance, such as recruitment guidance, profile feedback, or job matching suggestions.

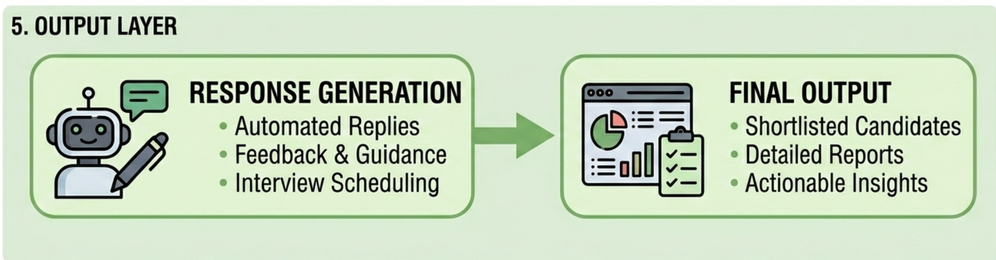

Figure 2.5: Output layer of the RequAI system.

2.1.3 Main Functional Modules

The proposed system is composed of several functional modules that cooperate to provide intelligent recruitment assistance. These modules are described at a conceptual level in this section. Their role is to show how the chatbot understands user requests, analyzes candidate information, supports matching, and generates useful responses.

2.1.3.1 Conversational Assistance and Natural Language Understanding

The conversational assistance and natural language understanding module is responsible for managing the interaction between the job seeker and the RequAI system. It allows the user to communicate with Requi using natural language instead of relying only on predefined forms, fixed menus, or exact keywords. This makes the interaction more flexible and closer to the way users normally express their needs.

This module helps the system understand the general purpose of the user request. For example, the job seeker may ask about available job offers, required documents, registration procedures, application steps, or guidance related to Wassit Online and ANEM services. By identifying the user's intention, the chatbot can decide whether the request requires an informational answer, a clarification question, profile analysis, or job search assistance.

In addition to identifying the user's intention, this module also supports the extraction of useful information from the message. Such information may include job titles, locations, skills, education level, experience, preferences, or document names. These elements help the system better understand the user's situation and prepare the necessary information for the following modules.

Overall, this module plays an important role in making the chatbot interaction more natural, guided, and useful. It acts as the first intelligent step between the user message and the rest of the system by transforming informal user input into structured information that can support guidance, profile analysis, and job matching.

2.1.3.2 CV/Profile Analysis and Skill Extraction

The CV/profile analysis and skill extraction module is responsible for analyzing the information provided by the job seeker in order to build a clearer representation of the candidate profile. This information may come from a CV, a profile form, or details shared during the conversation with the chatbot. The objective of this module is to identify relevant elements that can support recruitment guidance and job matching.

At the profile analysis level, the system considers information such as education, professional experience, field of study, previous positions, location, preferences, and general career objectives. Organizing these elements helps the chatbot better understand the user's background and provide more adapted assistance. For example, a user looking for a first job may need different guidance from a user who already has professional experience.

Skill extraction is an important part of this module because skills represent a central criterion in recruitment matching. The system aims to identify technical skills, professional competencies, and, when possible, soft skills mentioned in the user profile or CV. These extracted skills can then be used to compare the candidate profile with the requirements of available job offers.

Overall, this module helps transform unstructured or semi-structured candidate information into useful recruitment-related information. By identifying education, experience, preferences, and skills, it supports the following modules, especially job matching and recommendation, while helping the chatbot provide more personalized guidance to the job seeker.

2.1.3.3 Job Matching and Recommendation

The job matching and recommendation module is responsible for connecting the job seeker profile with relevant job offers. Its role is to compare the information extracted from the candidate profile, such as skills, education, experience, location, and preferences, with the requirements expressed in available job offers. This module aims to support more relevant job search assistance than simple keyword-based filtering.

In the proposed system, matching is considered as a decision-support function rather than an automatic recruitment decision. The objective is not to replace ANEM agents or employers, but to help organize and prioritize potentially suitable opportunities for the job seeker. By considering both candidate-related information and job-offer requirements, the system can provide recommendations that are more adapted to the user's profile.

The recommendation process may also help the chatbot guide the user when important information is missing. For example, if the user searches for a job without specifying a location, experience level, or professional domain, the chatbot can ask additional questions before suggesting offers. This makes the interaction more guided and improves the quality of the recommendations.

Overall, this module contributes to the main objective of RequAI by improving the relevance of job search results and supporting a more personalized recruitment assistance experience. It helps transform the extracted profile information into practical suggestions that can guide the job seeker toward suitable employment opportunities.

2.1.3.4 Knowledge-Based Response Generation

The knowledge-based response generation module is responsible for producing clear and reliable answers for the job seeker by using the information available in the system knowledge base. Its role is to support the chatbot when the user asks about recruitment procedures, required documents, platform guidance, application steps, or other employment-related information.

This module is important because recruitment assistance often requires accurate and domain-specific information. In the context of Wassit Online and ANEM-related services, users may need guidance that is consistent with official procedures and platform rules. Therefore, the chatbot should not depend only on general generated answers, but should rely on organized recruitment knowledge whenever possible.

The knowledge base may include frequently asked questions, information about Wassit Online services, ANEM-related procedures, required documents, registration guidance, and general employment-service information. When the user asks a question, the system can use this knowledge to prepare an answer that is more relevant to the user's request.

Overall, this module helps improve the reliability and usefulness of the chatbot responses. It supports the main objective of RequAI by allowing the system to provide guided, understandable, and context-aware answers to job seekers, especially when the request concerns procedures, documents, or platform-related information.

2.1.4 Proposed Methodology

The proposed methodology describes the general process followed by the RequAI system to transform user input into useful recruitment assistance. It explains how user requests are received, how candidate and job-related information is organized, how matching is supported, and how answers or recommendations are generated. This methodology is presented at a conceptual level, while technical implementation details are discussed later in the implementation and experimentation section.

2.1.4.1 User Input Processing

User input processing represents the first step in the proposed methodology. At this stage, the job seeker interacts with Requi through the web-based interface by sending a natural language message, describing a job search need, asking a question, or providing CV/profile-related information. This input represents the starting point of the system workflow.

The system first identifies the type of input provided by the user. A message may correspond to an informational request, such as asking about required documents or registration procedures. It may also correspond to a job search request, such as looking for a job in a specific location or domain. In other cases, the input may contain candidate-related information, such as skills, education, experience, or preferences.

After identifying the general type of request, the system prepares the input for further processing. The objective is to transform the user's message or profile information into useful elements that can be handled by the following modules. These elements may include the user intention, relevant keywords, job-related criteria, profile details, or missing information that requires clarification.

If the user request is incomplete or unclear, the chatbot may ask additional questions in order to collect the necessary information. For example, if the user asks for job offers without specifying the location or professional domain, the system can request more details before continuing. This step helps improve the relevance of the following analysis and recommendations.

Overall, user input processing allows the system to move from an informal user message to a more organized representation of the request. This step is important because the quality of the following stages, including profile analysis, matching, and response generation, depends on how clearly the initial input is understood and prepared.

2.1.4.2 Profile and Job Offer Representation

Profile and job offer representation is the step where the system organizes the information needed for recruitment assistance in a structured and meaningful way. After the user input is processed, the system needs to represent both the job seeker profile and the job offer requirements so they can be compared during the matching process.

The job seeker profile may include information such as education, professional experience, skills, location, preferences, and career objectives. This information can be provided through a CV, a profile form, or directly through the conversation with the chatbot. Representing the profile clearly allows the system to understand the user's background and identify the criteria that may influence job recommendations.

On the other side, job offers are represented through their main requirements and characteristics. These may include the job title, required skills, field of activity, location, experience level, education requirements, contract type, and other recruitment criteria. Organizing job offers in this way helps the system compare them with candidate profiles more effectively.

This representation step is important because job matching cannot depend only on exact keywords. A candidate may describe a skill or experience differently from the way it appears in a job offer. Therefore, organizing profile and job offer information around meaningful recruitment criteria helps improve the relevance of the matching process.

Overall, profile and job offer representation prepares the necessary information for the next stage of the methodology. It creates a common basis for comparing the job seeker's profile with available job opportunities and supports more personalized recommendations.

2.1.4.3 Matching and Recommendation Process

The matching and recommendation process represents the step where the organized job seeker profile is compared with available job offers. After the system has identified relevant profile elements such as skills, education, experience, location, and preferences, these elements are used to search for job opportunities that correspond more closely to the user's needs.

This process does not rely only on the presence of exact keywords. Instead, it aims to compare the candidate profile and the job offer requirements according to meaningful recruitment criteria. These criteria may include the required skills, professional domain, education level, experience, location, and other conditions mentioned in the job offer. This allows the system to provide recommendations that are more adapted to the user profile.

When the available information is incomplete, the recommendation process can also support the chatbot in asking clarification questions. For example, if the user asks for job offers without specifying a preferred location, field, or experience level, the system may request additional details before presenting recommendations. This helps avoid irrelevant suggestions and improves the usefulness of the interaction.

The output of this process is a set of job suggestions or guidance elements that can help the job seeker identify suitable opportunities. These recommendations remain a form of decision support and do not replace the role of ANEM agents, recruiters, or official recruitment procedures. Their purpose is to assist the user and improve the relevance of job search results.

2.1.4.4 Generation of Guidance and Responses

The generation of guidance and responses represents the final step of the proposed methodology. After the user request has been processed, the candidate profile has been represented, and the matching process has been performed when needed, the system prepares an answer adapted to the user's request.

The generated response depends on the type of request submitted by the job seeker. If the user asks for information about recruitment procedures, required documents, registration steps, or platform guidance, the system provides a clear answer based on the available recruitment knowledge. If the user asks for job search assistance, the response may include suggested offers or additional questions to refine the search. If the user provides CV or profile information, the response may include extracted profile elements, identified skills, or general feedback that helps the user better understand their profile.

This step is important because the usefulness of the chatbot depends not only on understanding the user request, but also on presenting the result in a simple and understandable way. The response should be organized, relevant, and adapted to the job seeker's situation. When necessary, the chatbot may also ask clarification questions before giving a final answer or recommendation.

Overall, the generation of guidance and responses transforms the results produced by the previous stages into practical support for the user. It allows RequAI to provide employment-related answers, recruitment guidance, profile feedback, and job recommendations through a conversational interface.

2.1.5 Data Requirements and Sources

The RequAI system requires different types of data in order to provide intelligent recruitment assistance. These data support user understanding, profile analysis, skill extraction, job matching, and knowledge-based response generation. Since the system aims to assist job seekers in the context of Wassit Online and ANEM-related services, the required data must include candidate-related information, job offer information, recruitment knowledge, and prepared examples for chatbot interaction.

2.1.5.1 Candidate Profile Data

Candidate profile data represents the information related to the job seeker using the RequAI system. This type of data is important because the chatbot cannot provide personalized assistance or relevant job recommendations without understanding the user's background, qualifications, and preferences.

In the proposed system, candidate profile data may include personal and professional information such as name, location, education, field of study, professional experience, skills, career objectives, and job preferences. Some of this information can be provided directly by the user through the web interface, while other elements can be extracted from a CV or collected progressively during the conversation with the chatbot.

Skills represent one of the most important parts of candidate profile data. They allow the system to identify the user's competencies and compare them with the requirements of available job offers. These skills may include technical skills, professional competencies, and, when possible, soft skills relevant to recruitment guidance.

Professional experience is also useful for building a clearer representation of the candidate profile. It may include previous job titles, companies, tasks, duration of experience, and descriptions of previous responsibilities. This information helps the system distinguish between first-time job seekers, young graduates, and candidates with professional experience.

In addition to profile and CV information, interaction history can also support the chatbot experience. Previous messages, user requests, and conversation context may help the system provide more coherent guidance and avoid asking the user to repeat the same information several times. However, since candidate data can contain sensitive personal information, it must be handled carefully, with attention to privacy, confidentiality, and responsible data use.

Overall, candidate profile data is necessary to support CV/profile analysis, skill extraction, personalized guidance, and job matching. It allows RequAI to move from general assistance toward more adapted recruitment support based on the real needs and characteristics of the job seeker.

2.1.5.2 Job Offer Data

Job offer data represents the information related to available employment opportunities. This type of data is necessary because the RequAI system needs to compare candidate profiles with job requirements in order to provide relevant job search assistance and recommendations.

In the proposed system, job offer data may include information such as the job title, company or employer, location, required skills, education level, experience requirements, contract type, salary range when available, and publication date. These elements help describe the characteristics of each job opportunity and provide the basis for comparing it with the job seeker profile.

The requirements of a job offer are particularly important for the matching process. They allow the system to identify the skills, qualifications, and professional conditions expected from the candidate. For example, a job offer may require a specific technical skill, a certain level of experience, or availability in a particular wilaya. By organizing this information, the system can provide recommendations that are more adapted to the user's profile and preferences.

Job offer data can also support conversational assistance. When a user asks about available jobs in a specific field or location, the chatbot needs access to structured job information in order to return useful results or ask additional questions when the request is incomplete. This makes job offer data important not only for matching, but also for guiding the user during the job search process.

Overall, job offer data is essential for connecting candidate information with employment opportunities. It supports job matching, recommendation generation, and user guidance by allowing the system to reason over the requirements and characteristics of available offers.

2.1.5.3 Recruitment Knowledge Base

The recruitment knowledge base represents the collection of domain-specific information used by the RequAI system to answer employment-related questions and guide job seekers. Unlike candidate profile data and job offer data, which are mainly used for personalization and matching, the knowledge base is used to provide reliable information about recruitment procedures, platform guidance, required documents, and frequently asked questions.

In the context of Wassit Online and ANEM-related services, users may ask different types of procedural questions. For example, they may need to know how to register, how to search for job offers, what documents are required, how to apply, how to update profile information, or how to understand certain platform services. These questions require answers that are clear, consistent, and connected to the recruitment context.

The knowledge base may therefore include frequently asked questions, guidance related to Wassit Online services, ANEM-related procedures, required documents, application steps, and general employment-service information. It can also include explanations that help users understand the role of the platform and the different steps of the recruitment process.

This type of data is important because the chatbot should not depend only on general generated responses when answering procedural or administrative questions. By relying on organized recruitment knowledge, the system can provide answers that are more relevant, more consistent, and more useful for job seekers.

Overall, the recruitment knowledge base supports the knowledge-based response generation module. It helps RequAI provide guided assistance, reduce repetitive questions, and improve access to recruitment-related information in the context of Wassit Online and ANEM services.

2.1.5.4 Generated and Prepared Data

In addition to candidate profile data, job offer data, and recruitment knowledge, the RequAI system also requires prepared data to support chatbot interaction, experimentation, and evaluation. This prepared data is important because real conversational datasets related to Wassit Online and ANEM services are not always publicly available, and access to real user interactions may be limited for privacy and administrative reasons.

Generated data can be used to simulate realistic user requests in the recruitment context. These examples may include questions about job search, required documents, registration steps, application procedures, platform guidance, CV information, skills, and job preferences. Preparing such examples helps cover different situations that job seekers may express during their interaction with the chatbot.

This type of data is also useful in a multilingual context. Since users in Algeria may communicate in Arabic, French, English, Algerian Darja, Arabizi, or mixed-language forms, generated and prepared examples can help represent different ways of asking the same question. This supports the design of a chatbot that is more adapted to real user language and not limited to formal expressions only.

Prepared data may also include structured examples of candidate profiles, skills, job offers, and frequently asked questions. These examples help test the different parts of the system, such as user request understanding, profile analysis, skill extraction, job matching, and knowledge-based response generation.

Overall, generated and prepared data is used to compensate for the limited availability of real domain-specific datasets, while allowing the proposed system to be designed and tested in conditions close to the target recruitment context. However, this type of data must be prepared carefully in order to remain realistic, relevant, and aligned with the objectives of the system.

2.1.6 Justification of the Proposed Approach

The proposed approach is justified by the need to improve job seeker assistance, user guidance, and recruitment matching in the context of Wassit Online and ANEM-related services. Although Wassit Online provides important digital employment services, users may still face difficulties when searching for information, understanding procedures, or finding job offers that correspond well to their profiles. For this reason, the proposed solution is designed as an intelligent support layer that complements the existing platform without replacing ANEM agents or official recruitment procedures.

The choice of a chatbot-based approach is motivated by the need for a more interactive and accessible form of assistance. Unlike static pages or fixed forms, a chatbot allows users to express their needs through natural language and receive guidance in a conversational way. This is especially useful for job seekers who may not know the exact keywords to use, the correct service to select, or the steps required to complete a recruitment-related procedure.

Natural language understanding is included in the proposed approach because user requests can be expressed in different ways. In the Algerian context, users may communicate in Arabic, French, English, Algerian Darja, Arabizi, or mixed-language forms. Therefore, the system should not depend only on rigid keywords or predefined forms. By analyzing user messages, the chatbot can identify the general intention of the request and extract useful information such as job title, location, skills, experience, education level, or preferences.

CV/profile analysis and skill extraction are also important choices because recruitment matching depends on more than job titles or simple keywords. A candidate profile may contain relevant information about education, experience, competencies, and preferences. Extracting and organizing these elements helps the system build a clearer representation of the job seeker and prepare the profile for comparison with job offer requirements.

The job matching and recommendation component is justified by the limitations of classical search and filtering mechanisms. Basic keyword-based search may miss relevant opportunities when the user profile and job offer use different terms, or when the candidate does not know how to formulate the search precisely. By comparing candidate-related information with job offer requirements, the system aims to provide more adapted recommendations and improve the relevance of job search assistance.

Finally, the use of a recruitment knowledge base is justified by the need for reliable and domain-specific answers. In public employment services, questions about procedures, documents, registration, and application steps must be answered clearly and consistently. A knowledge-based approach helps the chatbot provide responses connected to organized recruitment information instead of depending only on ungrounded generated text.

Overall, the proposed approach combines conversational assistance, natural language understanding, profile analysis, skill extraction, job matching, and knowledge-based responses in order to provide a more personalized and useful recruitment support experience. This combination is aligned with the objective of improving accessibility, guidance, and matching relevance for job seekers using Wassit Online and ANEM-related services.

2.2 Implementation and Experimentation

This section presents the practical implementation of the RequAI system. It describes the development tools, the first chatbot approach, and the main implementation choices used to transform the proposed architecture into a working prototype.

2.2.1 Development Environment and Tools

This subsection introduces the main environments, frameworks, libraries, and tools used during the development and experimentation phases of the RequAI system.

2.2.1.1 Integrated Development Environments

The development of the RequAI system required different development environments in order to design, implement, and test the main parts of the project. Since the system includes a web interface, backend services, data processing, and AI-based components, the use of appropriate development environments helped organize the implementation process and simplify testing.

Visual Studio Code was used as the main integrated development environment for writing and organizing the project source code. It provided a practical workspace for developing both frontend and backend components, managing project files, editing scripts, and running the application during development. Its integrated terminal and support for extensions made it suitable for full-stack development tasks.

Kaggle Notebook was used during the experimentation phase, especially for data exploration, preprocessing, model testing, and analysis of AI-related components. This environment was useful because it allowed the work to be organized step by step, combining code execution, intermediate results, and explanations in the same workspace. It also facilitated experimentation with datasets and model-related tasks before integrating the results into the complete system.

GitHub was used as a version control and collaboration platform during the development of RequAI. It facilitated the management of project files, source code updates, and different development versions. Since the system includes several components, including the web interface, backend services, data processing scripts, and AI-related experimentation, GitHub helped organize the project structure, track modifications, and support collaboration between team members.

In addition to these environments, a **web browser** was used to test the user interface and verify the interaction between the job seeker and the chatbot. Browser-based testing helped observe the behavior of the interface, check the chatbot interaction flow, and validate the communication between the frontend and backend components.

Overall, these development environments supported the implementation of RequAI by separating code development, experimentation, and interface testing. They helped organize the work and made it easier to test each part of the system before integration.

2.2.1.2 Frontend Development Tools and frameworks

The frontend part of RequAI was developed using a set of tools and frameworks that support the creation of a modern, responsive, and interactive web interface. This interface represents the main access point through which job seekers interact with the Requi chatbot, provide profile information, and receive answers or recommendations.

React was used as the main frontend framework for building the user interface. It allows the interface to be divided into reusable components, which makes the development process more organized and easier to maintain. In the RequAI system, React supports the creation of interface elements such as the chatbot conversation area, message components, profile forms, and other interactive elements.

Vite was used as the frontend development and build tool. It provides a fast development environment and simplifies the process of running and building the web application. This helped accelerate frontend development and testing during the implementation phase.

TailwindCSS was used for styling the interface. It allowed the design of a clean and responsive user interface using utility-based classes. This was useful for creating a simple, modern, and accessible chatbot interface adapted to different screen sizes.

Axios was used to manage communication between the frontend and backend services. Through HTTP requests, the frontend can send user messages, profile information, and other data to the backend, then display the returned chatbot responses or recommendations.

Additional frontend tools such as **React Hooks** and reusable UI components helped manage interface behavior and improve code organization. Overall, these frontend development tools contributed to building a user-friendly web interface that supports the interaction between job seekers and the RequAI system.

2.2.1.3 Backend Development Tools

The backend part of RequAI was developed using tools and frameworks that support server-side logic, data management, communication with the frontend, and integration with the intelligent modules of the system. The backend acts as an intermediate layer between the user interface, the database, and the AI-based services.

Flask was used as the main backend framework for building the web services of the application. It provides a lightweight and flexible environment for creating routes, handling user requests, and organizing the communication between the frontend interface and the system services. In RequAI, Flask supports the processing of chatbot messages, profile-related requests, and job recommendation operations.

SQLAlchemy was used to facilitate interaction with the database. It allows data models to be represented and manipulated in a more organized way, which helps manage information such as user profiles, skills, experience, chat sessions, messages, and job-related data.

Flask-CORS was used to manage cross-origin communication between the frontend and backend parts of the application. Since the frontend and backend may run on different ports during development, this tool helps allow controlled communication between them.

Flask-Session was used to support session management and maintain user-related interaction context during chatbot usage. This is useful for preserving continuity between user requests and supporting a more coherent interaction with the system.

Marshmallow was used for data validation and serialization. It helps verify the structure of exchanged data and supports cleaner communication between the frontend, backend, and database components.

Overall, these backend development tools helped organize the server-side part of RequAI and supported the implementation of the main services required by the system, including profile management, chatbot interaction, data access, and recommendation-related operations.

2.2.1.4 Artificial Intelligence and NLP Tools

The artificial intelligence and natural language processing part of RequAI relied on several Python libraries and machine learning tools during the experimentation and prototyping phases. These tools were mainly used to prepare recruitment-related data, process textual information, test intent understanding methods, and support early experiments related to job recommendation and information retrieval.

Data Manipulation and Preprocessing Libraries

Pandas and **NumPy** were used for data manipulation, numerical processing, and dataset preparation. They supported the organization of job offer data, candidate profile information, generated examples, and other structured data required during the development of the system.

Matplotlib was used to visualize data distributions and experimental results. It helped analyze intermediate outputs and better understand the behavior of the developed components during the experimentation phase.

Text Representation and Machine Learning Tools

Scikit-learn was used to support several machine learning and NLP-related experiments. In particular, **TF-IDF Vectorizer** was used to represent textual data in numerical form, while **cosine similarity** was used in similarity-based experiments, such as comparing candidate profiles, job offers, or knowledge elements.

Scikit-learn was also used during the intent classification experiments, where different text representation and classification methods were tested. These experiments helped evaluate how user requests could be categorized into predefined intent classes and later routed toward the appropriate system component.

2.2.1.5 Database and Data Management Tools

The RequAI system requires data management tools to store and organize the information used by the application. This includes user profiles, skills, professional experience, chat sessions, exchanged messages, and job-related information. Managing these data elements is necessary for supporting chatbot interaction, profile analysis, and job recommendation functionalities.

SQLite was used as the database during the development phase. It provides a lightweight relational database solution that is suitable for local development and testing. In RequAI, SQLite was used to store structured information such as user data, skills, experience records, chat history, and job-related data.

SQLAlchemy was used to facilitate communication between the backend application and the database. It allows database entities to be represented as models inside the backend, making data access and manipulation more organized. This helped manage the relationship between user profiles, skills, experience, messages, and job information.

Overall, the database and data management tools supported the persistence and organization of the information required by RequAI. They helped maintain user-related data, conversation history, and job offer information, which are necessary for personalized assistance and recruitment-oriented recommendations.

2.2.2 First Chatbot Approach

This subsection presents the first chatbot approach developed during the experimentation phase. It describes its objective, the data used, the preprocessing steps, the processing pipeline, and its role in the complete RequAI system.

2.2.2.1 Objective of the First Approach

The first chatbot approach was designed as an NLP-based recruitment assistant that relies on prepared datasets, text preprocessing, intent classification, and similarity-based matching. The objective of this approach is to build a controlled chatbot pipeline capable of understanding common job seeker requests, extracting useful information, retrieving appropriate responses, and supporting job recommendation based on candidate and job offer data.

This approach focuses on transforming user messages into structured information that can be processed by the system. When a job seeker sends a message, the chatbot aims to identify the intention behind the request, such as searching for job offers, asking about required documents, requesting registration guidance, or seeking platform-related assistance. This allows the system to choose the appropriate processing path and provide a response that corresponds to the user's need.

Another objective of this approach is to support job matching using prepared textual representations of job offers and candidate profiles. Instead of relying only on exact keyword search, the system prepares job-related text by combining important fields such as job title, role, required skills, job description, responsibilities, qualifications, and experience. This representation helps compare candidate information with job offers and supports the generation of more relevant job suggestions.

The first approach also aims to provide a reusable experimentation pipeline. The prepared data, trained intent classifier, vectorizers, response templates, and knowledge files can be saved and reused during testing and integration. This makes the approach suitable for evaluating the main chatbot functions before or alongside more advanced LLM-based interaction.

Overall, the objective of the first chatbot approach is to provide a structured and explainable baseline for recruitment assistance. It supports user request understanding, basic conversational guidance, profile-related processing, and job matching using classical NLP and machine learning techniques. This approach is useful because its behavior is easier to control, evaluate, and interpret during experimentation.

2.2.2.2 Data Used in the First Approach

The first chatbot approach relies on several prepared data sources that support the main NLP-based components of the system, including intent understanding, entity extraction, job recommendation, profile analysis, and knowledge-based responses. These data sources were organized in a structured way in order to allow the chatbot to process user requests and generate relevant answers.

The main data source used in this approach is a job offer dataset adapted to the Algerian recruitment context. The original job data was based on a public job description dataset obtained from Kaggle, then cleaned, localized, and enriched with information relevant to the Algerian employment environment. The final version contains job-related attributes such as job title, role, description, required skills, qualifications, experience, location, work type, responsibilities, company information, and salary values expressed in Algerian dinars.

A processed version of the job dataset was also prepared for natural language processing and recommendation tasks. In this version, textual job information was cleaned, normalized, and combined into a unified text representation. This representation includes important elements such as the job title, required skills, job description, responsibilities, qualifications, and experience. It was mainly used to compare candidate profiles with job offers and support text-based job matching.

In addition to job data, simulated candidate profile data was used to test the chatbot without relying on real private user information. These profiles include information such as education level, field of study, experience, skills, target job title, location, and CV-like textual content. This data helped evaluate the profile analysis, skill extraction, and recommendation components of the chatbot.

The first approach also includes recruitment-related knowledge data. This knowledge base contains frequently asked questions, procedural information, and employment guidance related to ANEM and job search processes. It was prepared in a structured and retrievable form so that the chatbot could answer informational questions and provide guidance based on predefined knowledge.

Finally, several configuration and saved component files were used to support the execution of the chatbot. These include resources for intent classification, text vectorization, entity extraction, response templates, and retrieval-based answering. Their role is mainly technical, as they allow the chatbot components to be reused, tested, and integrated more easily.

Overall, the data used in the first approach was prepared to support a classical NLP-based chatbot pipeline. It provides the necessary information for intent classification, entity extraction, knowledge retrieval, profile analysis, and job recommendation.

2.2.2.3 Data Preparation and Preprocessing

After identifying the data used in the first chatbot approach, several preparation and preprocessing steps were applied to make the data suitable for the chatbot pipeline. The objective of this phase was to transform raw and semi-structured data into organized representations that could support intent understanding, entity extraction, job matching, profile analysis, and knowledge retrieval.

The job offer dataset was first adapted to the Algerian recruitment context. Relevant recruitment fields were selected, including the job title, role, description, required skills, responsibilities, qualifications, experience, location, work type, and salary information. This selection helped keep only the information that could be useful for understanding job offers and comparing them with candidate profiles.

Text preprocessing was then applied to the main textual fields. This step included converting text to lowercase, tokenizing the text, removing common stop words, and normalizing some abbreviations and synonyms. For example, technical abbreviations such as `js`, `py`, `ml`, `ai`, and `nlp` were transformed into clearer forms such as `javascript`, `python`, `machine_learning`, `artificial_intelligence`, and `natural_language_processing`. This normalization helped reduce variation between different ways of expressing the same skill or technical concept.

In addition to text cleaning, categorical information was prepared for later processing. Attributes such as job title, experience level, qualification, work type, location, role, and skills were represented in a more structured form. This made the data easier to use during experimentation, analysis, and matching.

A combined textual representation was also created for each job offer. This representation groups the most important job information, such as the title, role, required skills, description, responsibilities, qualifications, and experience, into a single cleaned text field. The purpose of this step was to represent each job offer in a complete textual form that could be used for vectorization, similarity calculation, and recommendation.

Candidate-related data was also prepared to support profile analysis and recommendation testing. Simulated user profiles were organized using information such as wilaya, education level, field of study, experience, skills, target job title, and CV-like text. The use of simulated profiles allowed the chatbot components to be tested without relying on real private candidate data.

The recruitment knowledge base was prepared separately to support informational answers. Frequently asked questions and procedural information related to ANEM and job search processes were structured and divided into smaller knowledge units. This preparation made it easier for the chatbot to retrieve relevant information according to the user request.

Finally, additional configuration resources were prepared to support entity extraction and response generation. These resources included predefined lists of useful recruitment-related terms, such as wilayas, education levels, work types, and skills, as well as response patterns for common user requests. These elements helped the chatbot identify important information in user messages and generate more consistent answers.

Overall, the preprocessing phase transformed the collected data into structured and usable resources. These prepared resources support the main components of the first chatbot approach, including intent classification, entity extraction, job matching, knowledge retrieval, and profile-based assistance.

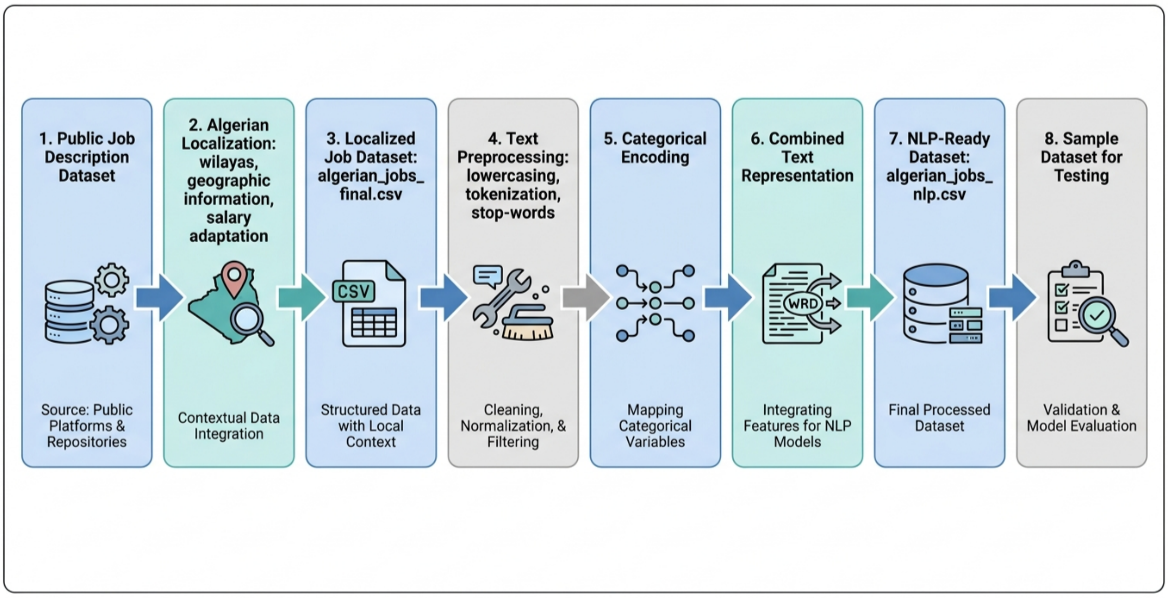

Figure 2.6: Data preparation and preprocessing pipeline used in the first chatbot approach.

2.2.2.4 Model and Processing Pipeline

The first chatbot approach follows a classical NLP-based processing pipeline. The pipeline is designed to process the user message step by step, starting from text preparation and intent understanding, then moving to entity extraction, knowledge retrieval, response selection, or job recommendation depending on the detected request.

The first step consists of receiving the user message and applying basic text preprocessing. The input text is normalized by converting it to lowercase and reducing unnecessary textual variation when possible. This preparation makes the message easier to compare with the training examples, knowledge-base content, and predefined response patterns.

After preprocessing, the chatbot applies intent classification in order to identify the purpose of the user request. In this approach, a supervised machine learning model based on Character TF-IDF representation and a Linear Support Vector Machine classifier was used. Character-level TF-IDF was selected because it is useful for handling spelling variation, informal expressions, multilingual input, and mixed-language messages, which are common in the Algerian context. The classifier assigns the user message to predefined intent categories such as job search, CV or profile assistance, registration help, required documents, agency information, platform guidance, or general conversation.

Once the intent is detected, the system applies entity extraction to identify useful information contained in the message. This step relies on predefined recruitment-related resources, including lists of Algerian wilayas, education levels, work types, job-related terms, and skill keywords. The extracted entities help the chatbot identify important details such as the preferred location, target job, required skill, education level, or type of request. These entities are then used to guide the next processing step.

If the user request is related to job search or recommendation, the system compares the user request or candidate profile with the prepared job offer representations. Each job offer is represented using a cleaned textual field that combines important information such as the job title, role, required skills, description, responsibilities, qualifications, and experience. The user profile or search request is transformed into a comparable textual representation, and cosine similarity is used to identify the most relevant job offers. The chatbot can then return a ranked list of possible job recommendations.

If the user request is informational or procedural, the system uses a retrieval-based mechanism over the prepared recruitment knowledge base. The knowledge base contains FAQ content and structured information related to ANEM procedures, required documents, registration steps, and platform guidance. The user message is compared with the available knowledge entries, and the most relevant information is retrieved to generate a domain-specific answer.

For frequent and predictable requests, the chatbot also uses predefined response templates. These templates help produce consistent answers for common situations, such as greetings, general guidance, fallback responses, and standard recruitment-related questions. This makes the chatbot behavior more controlled and easier to evaluate.

The prepared models and vectorization resources are saved and reused during execution. This allows the chatbot to load the intent classifier, text vectorizers, retrieval resources, and recommendation components without rebuilding them each time. This organization supports easier experimentation, testing, and integration of the first chatbot approach into the complete system.

Overall, the processing pipeline of the first approach can be summarized as follows: the user message is received, preprocessed, classified by intent, analyzed for entities, and routed to the appropriate module. According to the detected intent, the chatbot either retrieves knowledge, selects a predefined response, or recommends relevant job offers. This approach provides a structured and explainable chatbot behavior based on classical NLP and machine learning techniques.

2.2.2.5 Bias Analysis Consideration

In addition to the main NLP-based chatbot pipeline, the first approach included an exploratory bias analysis step. This analysis was not the central objective of the proposed system, but it was useful for observing whether recommendation scores could be influenced by identity-related signals that are not directly related to candidate competence.

The bias analysis was based on comparing two recommendation conditions. In the first condition, the candidate profile or CV was processed with visible identity signals, such as name, gender-related indicators, or location-related information. In the second condition, these identity signals were masked or removed, while keeping the same CV content, skills, and qualifications. The objective was to observe whether the match score changed when the candidate identity information was hidden.

This step is important in recruitment systems because recommendation results should mainly depend on relevant professional information such as skills, education, experience, and job requirements. If two profiles have the same qualifications but receive different recommendation scores because of identity-related information, this may indicate a possible fairness issue in the matching process.

Figure 2.7 illustrates an example of score recovery after bias masking. The comparison shows that some profiles obtained higher match scores after identity signals were removed. For example, the scores of the profiles represented as Mohamed and Fatima increased after masking, while the difference was smaller for other profiles. This suggests that identity-related signals may influence the recommendation score in some cases.

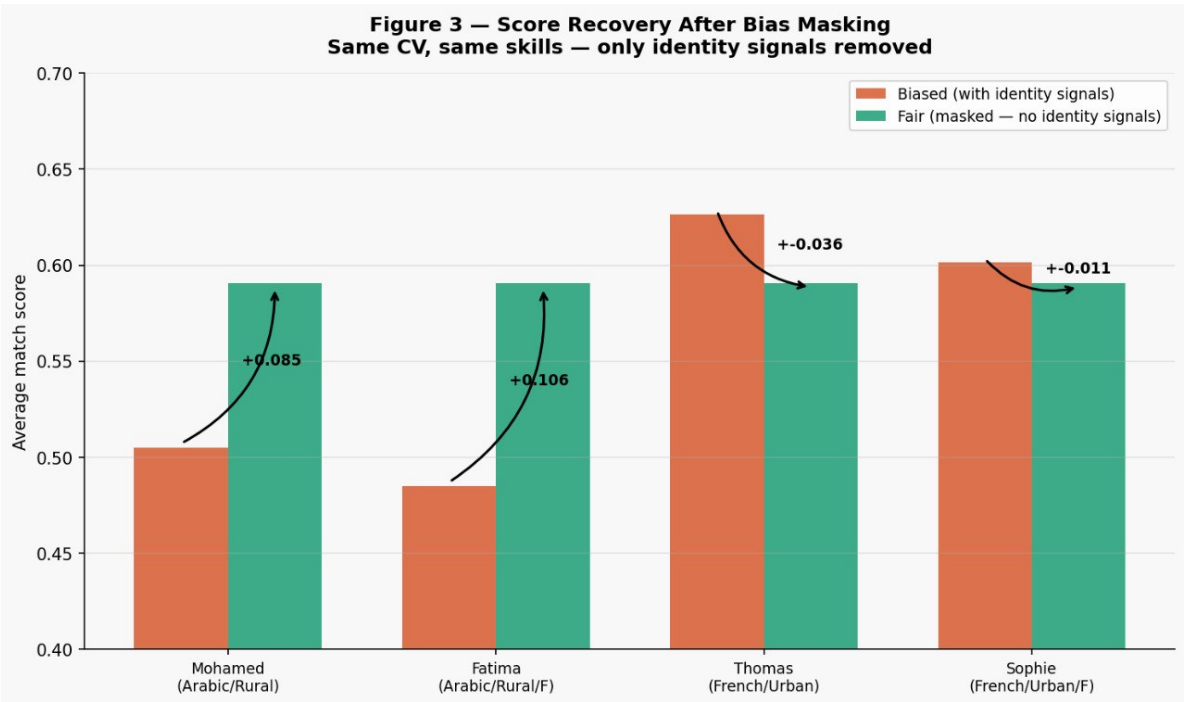

Figure 2.7: Score recovery after bias masking using the same CV and skills while removing identity-related signals.

The bias analysis was therefore treated as an exploratory diagnostic step rather than a complete fairness solution. Its role was to help identify possible differences in recommendation behavior and to show that masking sensitive or identity-related information can be useful when evaluating recruitment matching systems. However, a complete fairness evaluation would require more formal metrics, larger validation scenarios, and careful analysis of different candidate groups.

Overall, this bias analysis provided additional insight into the limitations of the first chatbot approach. It showed that recommendation systems in recruitment should not only be evaluated according to matching relevance, but also according to fairness, transparency, and responsible use.

2.2.2.6 Output and Use in the System

The output of the first chatbot approach is the result produced after the user message has been processed by the NLP-based pipeline. Depending on the detected intent and the information extracted from the message, the system can generate different types of outputs, including direct answers, procedural guidance, extracted entities, job recommendations, or clarification questions.

When the user request is related to general employment information, ANEM procedures, required documents, registration steps, or platform guidance, the chatbot uses the detected intent and the prepared response templates to return a structured answer. These responses are supported by files such as `response_templates.json`, `anem_faq_dynamic.csv`, and `anem_procedures.json`. This makes the output more controlled and more suitable for frequent user questions.

When the user request requires knowledge retrieval, the system uses the prepared knowledge chunks stored in `rag_chunks.csv`. The user message is compared with the available chunks, and the most relevant information is retrieved to support the generated answer. This allows the chatbot to provide more context-aware responses for questions related to recruitment procedures, documents, or employment services.

For job search and recommendation requests, the output consists of a ranked list of potentially relevant job offers. The recommendation is based on the comparison between the user request or candidate profile and the prepared job offer representation, especially the `job_text_clean` field. The system can use similarity scores to identify job offers that are closer to the candidate's skills, experience, location, and preferences.

The first approach can also return intermediate information that supports the interaction, such as the detected intent, extracted entities, or missing information. For example, if the user asks for job offers without specifying a location or domain, the chatbot can ask a clarification question before producing recommendations. This makes the interaction more guided and helps avoid irrelevant results.

In the complete RequAI system, the first chatbot approach can be used as a structured and explainable baseline. Its outputs are useful for testing the main recruitment assistance functions, such as intent classification, entity extraction, knowledge retrieval, and job matching. Since its components are based on saved classifiers, vectorizers, templates, and prepared datasets, this approach is easier to reload, evaluate, and debug during experimentation.

Overall, the first chatbot approach produces controlled outputs that support user guidance, recruitment-related question answering, and job recommendation. It contributes to the system by providing an interpretable NLP-based pipeline that can assist job seekers while remaining easier to analyze and evaluate than a fully generative approach.

2.2.3 Second Chatbot Approach

The second chatbot approach was developed as a multilingual and context-aware extension of the initial RequAI pipeline. While the first approach is mainly based on a classical NLP workflow with intent classification, entity extraction, templates, knowledge retrieval, and similarity-based job matching, the second approach focuses more on realistic multilingual user interaction. It introduces additional language understanding components, including language detection, multilingual intent classification, transformer-based entity extraction, and retrieval-supported response preparation.

This approach was designed to better handle the way Algerian job seekers may naturally interact with a chatbot. In practice, users may write in English, French, Arabic, Algerian Darija, Arabizi, or mixed-language forms. They may also use informal expressions, spelling variations, abbreviations, and incomplete sentences. For this reason, the second approach gives more importance to multilingual preprocessing, language identification, contextual intent understanding, and extraction of useful recruitment-related entities.

2.2.3.1 Objective of the Second Approach

The objective of the second chatbot approach is to improve the chatbot's ability to understand multilingual and informal recruitment-related messages. Instead of assuming that all user requests are written in one language or in a clean formal style, this approach considers the linguistic diversity of the Algerian context. It aims to support user messages written in English, French, Arabic, Algerian Darija, Arabizi, and mixed language.

Another objective is to make the chatbot more context-aware. The system does not only classify the message into a general intent, but also attempts to detect the language of the message, identify the user's request, extract useful entities, and prepare the necessary context for the response. These entities may include the desired job, location, skills, qualification, experience, salary, or work type.

This approach also aims to support more natural recruitment assistance. A job seeker may ask for a job in a specific wilaya, request information about ANEM procedures, ask about required documents, upload or analyze a CV, or ask for job recommendations based on skills and experience. The second approach was therefore designed to combine language detection, intent recognition, entity extraction, and contextual information retrieval in order to support these different interaction scenarios.

Overall, the objective of the second approach is to move RequAI from a simple controlled chatbot toward a more flexible multilingual recruitment assistant that can better understand realistic user messages and prepare more relevant responses.

2.2.3.2 Data Used in the Second Approach

The second chatbot approach relies on three main categories of data: language detection data, intent classification data, and entity extraction data. These datasets were prepared separately because each component of the chatbot requires a different type of input and annotation.

For language detection, several datasets were collected and combined in order to represent the main languages used by the target users. The data includes English examples from general and job-search datasets, French examples from general question datasets, Arabic examples from Arabic textual datasets, and Algerian Darija examples written in both Arabic script and Latin script. Additional Arabizi and mixed-language examples were also included to represent informal Algerian writing. These sources were used to build a language detection dataset with four final language labels: English, French, Arabic, and Darija. In this module, Darija includes Algerian dialect written in Arabic script as well as Arabizi or Latin-script Darija.

For intent classification, the data was prepared from initial chatbot intent examples, job-search examples, and multilingual recruitment-specific datasets. The final intent dataset contains user messages annotated with their corresponding intent and language. It includes several recruitment-related intents such as job search, job matching, CV upload help, CV analysis, application guidance, application status, profile update, registration help, required documents, agency location, agency hours, ANEM procedures, password or account help, bot identity, bot capabilities, greeting, goodbye, and fallback. The dataset was organized using three columns: `text`, `intent`, and `language`. It covers English, French, Arabic, Algerian Darija, Arabizi, and mixed-language messages.

For entity extraction, a specific dataset was prepared using recruitment-related entity aliases and generated examples. The entity aliases include Algerian wilayas, job titles, job fields, skills, qualifications, diplomas, work types, and job roles. These aliases were collected and adapted in several languages and writing styles, including English, French, Arabic, Darija, Arabizi, and mixed forms. Based on these resources, a multilingual entity extraction dataset was generated and annotated using BIO tagging.

In addition to these NLP datasets, the second approach also uses the prepared recruitment knowledge base and job-related data already used in the RequAI system. This includes job offer information, ANEM-related frequently asked questions, procedure data, and knowledge chunks. These resources support the retrieval and contextual response preparation part of the second approach.

Overall, the data used in the second approach was prepared to support multilingual understanding, intent detection, entity extraction, and contextual recruitment assistance.

2.2.3.3 Data Preparation and Preprocessing

The data preparation phase of the second approach was organized according to the needs of each NLP component. Since the approach includes language detection, intent classification, and entity extraction, each dataset required a specific preparation process.

For the language detection module, the collected datasets were unified into a common structure containing two columns: `text` and `label`. The labels were standardized into four classes: `en`, `fr`, `ar`, and `darja`. Texts were cleaned by removing empty values, trimming unnecessary spaces, removing duplicates, and keeping only valid labels. The cleaning process was intentionally light because language detection benefits from preserving useful linguistic signals such as Arabic characters, French accents, Latin-script Darja patterns, and informal spelling. The final language detection dataset was split into training, validation, and test sets using stratified splitting in order to preserve the distribution of each language class.

For the intent classification module, the preparation process focused on building a multilingual recruitment intent dataset. Each example was represented using three fields: `text`, `intent`, and `language`. The dataset was created by combining separate intent datasets related to job search, job matching, CV analysis, CV upload help, application guidance, application tracking, profile update, registration, required documents, agency information, ANEM procedures, account help, and general chatbot interaction. After merging, the data was cleaned by removing missing values, empty texts, and duplicate examples. A stratified split was then applied using both the intent and language information, so that each split preserves the distribution of the different intent-language combinations.

For entity extraction, the preparation started by creating an entity alias table. This table contains different ways of writing the same recruitment-related information. For example, a location may appear in French, Arabic, Darija, or Arabizi, and the same skill or work type may be expressed using several variants. The alias table was then used to generate multilingual recruitment messages with annotated entities. These examples were converted into token-level annotations using the BIO format, where `B-` marks the beginning of an entity, `I-` marks the continuation of an entity, and `O` marks tokens outside any entity.

The entity extraction dataset includes several entity types, such as `LOCATION`, `JOB_TITLE`, `JOB_FIELD`, `SKILL`, `EXPERIENCE`, `QUALIFICATION`, `SALARY`, and `WORK_TYPE`. The generated examples cover different recruitment scenarios, such as searching for a job in a specific location, requesting a job with a certain qualification, mentioning experience, asking for a salary range, or looking for a job that matches specific skills.

The prepared entity dataset was also split into training, validation, and test sets while preserving the language distribution. Additional verification was applied to ensure that the number of tokens matched the number of BIO tags for each example. This step was important because token classification models require each token to have a corresponding label.

Overall, the preprocessing phase transformed multilingual and informal recruitment data into structured datasets suitable for language detection, intent classification, and entity extraction.

2.2.3.4 Model and Processing Pipeline

The second chatbot approach follows a multilingual processing pipeline. The pipeline begins when the user sends a message to the chatbot. The message is first cleaned using light preprocessing, such as removing unnecessary spaces and normalizing the text format without removing important language-specific information.

The first model in the pipeline is the language detection model. This model identifies the language of the user message as English, French, Arabic, or Darija. Character-level TF-IDF representation was used because it is suitable for short and informal multilingual texts. It can capture useful patterns such as Arabic characters, French spelling, English word forms, and Darija or Arabizi character sequences. Logistic Regression and Linear SVM were both tested for this task. Logistic Regression was selected as the final model because it performed well and provides confidence scores, which are useful for detecting uncertain or mixed-language messages.

After language detection, the message is passed to the intent classification module. The objective of this module is to identify the user request, such as job search, CV analysis, registration help, required documents, agency location, agency hours, application guidance, or account help. Several intent classification models were tested, including Word TF-IDF with Linear SVM, Character TF-IDF with Linear SVM, and XLM-RoBERTa fine-tuning. The Character TF-IDF with Linear SVM model was selected as the main intent classifier because it achieved strong results while remaining simple, fast, and robust to spelling variation, Darija, Arabizi, and mixed-language messages.

Once the intent is detected, the entity extraction module is applied when the request requires additional information. This module was implemented using XLM-RoBERTa for token classification. The model receives the user message and predicts BIO labels for each token. It can extract recruitment-related entities such as location, job title, job field, skill, experience, qualification, salary, and work type. These extracted entities help the chatbot understand the details of the user's request.

After language detection, intent classification, and entity extraction, the system prepares the response context. If the request is procedural or informational, the system retrieves relevant information from the prepared recruitment knowledge base, such as ANEM procedures, required documents, registration steps, or Wassit Online guidance. If the request is related to job search or job matching, the system can use the extracted entities and candidate-related information to support the recommendation process.

The final response is then prepared according to the detected intent, extracted entities, and retrieved context. For simple and predictable requests, predefined response patterns can be used. For more complex requests, the response is prepared using the retrieved information and the user context. If the user message is incomplete, the chatbot can ask for missing information, such as the preferred location, target job, skills, or experience level.

The general pipeline of the second approach can be summarized as follows: the user message is received, lightly preprocessed, classified by language, classified by intent, analyzed for entities, enriched with retrieved context, and then used to prepare the chatbot response.

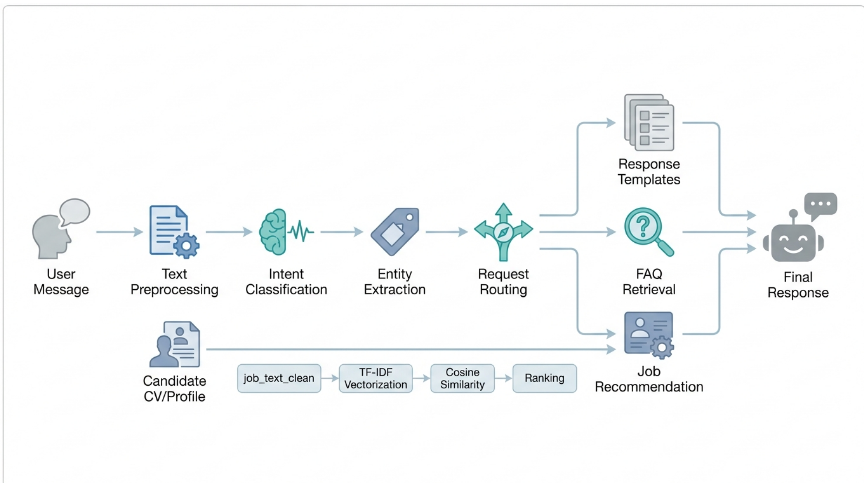

Figure 2.8: General processing pipeline of the second chatbot approach.

2.2.3.5 Output and Use in the System

The output of the second chatbot approach is not limited to a direct answer. It produces several intermediate outputs that help the chatbot understand the user request and prepare a more relevant response.

The first output is the detected language. This information helps the system identify whether the message is written in English, French, Arabic, or Darija. It can also help the chatbot adapt the response language or detect uncertain mixed-language cases.

The second output is the detected intent. This determines the main purpose of the user message. For example, the message may be classified as a request for job search, job matching, CV analysis, registration help, required documents, agency information, application guidance, or fallback. The detected intent is used to route the request to the appropriate processing module.

The third output is the set of extracted entities. These entities provide important details from the message, such as the desired job, preferred wilaya, required skill, qualification level, experience, salary, or work type. For example, from a message such as "I am looking for a driver job in Oran with no diploma", the system can extract the job field or job title, the location, and the qualification constraint.

The fourth output is the retrieved or prepared context. Depending on the intent, this context may come from the recruitment knowledge base, ANEM procedure data, FAQ data, job offer data, or candidate profile information. This context helps the chatbot produce answers that are connected to the prepared RequAI knowledge instead of generating unsupported responses.

Finally, the system produces the chatbot response. The response may be an explanation, a procedural answer, a clarification question, a job-search answer, or a profile-based recommendation. In the complete RequAI workflow, this approach can support multilingual interaction and improve the quality of user understanding before the response is generated.

Overall, the second approach is used to enrich the chatbot with stronger multilingual understanding. It supports language detection, intent recognition, entity extraction, context retrieval, and response preparation, making the chatbot more suitable for realistic recruitment conversations.

2.2.3.6 Implementation Status

The second chatbot approach was implemented through several experimental modules. These modules were developed and tested separately in order to evaluate the behavior of each part before integrating them into the complete chatbot workflow.

The language detection module was implemented using character-level TF-IDF representation with Logistic Regression and Linear SVM models. The final prepared dataset contains balanced examples across English, French, Arabic, and Darija. The final splits contain `6,383` training examples, `1,368` validation examples, and `1,368` test examples. A challenge test was also used to evaluate the behavior of the model on more difficult mixed and informal messages.

The intent classification module was implemented using a multilingual recruitment intent dataset. The final dataset contains `45,330` examples distributed across `18` intent classes and `6` language categories. The dataset was split into `36,264` training examples, `4,533` validation examples, and `4,533` test examples. Several models were tested, including Word TF-IDF with Linear SVM, Character TF-IDF with Linear SVM, and XLM-RoBERTa fine-tuning. The Character TF-IDF with Linear SVM model was selected as the main model because it provided strong performance while remaining efficient and robust for informal multilingual messages.

The entity extraction module was implemented using XLM-RoBERTa for token classification. A multilingual BIO-tagged dataset was generated from recruitment-related aliases and templates. The full entity dataset contains `6,000` examples, with `1,000` examples for each language category: English, French, Arabic, Darija, Arabizi, and mixed language. The dataset was split into `4,800` training examples, `600` validation examples, and `600` test examples. The model was trained to recognize recruitment-related entities such as location, job title, job field, skill, experience, qualification, salary, and work type.

The implementation also includes saved models and output files for reuse during experimentation. These include the language detection pipeline, the selected intent classification model, the XLM-RoBERTa entity extraction model, label encoders, classification reports, manual test outputs, and misclassification analysis files. These outputs make the second approach easier to test, compare, reload, and integrate with the wider RequAI system.

At this stage, the second approach represents an implemented multilingual NLP extension of RequAI. Its main understanding components have been prepared and evaluated at module level, while the full conversational workflow remains open to further refinement through broader scenario testing and user-based validation.

2.2.4 System Implementation

The implementation of RequAI follows a modular full-stack architecture that separates the presentation layer, backend services, data management, and intelligent chatbot components. This organization helps make the system easier to develop, test, maintain, and extend. The frontend is responsible for user interaction, the backend manages application logic and communication between components, the database stores user and conversation data, and the AI-related modules support natural language interaction, profile-based assistance, and job recommendation.

The system implementation is organized into five main parts: frontend implementation, backend implementation, database design, integration of the chatbot components, and deployment configuration. Each part contributes to transforming the conceptual architecture presented earlier into a working recruitment assistance system.

2.2.4.1 Frontend Implementation

The frontend part of RequAI was implemented as a web-based interface that allows job seekers to interact with the Requi chatbot and manage profile-related information. It represents the visible part of the system and serves as the main access point for users.

The interface was developed using **React** with **Vite** and organized into reusable components. These components include the chatbot interaction area, message display, user input, profile form, skills input, experience timeline, and shared interface elements. This organization separates the chat, profile, and common interface parts of the application.

The chatbot interface allows the user to write messages, ask employment-related questions, describe job search needs, and receive responses from the system. The profile-related interface allows the user to provide personal details, skills, and professional experience when needed. These elements are important because they provide the information required for personalized assistance and job recommendation.

At the interface level, **TailwindCSS**, **Headless UI**, and **Heroicons** were used to implement the visual elements of the chat and profile views. Communication with the backend was handled through **Axios**, which sends profile data and chat messages to the API endpoints and receives the generated responses or recommendations. Reusable components and hooks were used to organize interface behavior. Overall, the frontend implementation provides a simple and interactive environment through which job seekers can access the RequAI services.

2.2.4.2 Backend Implementation

The backend part of RequAI was implemented using **Flask**. It acts as the central layer responsible for receiving requests from the frontend, processing user actions, communicating with the database, and coordinating the intelligent modules of the system.

The backend is organized around routes, services, models, and AI-related components. The route layer handles incoming requests, such as chatbot messages, profile operations, job search, and recommendation requests. The service layer contains the main application logic, including chat management, profile handling, and recommendation-related operations. This separation helps keep the backend organized and makes each part easier to maintain.

For chatbot interaction, the backend receives the user message from the frontend and forwards it to the appropriate processing components. Depending on the user request, the backend may retrieve profile information, access conversation history, prepare context, call the AI-based service, or return job-related suggestions. This makes the backend an important coordination layer between the user interface, the database, and the intelligent processing modules.

At the backend level, validation, cross-origin communication, and session continuity were handled before requests reached the main services. **Marshmallow** was used to validate exchanged data, **Flask-CORS** supported controlled communication with the frontend during development, and **Flask-Session** helped preserve interaction context between user requests.

The backend exposes different groups of API endpoints that support profile management, chatbot interaction, job-related operations, and system monitoring. Table 2.1 summarizes the main endpoint groups used in the system.

Table 2.1: Main API endpoint groups of the RequAI backend

Endpoint Group	Examples	Purpose
Profile endpoints	`/api/profile`, `/api/profile/:id`, `/api/profile/:id/skills`	Create, update, retrieve, and manage job seeker profile information and skills.
Chat endpoints	`/api/chat`, `/api/chat/history/:userI` `/api/chat/:sessionId`	Send user messages to the chatbot, retrieve conversation history, and manage chat sessions.
Job endpoints	`/api/jobs/recommend/:use` `/api/jobs/search`, `/api/jobs/apply/:jobId`	Support job recommendation, job search, and job-related operations.
System endpoints	`/health`, `/metrics`	Verify system availability and monitor basic system status.

2.2.4.3 Database Design

The database design of RequAI supports the storage and organization of the main data required by the system. During development, **SQLite** was used as the database solution because it is lightweight and suitable for local development and testing.

The database stores several categories of information. User-related data includes identity and contact information such as name, email, phone number, and location. Skill-related data stores the skills associated with each user, while experience-related data stores professional background information such as previous job titles, companies, dates, and descriptions. These data elements help build a structured representation of the job seeker profile.

The database also stores conversation-related information. Chat sessions are used to group user interactions, while messages store the conversation turns between the user and the assistant. This makes it possible to maintain conversation history and support continuity during chatbot interaction.

In addition, job-related information is stored to support recommendation and matching tasks. Job records may include information such as job title, company, location, requirements, salary range when available, and posting date. This information provides the basis for comparing user profiles with available job opportunities.

Table 2.2 presents a summary of the main database tables used in the RequAI system.

Table 2.2: Database schema summary of the RequAI system

Table	Key Fields	Purpose
`users`	`id`, `name`, `email`, `phone`, `location`, `created_at`	Stores job seeker identity and contact information.
`skills`	`id`, `user_id`, `skill_name`, `proficiency_level`	Stores the skills associated with each user.
`experience`	`id`, `user_id`, `job_title`, `company`, `start_date`, `end_date`, `description`	Stores the professional experience of the job seeker.
`chat_sessions`	`id`, `user_id`, `session_token`, `created_at`, `last_active`	Groups messages into conversation sessions.
`messages`	`id`, `session_id`, `role`, `content`, `timestamp`	Stores the exchanged messages between the user and the chatbot.
`jobs_cache`	`id`, `title`, `company`, `location`, `requirements`, `salary_range`, `posted_date`	Stores job offer information used for recommendation and matching.

Overall, the database design supports the main functions of RequAI by organizing user profiles, skills, experience, chat history, and job-related data. This structured storage is necessary for personalized assistance, knowledge-based interaction, and job recommendation.

2.2.4.4 Model Data and Input Structure

The intelligent behavior of RequAI depends on several structured inputs prepared during user interaction. Instead of relying only on a user message, the system can combine the current request with profile information, conversation history, job-related data, and a prompt structure prepared for the language model. This organization helps the chatbot generate responses that are more contextual and connected to the available recruitment data.

The user profile context may include information such as the user's location, skills, professional experience, and previous interaction history. These elements help personalize the response and support more relevant job recommendations. Job-related information may include job title, company, location, requirements, salary range when available, and posting date. These fields provide the basis for matching user profiles with available opportunities.

The prompt sent to the LLM is built by combining fixed instructions with dynamic information retrieved from the system. This structure helps guide the chatbot behavior and connect its answers to the user profile, conversation history, and available recruitment data.

Table 2.3 summarizes the main data elements used to build the chatbot interaction context.

Table 2.3: Main data elements used in the chatbot interaction context

Data Element	Source	Role in the Interaction
User location	User profile	Helps provide geographically relevant job recommendations.
Skills	User profile / skills data	Used as a main signal for matching the user with job requirements.
Professional experience	Experience data	Provides context about the user's background and seniority level.
Conversation history	Messages data	Maintains interaction continuity and avoids unnecessary repetition.
Job listings	Jobs data	Provides available job opportunities for matching and recommendation.
Current user message	Chat interface	Represents the latest request that the chatbot must answer.

Table 2.4 presents the general prompt structure used for chatbot response generation.

Table 2.4: Prompt structure used for chatbot response generation

Role	Content Type	Description
System	Fixed instruction block	Defines the assistant identity, behavior rules, and response style.
System	User profile context	Includes user-related information such as location, skills, and experience.
System	Job or knowledge context	Provides relevant job listings or recruitment information retrieved by the system.
User/Assistant	Conversation history	Preserves previous turns to maintain continuity in the dialogue.
User	Current message	Contains the latest user request to be answered by the chatbot.

2.2.4.5 Integration of the Chatbot Components

The integration of the chatbot components connects the frontend, backend, database, and AI-related modules into a complete interaction flow. This integration allows the user to move from profile creation and message submission to receiving personalized guidance or job recommendations.

When the user submits profile information, the frontend sends the data to the backend. The backend validates and processes the information, then stores it in the database. This profile information can later be used to personalize chatbot responses and support job matching.

During chat interaction, the user sends a message through the chatbot interface. The frontend transmits the message to the backend, where the system retrieves useful context such as the user profile and recent conversation history. The message is then analyzed to identify the type of request and determine the appropriate processing path, such as answering a procedural question, updating profile information, or supporting job search assistance.

For AI-based interaction, the system prepares the necessary context before generating a response. This context may include the user profile, extracted skills, conversation history, and relevant job or knowledge-base information. The prompt engineering component helps organize this context before it is sent to the LLM service through **Groq Cloud**. The generated response is then returned to the frontend and displayed to the user through the chatbot interface.

This integration allows the different parts of RequAI to work together in a coordinated way. The frontend provides interaction, the backend manages logic and communication, the database stores required information, and the AI-related modules support natural language understanding and response generation. Together, these components support accessible, context-aware, and personalized recruitment assistance.

2.2.4.6 Deployment Configuration

The deployment configuration of RequAI is based on a containerized organization using **Docker** and **Docker Compose**. This configuration helps run the main parts of the system in separate environments and simplifies the management of the frontend, backend, and supporting services.

In this configuration, the **Flask** backend runs as a separate service and handles API requests, chatbot processing, profile management, and recommendation-related operations. The frontend runs as a separate service and provides the user interface developed with React and Vite. The use of separated services helps maintain a clear distinction between the client-side and server-side parts of the system.

Nginx is used as a reverse proxy to route requests between the frontend and backend services. In this setup, API requests can be directed to the backend service, while the remaining requests can be served through the frontend application. This organization helps centralize access to the application and simplify communication between its components.

Gunicorn is used as a WSGI process manager for serving the Flask backend in a more structured runtime environment. It helps manage backend execution and supports a more stable deployment configuration than the basic development server.

Persistent storage is also considered in the deployment configuration so that application data, such as the SQLite database, can remain available across service restarts. Overall, this deployment configuration supports a more organized execution of the RequAI system by separating services, managing communication through a reverse proxy, and preserving the data required by the application.

2.3 Evaluation and Result

This section presents the evaluation of the proposed system and the results obtained during experimentation. It focuses on assessing the chatbot approaches, the quality of the generated responses, the relevance of job matching, and the usability of the implemented interface.

2.3.1 Evaluation Methodology

This subsection explains how the evaluation was organized. It presents the main objectives of the evaluation, the test data and scenarios used, and the criteria selected to measure the performance and usefulness of the system.

2.3.1.1 Evaluation Objectives

The evaluation of RequAI aimed to assess the main components of the proposed chatbot system. First, it evaluated the ability of the intent classification module to identify user requests related to job search, ANEM information, CV assistance, salary questions, greetings, and other recruitment-related needs.

Second, the evaluation measured the capacity of the knowledge retrieval component to retrieve relevant information from the prepared recruitment knowledge base. This part focused mainly on ANEM procedures, Wassit Online guidance, required documents, registration steps, and employment-related information.

Third, the evaluation examined the job recommendation component, which compares candidate profiles with job offers using textual similarity. The objective was to observe whether the system could suggest relevant job offers based on skills, experience, target job, and profile information.

Finally, the evaluation included exploratory observations related to fairness and security. The fairness observation aimed to see whether identity-related information could influence recommendation scores, while the security validation tested how the chatbot reacts to suspicious or abnormal inputs.

2.3.1.2 Test Data and Evaluation Scenarios

The intent classification module was evaluated using a labeled test set containing several predefined intent classes. These classes included job search, CV upload, ANEM FAQ, bias check, salary inquiry, and greeting. The data was divided into training and testing parts in order to evaluate the classifier on unseen examples.

The knowledge retrieval component was tested using a prepared knowledge base containing ANEM-related FAQ entries and procedure information. The evaluated categories included registration, job search, unemployment benefits, youth programs, discrimination, salary rights, and training.

For job recommendation, candidate profiles and job offers were represented as textual data. The system compared candidate information with job descriptions in order to retrieve the most relevant job offers.

The fairness observation used different CV/profile variants while keeping the same professional content. The objective was to observe whether identity signals, such as name, gender, or location, could affect the recommendation score.

The security validation was tested using normal user messages, SQL injection examples, XSS-like inputs, oversized messages, and agitated user expressions.

2.3.1.3 Evaluation Criteria and Metrics

Several criteria were used during the evaluation. For intent classification, accuracy, precision, recall, and F1-score were used to measure the ability of the model to correctly classify user requests.

For knowledge retrieval and job recommendation, cosine similarity was used as the main relevance measure. It allowed the system to compare the textual representation of user queries, candidate profiles, job offers, and knowledge elements.

For the fairness observation, the comparison was based on the difference between the recommendation score obtained with visible identity signals and the score obtained after masking these signals. For security validation, the main criterion was the ability to detect suspicious inputs while allowing normal messages to continue through the chatbot pipeline.

2.3.2 Evaluation of the First Chatbot Approach

This subsection presents the evaluation of the first chatbot approach developed for RequAI. This approach represents the initial NLP-based pipeline of the system. It includes intent classification, knowledge retrieval, job recommendation, exploratory fairness observation, and basic security validation.

2.3.2.1 Intent Classification Results

The intent classification module was evaluated on a held-out test set. The classifier used TF-IDF-based text representation to transform user messages into numerical features. Character-level representation was useful for handling multilingual input, spelling variation, and informal expressions, especially in Algerian Darija.

Table 2.5 presents the obtained classification results.

Table 2.5: Intent classification performance on the held-out test set

Intent Class	Precision	Recall	F1-score	Support
job search	0.77	0.91	0.83	11
cv upload	1.00	1.00	1.00	7
faq anem	0.80	0.80	0.80	10
bias check	1.00	0.86	0.92	7
salary inquiry	0.62	0.71	0.67	7
greeting	1.00	0.60	0.75	5
Macro average	**0.87**	**0.81**	**0.83**	**47**
Weighted average	**0.85**	**0.83**	**0.83**	**47**

The classifier achieved an overall accuracy of approximately **83%**. The best result was obtained for the *cv upload* intent, while the lowest result was observed for the *salary inquiry* intent. This can be explained by the similarity between salary-related questions and job search requests, especially when users ask about salaries for specific job positions.

The multilingual tests also showed that the classifier could process user queries in English, French, Arabic, and Algerian Darija with acceptable confidence. The results suggest that character-level TF-IDF is useful for multilingual and informal chatbot input.

2.3.2.2 Knowledge Retrieval and Response Results

The knowledge retrieval component was evaluated using the prepared ANEM and recruitment knowledge base. The knowledge base was divided into smaller chunks, and each user query was compared with these chunks using TF-IDF representation and cosine similarity.

Table 2.6 presents the retrieval results by category.

Table 2.6: Knowledge retrieval results by category

Category	Avg. Retrieval Score	Avg. Chunks Retrieved	Answer Found
Registration	0.0234	3	100%
Job search	0.0189	2	100%
Unemployment benefits	0.0212	3	100%
Youth programs	0.0198	2	100%
Discrimination	0.0241	3	100%
Salary rights	0.0223	2	100%
Training	0.0204	3	100%
Overall	**0.0214**	**2.6**	**100%**

The results show that the retrieval component was able to find relevant knowledge elements for the evaluated categories. This confirms that organizing the knowledge base into smaller chunks can help the chatbot retrieve useful information for procedural and informational questions. However, some informal or Arabic expressions may still require better normalization to reduce retrieval confusion.

2.3.2.3 Job Recommendation Results

The job recommendation component used TF-IDF representation and cosine similarity to compare candidate profiles with job offers. The comparison was based on important textual information such as skills, experience, target job, job title, job description, responsibilities, qualifications, and location.

A real CV example was tested to observe the behavior of the recommendation component in a more realistic case. The system returned software engineering as the top recommended job category, which matched the technical profile used during the test.

Table 2.7: Real CV recommendation results

Mode	Average Similarity	Top Recommended Job	Salary Range
Original profile	0.3060	Software Engineer	79,799 – 239,399 DZD/month
Masked profile	0.3192	Software Engineer	79,799 – 239,399 DZD/month
Delta	**-0.0132**	Score reduced in original profile	

These results show that the similarity-based recommendation mechanism can provide relevant job suggestions. However, the recommendation quality depends on the quality of the candidate profile text and the completeness of the job offer representation.

2.3.2.4 Fairness and Security Observations

An exploratory fairness observation was conducted to examine whether identity-related signals could influence recommendation scores. Several CV variants were compared while keeping the same professional content. The results showed that visible identity information, such as name, gender, or location, could change the similarity score in some cases. After masking these signals, the scores became closer to the baseline. This observation highlights the importance of fairness and responsible use in recruitment recommendation systems.

The security validation showed that the chatbot could detect suspicious inputs, such as SQL injection examples and abnormal user messages. Normal messages were still processed, which indicates that a basic validation layer can help protect the chatbot interaction without blocking legitimate users.

2.3.2.5 Summary of Findings

Overall, the evaluation of the first chatbot approach showed that the initial NLP-based pipeline can support the main functions of RequAI. The intent classification module achieved acceptable results, with an overall accuracy of approximately **83%**. The knowledge retrieval component was able to retrieve relevant information from the prepared recruitment knowledge base, and the recommendation component produced relevant job suggestions using TF-IDF and cosine similarity.

The evaluation also highlighted some limitations. Salary-related questions may be confused with job search requests, informal multilingual expressions still require better normalization, and recommendation systems must be carefully evaluated to avoid the influence of non-professional identity signals. These observations are useful for improving the next versions of the chatbot and enriching the experimentation phase.

2.3.3 Evaluation of the Second Chatbot Approach

The evaluation of the second chatbot approach follows a module-level organization because the approach is composed of several NLP components that work together to support multilingual and context-aware interaction. The evaluation therefore focuses on three main components: language detection, multilingual intent classification, and entity extraction. In addition to quantitative evaluation, manual scenario testing was used to observe the behavior of the approach on realistic recruitment-related messages.

This evaluation is important because the second approach targets realistic user interaction in the Algerian recruitment context, where users may write in English, French, Arabic, Algerian Darija, Arabizi, or mixed-language forms. For this reason, the evaluation does not only measure general accuracy, but also observes how the system behaves with informal expressions, spelling variation, short messages, and multilingual inputs.

2.3.3.1 Language Detection Results

The language detection module was evaluated using four language classes: English, French, Arabic, and Darija. In this work, the Darija class includes Algerian dialect written in Arabic script as well as Arabizi or Latin-script Darija. The final dataset was divided into training, validation, and test sets. The training set contains `6,383` examples, while the validation and test sets contain `1,368` examples each.

Two models were evaluated for this task: Logistic Regression and Linear SVM. Both models used character-level TF-IDF representation. Character-level representation was selected because it is useful for language detection in short and informal messages. It can capture spelling patterns, Arabic characters, French accents, English word forms, and Arabizi sequences.

Table 2.8 presents the main results obtained on the standard validation and test sets.

Table 2.8: Language detection performance

Model	Validation Accuracy	Validation F1	Test Accuracy	Test F1
Logistic Regression	0.9971	0.9971	0.9934	0.9934
Linear SVM	0.9985	0.9985	0.9971	0.9971

The results show that both models achieved strong performance on the standard test set. Linear SVM obtained the best standard test accuracy with `0.9971`, while Logistic Regression also achieved high performance with `0.9934`. A second evaluation was also carried out using a challenge test set containing more difficult examples, including informal, short, and mixed-language messages.

Table 2.9 presents the challenge test results.

Table 2.9: Language detection challenge test results

Model	Challenge Accuracy	Number of Errors
Logistic Regression	0.9715	39
Linear SVM	0.9613	53

Although Linear SVM achieved better results on the standard test set, Logistic Regression performed better on the more realistic challenge test. It also provides confidence scores, which are useful in a chatbot context because they can help detect uncertain predictions. For this reason, Logistic Regression was selected as the final language detection model.

The error analysis showed that the most difficult cases were mainly mixed French/Darja and Arabic/Darja messages. For example, messages combining French expressions with Darija words, or Arabic script with foreign job-related terms, were sometimes confused. This is expected because mixed-language inputs are naturally difficult to assign to a single language class. These observations show the importance of confidence scores and clarification strategies when the detected language is uncertain.

2.3.3.2 Intent Classification Results

The intent classification module was evaluated using a multilingual recruitment intent dataset. The dataset contains `45,330` examples distributed across `18` intent classes and

6 language categories: English, French, Arabic, Algerian Darija, Arabizi, and mixed language. The data was divided into 36,264 training examples, 4,533 validation examples, and 4,533 test examples.

The intent classes cover the main types of user requests expected in the RequAI chatbot. These include job search, job matching, CV analysis, CV upload help, application guidance, application status, profile update, registration help, required documents, agency location, agency hours, ANEM procedures, password or account help, bot identity, bot capabilities, greeting, goodbye, and fallback.

Several models were evaluated for this task. The first model used Word TF-IDF with Linear SVM, while the second model used Character TF-IDF with Linear SVM. A transformer-based XLM-RoBERTa model was also tested as an advanced multilingual alternative.

Table 2.10 presents the main comparison between the Word TF-IDF and Character TF-IDF models.

Table 2.10: Intent classification model comparison

Model	Validation Accuracy	Test Accuracy	Validation Macro F1	Test Macro F1
Word TF-IDF + Linear SVM	0.9956	0.9965	0.9876	0.9906
Character TF-IDF + Linear SVM	0.9963	0.9974	0.9905	0.9937

The Character TF-IDF with Linear SVM model achieved the best overall result among the classical models, with a test accuracy of 0.9974 and a test macro F1-score of 0.9937. It also produced fewer validation errors than the Word TF-IDF model. This result is important because macro F1 gives more importance to all classes equally, including smaller classes such as greeting, goodbye, fallback, bot identity, and bot capabilities.

The per-language evaluation also showed that the Character TF-IDF model performed well across the different language categories. It obtained strong results on English, French, Arabic, Arabizi, Darija, and mixed-language messages. The mixed-language category achieved the highest score, while Darija remained more challenging compared to the other categories. This is expected because Darija has no single standardized written form and may appear in Arabic script, Latin script, or mixed spelling forms.

Table 2.11 presents the per-language results for the selected Character TF-IDF with Linear SVM model on the validation set.

Table 2.11: Per-language intent classification results using Character TF-IDF + Linear SVM

Language	Samples	Accuracy	Macro F1
Arabic	757	0.9974	0.9907
Arabizi	758	0.9947	0.9871
Darija	752	0.9920	0.9777
English	754	0.9973	0.9962
French	757	0.9960	0.9914
Mixed	755	1.0000	1.0000

The error analysis showed that most intent classification errors occurred between semantically close or short conversational intents. For example, some confusion appeared between `bot_identity` and `bot_capabilities`, between `greeting` and `goodbye`, and between some general or unclear requests and the `fallback` class. This type of confusion is understandable because short messages often contain limited information and may be ambiguous without dialogue context.

The final selected model for the intent classification module was Character TF-IDF with Linear SVM. This choice was based on its strong test performance, robustness to informal spelling, good behavior across languages, and lower computational cost compared to transformer fine-tuning.

2.3.3.3 Entity Extraction Results

The entity extraction module was implemented using XLM-RoBERTa for token classification. The objective of this module is to extract useful recruitment-related information from user messages. The model was trained using BIO tagging, where `B-` indicates the beginning of an entity, `I-` indicates the continuation of an entity, and `O` indicates tokens outside entities.

The entity extraction dataset contains `6,000` multilingual examples. It was divided into `4,800` training examples, `600` validation examples, and `600` test examples. The dataset covers English, French, Arabic, Algerian Darija, Arabizi, and mixed-language messages.

The model was trained to recognize several recruitment-related entity types, including `LOCATION`, `JOB_TITLE`, `JOB_FIELD`, `SKILL`, `EXPERIENCE`, `QUALIFICATION`, `SALARY`, and `WORK_TYPE`. These entities are important because they help the chatbot understand user constraints and preferences during job search, CV analysis, and recommendation scenarios.

Table 2.12 presents the main validation and test results of the XLM-RoBERTa entity extraction model.

Table 2.12: XLM-RoBERTa entity extraction results

Dataset	Accuracy	Macro F1	Weighted F1
Validation	1.0000	1.0000	1.0000
Test	0.9998	0.9998	0.9998

The results show very high performance on both validation and test data. The test accuracy reached `0.9998`, while the test macro F1-score reached `0.9998`. This indicates that the model learned the structure of the entity dataset very well.

The detailed classification report showed strong performance across all entity labels, including beginning and inside labels for location, job title, job field, skill, experience, qualification, salary, and work type. This means that the model was able to detect both single-token and multi-token entities.

The high scores reflect the quality and consistency of the prepared BIO-tagged dataset. Since the examples were generated from recruitment-related aliases and templates, the evaluation provides a clear view of how well the model recognizes the targeted entity categories within the designed RequAI scenarios.

Manual tests showed that the model can extract useful entities from multilingual examples. For example, from the sentence *"I am looking for a driver job in Oran with no diploma"*, the model extracted `driver` as a job field, `Oran` as a location, and `no diploma` as a qualification. From the sentence *"Je veux un stage en informatique à Alger"*, it extracted `stage` as a work type, `informatique` as a job field, and `Alger` as a location. Similar behavior was observed for Arabic, Darija, Arabizi, and mixed-language examples.

Some boundary variations were observed in informal messages. For example, in Darija sentences, the model may include an additional word with a location or qualification because of the informal structure of the sentence. These cases are handled as part of the normal error analysis of token classification tasks and show the importance of manual testing in addition to numerical metrics.

2.3.3.4 Manual Scenario Testing

In addition to quantitative evaluation, manual scenario testing was used to observe how the second approach behaves on realistic chatbot messages. This testing was applied to the language detection, intent classification, and entity extraction modules.

For language detection, manual examples included formal English and French messages, Arabic messages, Darija written in Arabic script, Arabizi messages, and mixed-language inputs. The model correctly identified many realistic examples such as English job search messages, Arabic job search requests, and Darija expressions such as *"rani nqaleb ala khedma"* or *"nheb nkhdem f oran"*. The most difficult examples were mixed inputs such as French sentences containing Darija words or Arabic sentences containing English recruitment terms.

For intent classification, the model was tested using new user messages that were not directly taken from the dataset. The results showed that the selected model correctly classified most common recruitment scenarios. For example, *"How can I upload my CV?"* was classified as `cv_upload_help`, *"Find jobs that match my profile"* was classified as `job_matching`, *"Where is the ANEM agency in Setif?"* was classified as `agency_location`, and *"What documents do I need for registration?"* was classified as `required_documents`.

The model also showed good behavior on Arabic and Arabizi examples. For example, " " was classified as `registration_help`, " " was classified as `required_documents`, and *"kifach nsajel f Wassit?"* was classified as `registration_help`.

For entity extraction, manual testing confirmed that the model can extract recruitment-related entities from different languages. Table 2.13 presents examples of manual entity extraction results.

Table 2.13: Examples of manual entity extraction results

User message	**Extracted entities**
I am looking for a driver job in Oran with no diploma	driver: `JOB_FIELD`; Oran: `LOCATION`; no diploma: `QUALIFICATION`
I know Python and SQL and I want a job in Constantine	Python and SQL: `SKILL`; Constantine: `LOCATION`
Je veux un stage en informatique à Alger	stage: `WORK_TYPE`; informatique: `JOB_FIELD`; Alger: `LOCATION`
nheb khedma chauffeur f oran bla diplome	chauffeur: `JOB_TITLE`; oran: `LOCATION`; diplome: `QUALIFICATION`
nheb salaire 7wali 60000 da f khedma security	60000 da: `SALARY`; security: `JOB_TITLE`

These manual tests show that the second approach can handle many realistic recruitment messages. They also confirm the importance of combining language detection, intent classification, and entity extraction. Together, these modules allow the chatbot to identify the language of the message, understand the user need, and extract useful details for the response.

2.3.3.5 Discussion of the Second Approach

The evaluation results show that the second chatbot approach provides strong multilingual understanding capabilities. The language detection module achieved high performance and showed useful behavior on both standard and challenge test sets. The use of character-level TF-IDF was suitable for short, informal, and multilingual messages. Logistic Regression was selected because it performed better on the challenge test and provides confidence scores, which are useful for chatbot decision-making.

The intent classification module also achieved strong results. Character TF-IDF with Linear SVM was selected as the main intent classifier because it obtained the best test macro F1-score among the classical models and performed well across the different language categories. This model is also efficient and easier to deploy compared to a transformer-based model, while still being robust to spelling variation and informal expressions.

The entity extraction module achieved very high results using XLM-RoBERTa. It was able to identify important recruitment-related entities such as locations, skills, job titles, job fields, qualifications, salary expressions, experience, and work types. These entities are useful for job search, job recommendation, CV analysis, and personalized guidance.

The analysis also highlighted some important observations. First, mixed-language messages are more complex than single-language messages because they may combine French, Darija, Arabic, and English recruitment terms in the same sentence. Second, short messages may be ambiguous when they contain very limited information. For example, greetings, fallback messages, bot identity questions, and bot capability questions can sometimes overlap. Third, entity boundaries in informal Darija messages may vary because users do not always follow a fixed grammatical structure.

Overall, the second approach improves the multilingual and contextual understanding of RequAI. It shows that language detection, multilingual intent classification, and transformer-based entity extraction can work together to support a realistic recruitment chatbot. The module-level evaluation demonstrates that the main NLP components are functional, effective, and suitable for integration into the RequAI system.

2.3.4 Comparative Analysis of the Two Chatbot Approaches

The two chatbot approaches developed for RequAI have different objectives and technical characteristics. The first approach represents a structured classical NLP-based pipeline, while the second approach extends the system with stronger multilingual understanding and more advanced entity extraction.

The first approach is more controlled and explainable. It relies on predefined intents, response templates, rule-based entity extraction, retrieval resources, and similarity-based job recommendation. This makes it easier to test, interpret, and integrate into the chatbot workflow. It is especially useful for structured tasks such as FAQ answering, job recommendation, and predefined recruitment guidance.

The second approach focuses more on realistic multilingual interaction. It introduces language detection, multilingual intent classification, and transformer-based entity extraction. This makes it more suitable for user messages written in English, French, Arabic, Algerian Darija, Arabizi, or mixed-language forms. It also improves the chatbot's ability to extract important recruitment-related information such as location, skills, job title, qualification, salary, experience, and work type.

Table 2.14 summarizes the main differences between the two approaches.

Table 2.14: Comparison between the first and second chatbot approaches

Aspect	First Approach	Second Approach
Main objective	Build a structured NLP-based chatbot pipeline	Improve multilingual and contextual understanding
Input handling	Mainly controlled user requests	More realistic multilingual and informal messages
Language support	Limited multilingual handling	English, French, Arabic, Darija, Arabizi, and mixed language
Intent classification	Classical intent detection	Multilingual intent classification with model comparison
Entity extraction	Rule-based extraction using predefined lists and configuration files	XLM-RoBERTa token classification with BIO labels
Knowledge answering	Retrieval from prepared FAQ and procedure data	Retrieval-supported contextual response preparation
Job recommendation	Similarity-based matching using job and profile text representations	Can use extracted entities to support job search and matching scenarios
Explainability	High, because the pipeline is structured and controlled	Moderate, because transformer-based entity extraction is less directly interpretable
Strength	Simple, controlled, explainable, and easier to integrate	Stronger multilingual understanding and better handling of informal messages
Main limitation	Less flexible with informal and mixed-language input	Requires more prepared multilingual data and module coordination

Tthe first approach provides the main structured baseline of RequAI, while the second approach enriches the system with stronger multilingual understanding. The two approaches are therefore complementary. The first approach supports controlled and explainable chatbot behavior, while the second approach improves the ability of the chatbot to understand realistic user messages in the Algerian recruitment context.

2.3.5 Discussion

The evaluation results show that RequAI can benefit from combining structured NLP processing with multilingual language understanding. The first chatbot approach demonstrated the usefulness of a classical pipeline based on intent classification, entity extraction, knowledge retrieval, response templates, and similarity-based job recommendation. This approach is practical because it is explainable, modular, and easier to control during implementation.

The second chatbot approach showed stronger performance in multilingual understanding. The language detection module achieved high accuracy, the multilingual intent classifier performed well across several language categories, and the XLM-RoBERTa entity extraction model achieved strong results on recruitment-related entities. These results are important because the target users may communicate using different languages and informal writing styles, especially in Algerian Darija and Arabizi.

The comparison between the two approaches shows that they should not be seen as contradictory. Instead, they represent two complementary levels of the same chatbot system. The first approach provides the structured foundation of the chatbot, while the second approach improves the understanding layer and makes the interaction more suitable for realistic user messages.

From an implementation perspective, the best direction for RequAI is to use the strengths of both approaches. The multilingual components of the second approach can be used to improve the input understanding stage, while the structured modules of the first approach can be used for retrieval, response control, and job recommendation. This combination allows the chatbot to remain explainable while becoming more flexible and adapted to the Algerian recruitment context.

Overall, the experimentation confirms that RequAI can support recruitment assistance through a combination of classical NLP, multilingual classification, entity extraction, knowledge retrieval, and similarity-based job matching. The results also show that multilingual support is not an optional feature in this context, but an important requirement for making the chatbot usable by Algerian job seekers.

2.3.6 Limitations of the Current Implementation

Although the implemented version of RequAI demonstrates the main idea of an intelligent recruitment assistant, several limitations remain in the current implementation.

These limitations are related to data availability, model maturity, evaluation scope, system integration, and the experimental nature of some components.

First, part of the data used during experimentation was generated or simulated due to the limited availability of real recruitment-related conversational datasets. This was useful for testing chatbot behavior, intent classification, profile analysis, and job recommendation scenarios. However, generated data may not fully represent all real user expressions, especially informal language, mixed-language messages, and complex recruitment situations.

Second, the system was not fully connected to the official live Wassit Online platform or to real-time ANEM databases. The job offer data and recruitment knowledge used in the prototype were prepared and adapted for experimentation. Therefore, the current implementation should be considered as a prototype that demonstrates the proposed approach rather than a fully deployed official recruitment service.

Third, the first chatbot approach was more stable and easier to evaluate because it relies on classical NLP and machine learning techniques such as TF-IDF representation, intent classification, entity extraction, response templates, and cosine similarity. In contrast, the second chatbot approach was introduced as an experimental enrichment of the initial pipeline and is still under refinement. For this reason, its evaluation remains more qualitative and less complete than the evaluation of the first approach.

Another limitation concerns multilingual and informal language handling. Although the system considers several languages used in Algeria, including Arabic, French, English, Algerian Darija, and mixed-language messages, informal expressions remain difficult to process. Darija and Arabizi can vary greatly between users, which may affect intent detection, entity extraction, and knowledge retrieval.

The job recommendation component also has limitations. Since it mainly relies on textual similarity, the quality of recommendations depends on the quality of the candidate profile and job offer descriptions. If the profile is incomplete or if the job offer lacks clear information about skills, experience, or requirements, the matching result may become less accurate. Moreover, similarity-based matching does not fully capture deeper recruitment criteria such as personality, motivation, soft skills, or employer-specific preferences.

The evaluation of the system was also limited. The first approach was evaluated using prepared test scenarios and experimental data, but a larger evaluation with real job seekers, ANEM agents, or recruitment specialists would be necessary to better assess usability, response quality, recommendation relevance, and practical usefulness.

Finally, the current implementation mainly focuses on the job seeker side. Employer-side functionalities, such as publishing job offers, managing applications, communicating with candidates, or supporting employer decision-making, were not fully covered in this version. These features remain possible extensions for future work.

Overall, the current implementation of RequAI provides a functional and experimental foundation for intelligent recruitment assistance. However, further work is required to improve data quality, strengthen multilingual understanding, complete the second chatbot approach, expand evaluation with real users, and integrate the system more deeply with real recruitment services.

Conclusion

This chapter presented the proposed contribution of this work, which consists of RequAI, an intelligent recruitment chatbot designed to support job seekers in the context of Wassit Online and ANEM-related services. The chapter first introduced the general idea of the proposed solution, its objectives, and its role in improving user guidance, access to recruitment information, CV/profile analysis, skill extraction, and job matching.

The proposed architecture was then described through its main layers, including the user and interface layer, the intelligent processing layer, the data and knowledge layer, and the output layer. These layers show how the system receives user input, processes recruitment-related information, accesses prepared data and knowledge sources, and returns guidance or recommendations through the chatbot interface.

The chapter also presented the main functional modules of the system, including conversational assistance, natural language understanding, CV/profile analysis, skill extraction, job matching, and knowledge-based response generation. These modules were designed to work together in order to provide a more interactive and personalized recruitment assistance experience for job seekers.

From the implementation side, the chapter described the development environment, frontend and backend tools, database organization, AI and NLP tools, and the integration of the chatbot components. It also presented two chatbot approaches. The first approach represents the initial structured NLP-based pipeline, relying on intent classification, entity extraction, similarity-based retrieval, response templates, and job recommendation. The second approach was introduced as an experimental enrichment of the initial pipeline, aiming to support more flexible and context-aware interaction.

Finally, the evaluation section presented the main results obtained from the first chatbot approach. The results showed that the initial pipeline can support intent classification, knowledge retrieval, and job recommendation with acceptable performance. The chapter also discussed the preliminary status of the second approach, interface testing, and the main limitations of the current implementation. These limitations mainly concern data availability, multilingual informal language processing, evaluation scope, system integration, and the ongoing refinement of the second approach.

Overall, this chapter demonstrated that RequAI provides a functional and experimental foundation for intelligent recruitment assistance. The proposed system shows how conversational AI, NLP techniques, knowledge retrieval, and profile-based matching can be combined to improve job seeker support in the Algerian employment context. The next chapter presents the general conclusion of this work and discusses possible future improvements.

General Conclusion

This work focused on the development of RequAI, an intelligent recruitment chatbot designed to support job seekers in the context of Wassit Online and ANEM-related employment services. The main objective was to propose a conversational assistance system that improves access to recruitment information, guides users through employment procedures, and supports profile-based job search through natural language interaction.

The study showed that digital employment platforms are important for improving access to job opportunities, but they may still face limitations related to rigid filtering, keyword-based search, manual processing, multilingual interaction, and limited personalized guidance. These limitations motivated the proposal of an intelligent chatbot layer that complements existing recruitment services and helps job seekers interact with the system in a simpler and more natural way.

The proposed solution was designed as a web-based recruitment assistant with a chatbot interface called Requi. It allows users to ask questions, understand recruitment procedures, express job search needs, provide profile or CV-related information, and receive adapted guidance. The system combines conversational assistance, natural language processing, recruitment knowledge, profile analysis, and job matching in order to provide more useful support for job seekers.

The implementation and evaluation showed that RequAI can support the main functions expected from an intelligent recruitment assistant. The system was able to process user requests, retrieve relevant recruitment information, and provide job-related assistance based on candidate profiles and available data. The work also highlighted the importance of multilingual understanding, fairness, security, and reliable knowledge-based responses in recruitment chatbot systems.

However, the current implementation still has some limitations. Some data used during experimentation was generated or simulated, and the system was not fully connected to live official employment platforms or real-time ANEM databases. In addition, informal multilingual language, especially Darija, Arabizi, and mixed-language messages, still requires further improvement. A larger evaluation with real job seekers, ANEM agents, and recruitment specialists would also be useful to better assess the system in real conditions.

Overall, RequAI provides a functional and experimental foundation for intelligent recruitment assistance in the Algerian employment context. It shows how conversational AI and recruitment-oriented data processing can be combined to improve job seeker guidance and support more relevant access to employment opportunities.

Future improvements can focus on connecting the system to real and updated job offer data from official employment platforms, improving multilingual support for Darija, Arabizi, and mixed-language messages, and enriching job matching by considering soft skills, personality suitability, communication style, motivation, and professional behavior. This is important because a candidate may appear suitable based on a CV, but may not always match workplace expectations during later recruitment stages. The system can also be extended to include employer-side functionalities, such as publishing job offers, managing applications, communicating with candidates, and supporting candidate profile analysis.

Bibliography

[1] S. Bouguetaia and E. Mezouri. E-governance in the public sector in algeria: Challenges and progress. *Journal of Contemporary Issues in Business and Government*, 30(1), 2024.

[2] World Bank. Algeria economic monitor: Accelerating reforms to protect the algerian economy. Technical report, World Bank, 2021.

[3] P. Shrivastav, N. Shrivastav, and P. Ranjan. A study of role of online platforms in modern recruitment process. *Journal of Informatics Education and Research*, 5(1), 2025.

[4] R. K. Dahal and S. P. Joshi. The application of e-recruitment by job seekers and the impact of perceived values in nepal. *Journal of Management Changes in the Digital Era*, 2024.

[5] R. Deepthi, P. Mandloi, R. Kakhandaki, S. Fiza, and S. S. Teja. An intelligent skill-based matching framework for automated recruitment using web and machine learning technologies. *International Journal of Research in Engineering and Science*, 14(1):120–125, 2026.

[6] OECD. A new dawn for public employment services. Technical report, OECD, 2024.

[7] J. Bersin. Conversational ai in recruiting. Technical report, Josh Bersin Company, 2021.

[8] C. Amanova, S. B. Ozgur, F. Msilu, and F. Demir. Demystifying the user experience: A case study on online job search engines. *American Journal of Qualitative Research*, 5(2):233–250, 2021.

[9] Agence Nationale de l'Emploi. Wassit online, 2026. Official Wassit Online platform. Available at: `https://wassitonline.anem.dz/`.

[10] Ministère du Travail, de l'Emploi et de la Sécurité Sociale. Politique nationale de l'emploi, 2026. Available at: `https://www.mtess.gov.dz/fr/politique-nationale-de-lemploi/`.

[11] Agence Nationale de l'Emploi. Offres disponibles – wassit online, 2026. Available at: `https://wassitonline.anem.dz/SearchOffers`.

[12] Agence Nationale de l'Emploi. Demandeur – wassit online, 2026. Official Wassit Online job seeker page. Available at: `https://wassitonline.anem.dz/Account/RegisterApplicant`.

[13] Agence Nationale de l'Emploi. S'authentifier – wassit online, 2026. Available at: `https://wassitonline.anem.dz/account/loginApplicant`.

[14] Agence Nationale de l'Emploi. Nouvelle inscription employeur – wassit online, 2026. Available at: `https://wassitonline.anem.dz/Account/RegisterEmployee`.

[15] Agence Nationale de l'Emploi. Forsati – demandeur d'emploi autonome, 2026. Available at: `https://wassitonline.anem.dz/postulation/LandingPage`.

[16] G. Caldarini, S. Jaf, and K. McGarry. A literature survey of recent advances in chatbots. *Information*, 13(1), 2022.

[17] O. Dogan and O. F. Gurcan. Enhancing e-business communication with a hybrid rule-based and extractive-based chatbot. *Journal of Theoretical and Applied Electronic Commerce Research*, 19(3):1984–1999, 2024.

[18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

[19] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020.

[20] E. Senger et al. A survey on skill extraction and classification from job posts and resumes. In *Proceedings of NLP4HR*, 2024.

[21] X. Yu et al. Confit v2: Improving resume-job matching using data augmentation and contrastive learning. In *Findings of ACL*, 2025.

[22] T. Pires, E. Schlinger, and D. Garrette. How multilingual is multilingual bert? In *Proceedings of ACL*, pages 4996–5001, 2019.

[23] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

[24] OECD. Ai language models: Technological, socio-economic and policy considerations. Technical report, OECD, 2023.

[25] Y. Zhang, Y. Li, T. Fu, F. Shi, J. Sun, Y. Zhang, et al. A survey on hallucination in large language models. *Computational Linguistics*, 51(4), 2025.

[26] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

[27] M. Miranda, E. S. Ruzzetti, A. Santilli, F. M. Zanzotto, S. Bratières, and E. Rodolà. Preserving privacy in large language models: A survey on current threats and solutions. *arXiv preprint arXiv:2408.05212*, 2024.

[28] Amazon Web Services. What is rag? – retrieval-augmented generation ai, 2026. Available at: `https://aws.amazon.com/what-is/retrieval-augmented-generation/`.

[29] SmartRecruiters. How generative ai recruitment chatbots supercharge hiring teams, 2025.

[30] Manish Raghavan, Solon Barocas, Jon Kleinberg, and Karen Levy. The myth of ai fairness in hiring. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAccT)*. ACM, 2020.

[31] OECD. Artificial intelligence and labour market matching. Technical report, OECD, 2023.

[32] H. Kavas et al. Multilingual skill extraction for job vacancy–job seeker matching. In *Proceedings of GenAIK*, 2025.

[33] A. Afzal et al. Candidate profile summarization: A rag approach with synthetic data generation for tech jobs. In *Proceedings of RANLP*, 2025.

[34] L. Ranaldi, F. Ranaldi, F. M. Zanzotto, B. Haddow, and A. Birch. Improving multilingual retrieval-augmented language models through dialectic reasoning argumentations. In *Proceedings of EMNLP*, pages 9064–9085, 2025.

Acronyms

AI Artificial Intelligence

ANEM Agence Nationale de l'Emploi

BERT Bidirectional Encoder Representations from Transformers

CNAS Caisse Nationale des Assurances Sociales

CORS Cross-Origin Resource Sharing

CSS Cascading Style Sheets

CV Curriculum Vitae

FAQ Frequently Asked Questions

GPT Generative Pre-trained Transformer

HR Human Resources

HTTP HyperText Transfer Protocol

IDE Integrated Development Environment

LLM Large Language Model

mBERT Multilingual Bidirectional Encoder Representations from Transformers

NLP Natural Language Processing

NTIC New Technologies of Information and Communication

OECD Organisation for Economic Co-operation and Development

PDF Portable Document Format

RAG Retrieval-Augmented Generation

SQL Structured Query Language

TF-IDF Term Frequency–Inverse Document Frequency

UI User Interface

UML Unified Modeling Language

Startup Project Annex

People's Democratic Republic of Algeria

Ministry of Higher Education and Scientific Research

University of Abdelhamid Mehri - Constantine 2

Business name: **RequAI**

Trademark image:

Project title

RequAI: Smart Recruitment Assistant Chatbot Extension for Wassit Online

Project for obtaining a startup certificate under Ministerial Decree 1275

Supervisor

Supervisor	Specialization
Samira Telghamti	Faculty of New Information and Communication Technologies

Work Team

Student Name	Specialization
Benameur Khaoula	Faculty of New Information and Communication Technologies
Nedjem Mohamed Anis	Faculty of New Information and Communication Technologies

Project description

Field of Activity: Digital services and innovative AI-based applications in the field of recruitment and employment assistance.

The Initial Idea

Project Title: RequAI: Smart Recruitment Assistant - Chatbot Extension for Wassit Online

RequAI is an innovative digital project in the field of recruitment assistance and employment services. The idea started from the observation that many job seekers in Algeria face difficulties when searching for jobs or using online employment platforms such as Wassit Online, which is related to ANEM. Even though these platforms contribute to the digitalization of employment services, users may still struggle to find clear information, understand application steps, prepare required documents, and identify job offers that match their profiles.

To respond to this problem, RequAI proposes a smart assistant called Requi. It is a chatbot designed to guide job seekers in a simple and interactive way. Through the platform, users will be able to ask questions, search for job offers, understand recruitment procedures, and receive direct answers related to employment services. The objective is to make the job search process easier, faster, and more accessible, especially for young graduates, first-time job seekers, and people who need guidance during their professional integration.

The platform will also help improve job recommendations by taking into account the user's profile, skills, preferences, and the requirements of available job offers. Instead of depending only on basic search filters, RequAI aims to provide more personalized support. The system may also include features such as CV analysis, skill identification, job matching, and soft-skill evaluation in order to better assist users and improve the quality of recruitment guidance.

RequAI will be developed as a web-based platform that combines a chatbot interface, employment-related information, and intelligent assistance tools. The project will be carried out by a team with skills in computer science, software engineering, and data processing, while also being guided by practical knowledge from the recruitment field. The development will take place within the Algerian employment context, with particular attention to ANEM services and the Wassit Online platform. In future versions, RequAI may also support employers by helping them publish job offers, manage recruitment requests, and communicate more effectively with job seekers and employment agencies.

2. The Proposed Values

RequAI, as a smart recruitment assistance platform, offers practical value by making the job search process easier, faster, and more organized for job seekers. It also supports the modernization of employment services by providing a digital assistant adapted to the Algerian recruitment context.

- **Easy Access to Employment Information:** RequAI helps job seekers quickly access information about job offers, application steps, required documents, and recruitment procedures in one place.
- **Clear Guidance for Job Seekers:** The platform guides users step by step during their job search journey, especially those who do not clearly understand recruitment procedures or how to use online employment services.
- **Interactive Assistance:** Through the chatbot Requi, users can ask questions and receive simple, direct, and understandable answers instead of searching through long pages or different sources.
- **Personalized Job Support:** RequAI helps users find job opportunities that are more suitable to their profile, skills, preferences, and professional goals.
- **CV and Profile Improvement:** The platform can help users better understand their CV, identify their strengths, and improve the way they present their skills to employers.
- **Time and Effort Reduction:** RequAI reduces the time and effort needed to search for job information, understand procedures, and identify suitable opportunities.
- **Support for Young Graduates and First-Time Job Seekers:** The platform is especially useful for users who are entering the job market for the first time and need simple guidance to understand the recruitment process.
- **Better Recruitment Orientation:** RequAI helps reduce confusion by directing users toward more relevant job offers and clearer employment steps, which can improve their chances of making better job search decisions.
- **User-Friendly Digital Experience:** The platform is designed to be simple, accessible, and easy to use, even for users who are not highly experienced with digital tools.
- **Support for Employment Service Digitalization:** RequAI contributes to the digital transformation of recruitment services in Algeria by offering a modern solution aligned with the needs of job seekers, ANEM services, and the Wassit Online environment.

3. Work Team

The RequAI project is carried out by a project team supported by academic supervision. The team is responsible for developing the project idea, organizing the work, and preparing the prototype, while the supervisor provides guidance and methodological support throughout the project development.

1. Telghamti Samira

Position: Researcher and university lecturer

Specialization: Software Engineering and Artificial Intelligence

Function: Supervisor

Role:

- Supports the team in structuring and developing the startup project idea. - Provides methodological and strategic guidance throughout the project. - Helps the team define clear objectives and organize the main stages of the work. - Guides the validation of the proposed solution and its relevance to the targeted problem. - Encourages autonomy, innovation, and effective collaboration among team members. - Ensures that the project remains coherent, realistic, and aligned with the expected startup framework.

2. Benameur Khaoula

Position: Master 2 computer science student

Specialization: Software Engineering and Intelligent Systems

Background:

- Master's student in Computer Science, specializing in Software Engineering and Intelligent Systems.

- Academic background in software development, databases, information systems, artificial intelligence, and data processing.

- Practical experience with chatbot development, Natural Language Processing, and machine learning models.

- Knowledge of recruitment platforms and employment services, especially in the Algerian context.

- Interest in developing intelligent digital solutions that support users and improve online services.

Role:

- Worked on refining the project idea, objectives, functional needs, and expected services.
- Worked on the chatbot concept and recruitment assistance features.
- Contributed to the market study, value proposition, and Business Model Canvas preparation.
- Worked on the chatbot interface design and user experience.
- Developed and tested a chatbot approach to support job seeker guidance and recruitment-related questions.
- Worked on the technical structure of the RequAI platform, including the organization of the main system components.
- Worked on fieldwork, collecting and organizing recruitment-related information for the chatbot.
- Worked on the theoretical writing, thesis structure, project documentation, and organization of the written content.
- Collected practical information for cost estimation and startup-related financial planning.
- Contributed to prototype testing by checking platform behavior and correcting implementation issues.

3. Nedjem Mohamed Anis

Position: Master 2 computer science student

Specialization: Networks and Distributed Systems

Background:

- Master 2 computer science student specializing in Networks and Distributed Systems.
- Familiar with full-stack development, including frontend, backend, and system integration.
- Skilled in developing interactive web platforms and managing communication between system components.
- Interested in AI-based applications, chatbot systems, and intelligent recruitment platforms.

Role:

- Worked on refining the project idea, objectives, functional needs, and expected services.
- Worked on the chatbot concept and recruitment assistance features.
- Contributed to the market study, value proposition, and Business Model Canvas preparation.
- Worked on the chatbot interface design and user experience.
- Developed and tested a chatbot approach to support job seeker guidance and recruitment-related questions.
- Developed the frontend and backend of the web-based prototype.
- Connected the main platform components and implemented the chatbot interaction interface.
- Worked on the final UI/UX implementation used in the prototype.
- Collected practical information for cost estimation and startup-related financial planning.
- Contributed to prototype testing by checking platform behavior and correcting implementation issues.

Timeline for Project Realization

Tasks	Months					Activities
	1	2	3	4	5	
Understanding the Project Context	X	X				Studying the Algerian recruitment context, especially ANEM services and Wassit Online, and identifying the main difficulties faced by job seekers when searching for jobs or understanding recruitment procedures.
Information and Needs Collection	X	X	X			Collecting information about job search steps, application procedures, required documents, common questions, user needs, and practical recruitment problems in order to define the main services that RequAI should provide.
Project Planning and Service Design		X	X			Defining the main objectives of the platform, the target users, the expected services, and the general user journey, including how job seekers will interact with the chatbot and receive guidance.
Preparation of Recruitment Content		X	X	X		Organizing employment-related information, frequently asked questions, recruitment steps, profile information, and job offer details in a clear structure that can be used by the platform to support job seekers.
Prototype Development			X	X		Developing the first version of the RequAI web platform, including the chatbot interface, user profile support, job search assistance, and basic recruitment guidance services.
Personalized Assistance Features			X	X	X	Adding features that help users receive more suitable job guidance, such as profile-based recommendations, CV and skills support, and better orientation toward relevant job opportunities.
Testing and Improvement					X	Testing the platform with different user situations, checking if the answers are clear, verifying if the guidance is useful, identifying problems, and improving the user experience.
Final Validation and Presentation					X	Preparing the final prototype demonstration, organizing the project documentation, validating the main results, and preparing the project for presentation within the startup framework.
Protection and Future Development					X	Studying the possible protection of the project idea, name, and digital content through appropriate legal or institutional procedures, while defining future improvements such as employer-side services.

Market Segment Overview for RequAI

Potential Market

The potential market for RequAI includes all individuals and organizations that may need support in job search, recruitment guidance, and employment services. Since the project is designed for the Algerian employment context, this market includes job seekers, students, young graduates, universities, career clubs, training centers, employment agencies, and recruitment support services.

In future versions, the potential market may also include employers and companies that need smarter tools to publish job offers, manage recruitment requests, and communicate more effectively with candidates.

Target Market

The main target market for RequAI is job seekers in Algeria, especially young graduates, first-time job seekers, and students preparing to enter the job market. This category was selected because it is directly affected by the problems that RequAI aims to solve, such as difficulty finding suitable job offers, lack of clear information about recruitment procedures, and limited guidance during the application process.

RequAI is particularly useful for users who need simple explanations, direct answers, and personalized support to better understand the job search process and improve their chances of finding suitable employment opportunities.

Justification for Choosing the Target Market

The target market was chosen because job seekers represent the main users of the proposed solution. Many candidates, especially young people entering the job market for the first time, face difficulties in understanding how recruitment platforms work, how to apply correctly, and how to present their profiles to employers.

By focusing on this segment, RequAI can provide real value through accessible guidance, faster access to employment information, and better orientation toward suitable job opportunities. This makes the project relevant to the needs of the Algerian employment market and aligned with the digital transformation of employment services.

Possibility of Agreements with Important Clients

RequAI may establish future partnerships or service agreements with important institutions and organizations, such as employment agencies, universities, career clubs, training centers, recruitment support services, and companies.

These agreements could help the project reach more users, improve the quality of employment guidance, and support institutions that already work with job seekers. In future versions, partnerships with employers may also allow the platform to provide recruitment support services for companies looking for suitable candidates.

Competition Intensity

The competition in the Algerian recruitment assistance market can be considered moderate to strong. It is moderate because the number of well-known Algerian online recruitment platforms is limited. However, it is also strong because some platforms already have visibility, experience, and an existing user base in the employment market.

Exact market share data for Algerian recruitment platforms is not publicly available. For this reason, the competition intensity is evaluated based on visible market presence, type of services offered, strengths, and weaknesses.

Direct Competitors

The main digital competitors and online alternatives of RequAI include Algerian recruitment platforms such as Emploitic and EmploiPartner, international job platforms, and social media job groups. These solutions are used by job seekers to search for employment opportunities and by recruiters or companies to find suitable profiles.

Emploitic is one of the most visible employment platforms in Algeria. Its main strengths are its strong presence in the Algerian market, the availability of many job offers, and its services for both candidates and recruiters. However, Emploitic mainly focuses on job search, job posting, and connecting candidates with recruiters. Based on its public services, it does not clearly provide a complete interactive assistant that guides job seekers step by step, explains recruitment procedures, answers employment-related questions, and supports profile improvement in a conversational way.

EmploiPartner is also an Algerian e-recruitment platform. Its main strengths are its recruitment experience, job listings, profile creation, and support for companies looking for candidates. However, it is mainly oriented toward job listings, sourcing, and recruitment connection. Based on its public services, it does not clearly offer the same type of job-seeker assistance proposed by RequAI, especially chatbot guidance, simple explanations of recruitment procedures, and personalized support for young graduates and first-time job seekers.

International job platforms can offer access to many opportunities and professional visibility. However, they are not specifically adapted to the Algerian recruitment context and do not provide local guidance related to Algerian employment procedures or the specific needs of local job seekers.

Social media job groups are widely used because they are free, familiar, and easy to access. However, the information shared in these groups may be unorganized, repeated, incomplete, or unreliable. Job seekers may also find it difficult to know which offers are serious and which steps they should follow.

Therefore, RequAI can differentiate itself by offering more than a job search platform. It aims to provide guidance, explanation, orientation, and personalized support during the job search journey. This gives RequAI a strong advantage for users who need help understanding what to do, how to apply, and how to improve their chances of finding suitable employment.

Indirect Competitors

Indirect competitors include non-digital or less structured alternatives that job seekers may use to find employment opportunities. These include traditional private recruitment agencies, personal networks, university career events, and direct contact with companies.

Traditional private recruitment agencies provide human contact and professional recruitment support. Their strengths are experience, direct communication, and personalized contact with some candidates and employers. However, they may not be available at all times, may require more time, and may not provide continuous support for all job seekers.

Personal networks are also commonly used to find job opportunities. Their main strength is trust, because they are based on recommendations and direct relationships. However, this method can limit access to opportunities, since not all job seekers have the same contacts or the same professional environment.

University career events and job fairs can help students and young graduates meet employers and discover opportunities. However, these events are occasional and limited in time and place. They do not provide continuous guidance during the whole job search process.

Compared to these alternatives, RequAI aims to offer continuous, organized, and accessible support. The platform can help job seekers at different stages of their journey, from understanding procedures to improving their profile and finding more suitable opportunities.

Strengths and Weaknesses of the Competitive Environment

The main strength of existing competitors is that they already have market visibility and experience in connecting candidates with job opportunities. Some of them are already known by Algerian job seekers and recruiters, which gives them an advantage in terms of trust and audience.

Their main weakness, compared to RequAI, is that they generally focus on displaying job offers, collecting candidate profiles, or connecting recruiters with applicants. They do not clearly provide full interactive guidance for the job seeker. This creates an opportunity for RequAI to position itself as a smart assistant that helps users understand recruitment steps, ask questions, improve their profile, and receive more personalized orientation.

For this reason, RequAI does not compete only by offering job search features. Its competitive advantage is based on assistance, simplicity, personalization, and adaptation to the Algerian employment context.

Initial Costs and Burdens

Description	Cost	Details
Administrative and legal setup	30.000 DA	Initial costs related to preparing administrative documents and basic registration procedures.
Project name and logo protection	28.000 DA	INAPI protection for the RequAI name and logo.
Workspace domiciliation / coworking space	15.000 DA	First month of domiciliation or coworking space.
Workspace security deposit	/	No caution required according to the coworking option.
Internet installation and first subscription	10.500 DA	Internet installation and first subscription for development and online demonstrations.
Domain name reservation	10.000 DA	Estimated domain name reservation cost. The price may vary between 5.000 DA and 10.000 DA.
Web hosting VPS	500 DA	First month of VPS web hosting.
Platform development fees	/	Depends on the complexity of the system. This cost is not included in the total.
User interface and user experience design	35.000 DA	Design of the platform interface and user experience.
Development and collaboration tools	20.000 DA	Online tools, storage, testing services, and collaboration resources.
Data collection and recruitment content preparation	30.000 DA	Collecting and preparing employment information, FAQs, and recruitment guidance content.
Data cleaning, organization, and processing	30.000 DA	Cleaning, organizing, and preparing data/content to be used by the platform.
Prototype testing and validation	20.000 DA	Testing the prototype, collecting feedback, and improving the user experience.
Unexpected initial expenses	20.000 DA	Reserve budget for small additional expenses during the first development phase.
Total of initial costs and charges	**249.000 DA**	Development fees are not included because they are variable.

Additional Initial Costs and Burdens

Description	Cost	Details
Legal consultation	50.000 DA	Legal advice related to startup creation, contracts, and project protection.
Communication materials	10.500 DA	Affiches, flyers, and presentation supports. Estimated as three basic lots at 3.500 DA each.
Branding and presentation materials	25.000 DA	Visual identity refinement, presentation design, and basic promotional materials.
Computer and technical equipment	300.000 DA	Laptops, storage devices, accessories, and technical equipment needed for development and demonstrations.
Office equipment	40.000 DA	Basic office material and small equipment for workspace organization.
Additional recruitment content review	15.000 DA	Reviewing and improving recruitment guidance content before final presentation or launch.
External development support, if needed	/	Optional cost depending on the technical needs of the platform. Not included in the total.
External AI/data support, if needed	/	Optional support for advanced AI or data processing tasks. Not included in the total.
Training or project support	25.000 DA	Short training sessions, workshops, or support related to entrepreneurship and project development.
Transport and field meetings	30.000 DA	Meetings with potential partners, users, employment support structures, or startup support organizations.
Total of additional initial costs and burdens	**495.500 DA**	Variable external support costs are not included.

Fixed Costs per Year

Description	Cost	Details
Annual workspace domiciliation / coworking space	180.000 DA	15.000 DA per month for 12 months.
Internet and telephone	75.000 DA	Annual communication costs for development, meetings, support, and online demonstrations.
VPS web hosting	6.000 DA	500 DA per month for 12 months.
Domain name renewal	10.000 DA	Annual renewal of the RequAI domain name.
Software and collaboration tools	120.000 DA	Digital tools, online storage, testing services, and team collaboration tools.
Platform maintenance and updates	200.000 DA	Maintaining the platform, correcting problems, and improving the service during the year.
Data update and content maintenance	60.000 DA	Updating recruitment information, FAQs, job-related content, and platform knowledge content.
Technical support	120.000 DA	Technical assistance, bug fixing, and support for platform users or partner organizations.
Marketing and communication	120.000 DA	Social media communication, awareness campaigns, posters, flyers, and promotion of the platform.
Legal and accounting follow-up	70.000 DA	Legal, accounting, and administrative follow-up after launching the startup activity.
Transport and meetings	60.000 DA	Transport costs for meetings with partners, clients, users, and support institutions.
Office supplies	50.000 DA	Basic office materials and small supplies used during the year.
Team compensation or part-time support	720.000 DA	Estimated annual compensation for part-time support or operational assistance.
Unexpected yearly expenses	70.000 DA	Reserve budget for additional costs that may appear during the year.
Total of fixed costs	**1.861.000 DA**	

Prototype

The following figures present the first prototype interfaces of the RequAI chatbot. The prototype includes a web version and a mobile version designed to provide job seekers with simple, interactive, and accessible recruitment assistance.

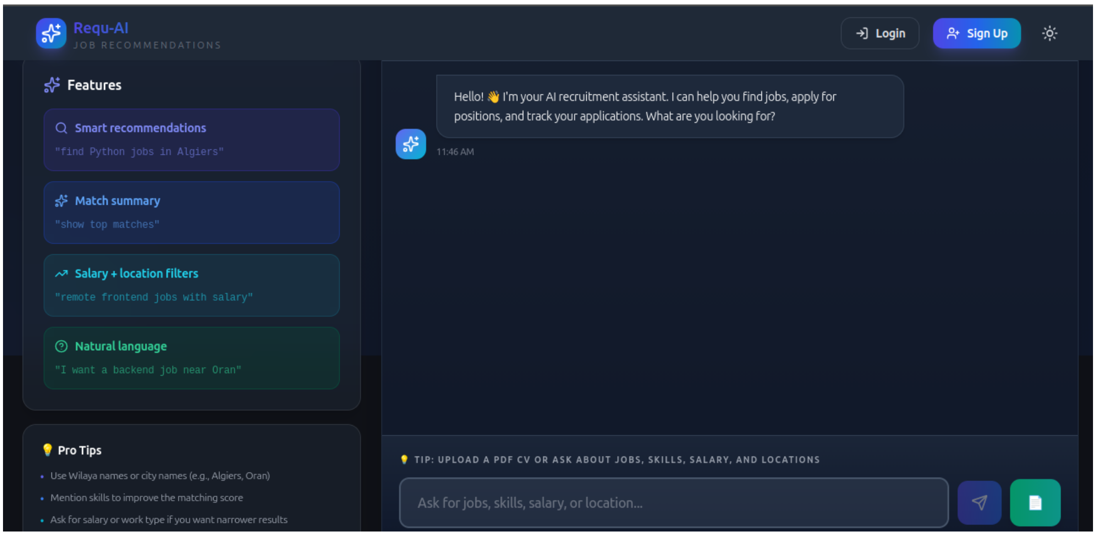

Figure 1: Web interface prototype of the RequAI chatbot.

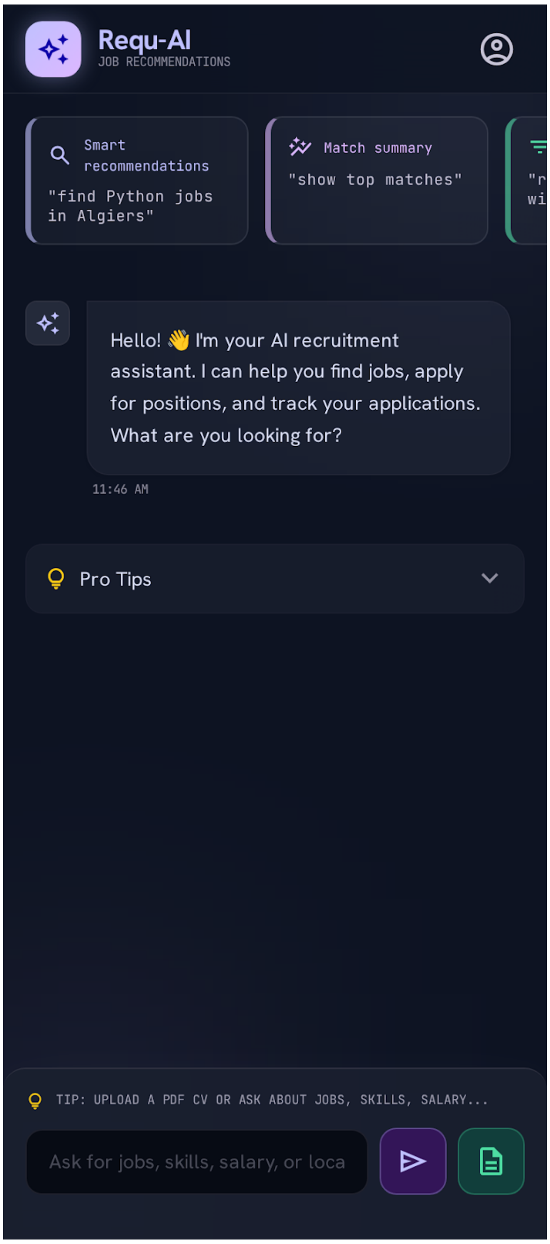

Figure 2: Mobile interface prototype of the RequAI chatbot.

Business Model Canvas

Key Partners:

- ANEM and Wassit Online ecosystem as a potential collaboration environment
- Local employment agencies and recruitment support services
- Training centers and professional development organizations
- entrepreneurship support programs

Key Activities:

- Development and improvement of the RequAI platform
- Design and maintenance of the Requi chatbot
- Recruitment content preparation and organization
- Data collection, cleaning, organization, and processing
- CV support, skill identification, and job matching assistance
- Prototype testing, validation, and user feedback analysis
- Partnership and communication management

Key Resources:

- Web-based platform and chatbot interface
- Recruitment knowledge base and employment guidance content
- AI/NLP tools and job matching support modules
- Recruitment-related data and user profile information
- Skilled team in software engineering, AI, and data processing
- practical recruitment insights
- Financial resources and workspace

Value Proposition:

- Easy access to employment information
- Clear guidance for job seekers
- Interactive assistance through the Requi chatbot
- Personalized job support based on user profiles and skills
- CV and profile improvement support
- Time and effort reduction during job search
- User-friendly digital experience
- Support for employment service digitalization in Algeria

Customer Relationships:

- Self-service assistance through the Requi chatbot
- Personalized guidance based on user needs and profiles
- Continuous support for job search and recruitment procedures
- Simple answers to employment-related questions
- Feedback collection to improve the platform
- Regular updates to improve guidance and recommendations

Distribution Channels:

- Web-based RequAI platform
- Mobile-friendly chatbot interface
- Social media and digital communication channels
- University events, career clubs, and workshops
- Partnerships with employment agencies and training centers
- Future collaboration with recruitment institutions

Customer Segments:

- Job seekers in Algeria
- Young graduates entering the job market
- First-time job seekers needing guidance
- Students preparing for professional integration
- career clubs, and training centers
- Employment agencies and recruitment support services
- Employers and companies in future versions

Cost Structure:

- Platform development, chatbot design, and maintenance costs
- Data collection, processing, and recruitment content preparation costs
- Hosting, domain name, workspace, internet, and technical equipment costs
- Marketing, legal, administrative, support, and update costs

Revenue Streams:

- Free basic access for job seekers with paid advanced features
- Premium subscription for personalized job seeker support
- Paid CV and profile improvement services
- Institutional subscriptions for training centers, and recruitment support services
- Training and workshops related to job search and platform use
- Custom support or integration for partner organizations
- Future employer-side service packages